A guide to the Global Macroeconomic Analysis Projections and Scenarios:

conditional forecasts, global scenarios, and policy counterfactuals in the browser[1]

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Source code (private repository at this writing):
https://github.com/matyasfarkas/BVARScenarioTool

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Abstract

This document describes GMAPS, the Global Macroeconomic Analysis Projections and Scenarios, a browser-based and desktop tool for quarterly macroeconomic projection and scenario analysis covering about fifty economies. Each economy is a Bayesian vector autoregression estimated with the hierarchical prior of Giannone et al. (2015) under a COVID-as-missing data-augmentation treatment; thirty-five of them are coupled into a global system with an endogenous United States, solved as a linear fixed point at display time. The user conditions any subset of variables on hard or soft paths — a mouse drag, an imported outside forecast with a chosen degree of trust — and a Durbin–Koopman smoother (Durbin and Koopman, 2002) propagates the constraints to every other variable and, on request, across borders. Public debt and the external position are derived from the estimated drivers by their accounting identities; long-run anchors shift the companion drift exactly and instantly; and monetary or fiscal counterfactuals are evaluated with impulse responses from the Macroeconomic Model Data Base (Wieland et al., 2012). We state the methods with their exact implemented conventions, show how each is operated from the interface, and document the data, the out-of-sample evaluation, and the replication path. All computation runs client-side; there is no server.

Keywords: BVARs, Conditional Forecasting, GVAR, Missing Values, Scenario Analysis, Kalman Filter, Policy Counterfactuals, DSGE Models

JEL classification: C32, C53, E37, E52, E58, F47

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1 Introduction

The tool described in this document comes in two forms: a web page and a desktop application (an installer for Windows; macOS builds from source). Both load the estimated models of fifty-two economies and let the user read, bend, and export their quarterly forecasts. There is no server-side computation: every number on the screen is produced client-side, so the web version is a static website, and the desktop application runs fully offline on the data pack it carries (Section 11.2). The source code, the estimation pipeline, and the data files are in one repository (private at this writing),

`https://github.com/matyasfarkas/BVARScenarioTool`

with the deployed application served from its `app/` static export. The tool comes as is; it is a research platform, not an official forecast of any institution.

The economic machinery is organized in three layers, applied in order:

(i) **Baseline.** Each economy is an n-variable Bayesian vector autoregression (BVAR) estimated on quarterly data with the hierarchical Minnesota prior of Giannone et al. (2015), treating the pandemic quarters as missing data (Section 3). Thirty-five economies are additionally coupled into a global VAR (GVAR) with a fully endogenous United States (Section 5). The baseline forecast is computed once, offline, and shipped with the model.

(ii) **Scenario.** The user conditions any subset of variables on desired paths. Conditioning is a partial-observation problem in the companion-form state space, solved by a Kalman filter and Durbin–Koopman smoother (Bańbura et al., 2015; Durbin and Koopman, 2002); a pinned value can be *hard* (exact) or *soft* (an outside forecast held with a chosen degree of trust), and it propagates to the other variables — and, on request, to the other economies — through the estimated covariance and trade structure (Sections 4 and 5).

(iii) **Counterfactual.** A monetary or fiscal policy counterfactual is overlaid on the scenario using impulse responses from a DSGE model of the user's choice, drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012), and re-conditioned through the BVAR so the remaining variables stay consistent with the estimated co-movements (Section 9).

On top of these, two accounting blocks derive what should not be estimated freely: public debt from the budget identity (Section 6) and the net external position from the balance-of-payments identity (Section 7); and a long-run anchor mechanism shifts the far end of any forecast onto a user-chosen value, exactly and instantly (Section 8).

Three display rules govern everything on screen, and we state them here because the rest of the document relies on them. *Data are data*: observed history is drawn solid; any model-filled value — a quarter not yet published, or one the publisher skipped, estimated from everything already observed — is drawn in amber and labelled, never disguised as an observation. *The model is frozen within the year*: quarterly data updates advance the forecast origin but never silently change the estimated

coefficients; re-estimation happens once a year, on all data published by then (Section 11). The data and the estimated model travel together as a *data pack*, separate from the program, so either can be updated without a new version of the tool (Section 11.2). *Judgment is visible*: whenever a forecast is moved away from the model, the screen says so — a conditioned chart is framed in blue and reads **CONDITIONED**, a long-run target puts an amber **judgement** badge on its chart and is counted in the line above the grid, and a counterfactual is drawn as its own dashed line. This includes the default long-run targets the tool ships, which are on at start (Section 8).

This document is both the user guide and the methodological documentation. It is organized so that each section states a method with its exact implemented conventions — the equations below were checked against the code, and file names are cited where the correspondence matters — and then shows how that method is operated from the interface, in numbered Examples that name the on-screen controls exactly. Section 2 gets a new user from a blank page to a first scenario. Sections 3–4 cover the estimation and the conditional-forecasting engine, Section 5 the cross-country system, Sections 6–7 the accounting blocks, Section 8 the long-run anchors, Section 9 the policy counterfactuals, and Section 10 the experimental decomposition of a scenario path into a DSGE model's shocks. Section 11 documents the data, the data packs and the vintage policy, Section 12 the out-of-sample evaluation, Section 13 the implementation and the replication path, and Section 14 concludes with limitations and known issues. Notation is collected in Appendix A; the per-economy variable tables — generated directly from the deployed data files — are Appendix B; a glossary is Appendix C.

2 Getting started

2.1 The screen

Open the tool's address in a modern browser. The website first asks for a user name and password (**Sign in**); the desktop application opens directly. The screen then reads *Loading data pack...* while the data pack is checked (Section 11.2), and the first economy's charts follow a few seconds later. The screen has five parts:

- the *top bar*: the tool's title with a small **i** button that opens **About GMAPS** (version, the data pack in use, how to cite, licence); the **Country** dropdown and the **Data & coverage** dialog; links to this manual (**User Manual**) and to the online documentation (**Docs**); the **Horizon** toggle (**5yr**, the default, or **3yr**, which zooms the charts to the first three years); the display **Units** toggle; the **Uncertainty** switch (**None**/**Quick**/**Full**); and the **Compute** button;
- the *left control panel*: the **Scenario** block (with **Reset**, **Reset all**, **Save**, **Load**, the **Cookbook** of ready-made scenarios, and **Export session** / **Import session**, which save and restore everything — every economy's pins, long-run targets and imported forecasts — as one file), **Long-run targets**, **External forecasts**, **Cross-border spillovers**, the **Policy Coun-**

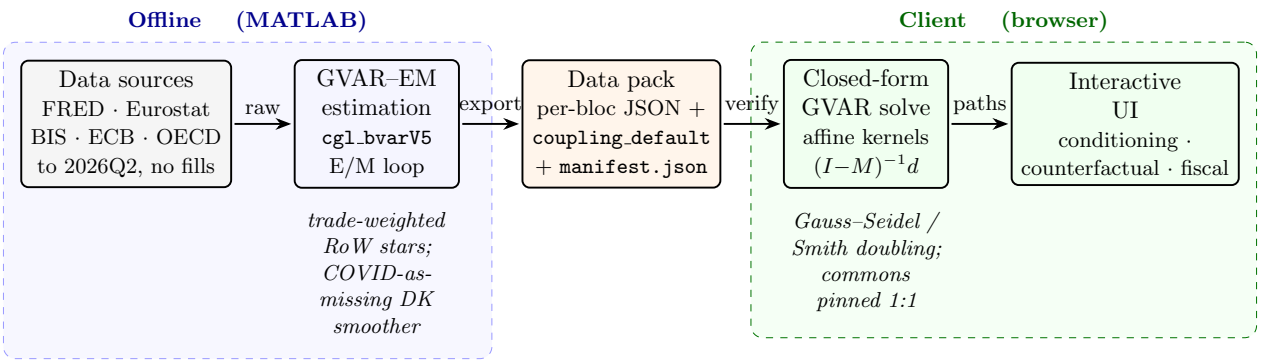


Figure 1: End-to-end system architecture of the EM-Bayesian Global VAR platform. Macro-financial data from published sources (FRED, Eurostat, BIS, ECB, OECD and national sources; every series to its latest published quarter, 2026Q2 at this writing, with no filled values, §11) feed the *offline* MATLAB GVAR–EM estimator: per-bloc Bayesian VARs (`cgl_bvarV5`) are re-fit in an EM loop that, at each sweep, rebuilds the trade-weighted rest-of-world "star" variables (ROW-DEM/ROWINF/ROWFIN) from partners' Durbin–Koopman-completed panels and re-estimates every bloc with the 2020Q2–Q4 *macro* columns masked (the foreign-rate star ROWFIN and all domestic financials kept observed; COVID-as-missing), iterating to max $|\Delta\text{star}| < 0.03$ (cf. §5.2). Each converged bloc is serialized to the eight-file JSON contract (`spec`, `companion_mean`, `companion_draws`, `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations`; the draws ship gzipped), plus the `estimation_mask` of model-drawn cells for every economy, `fiscal_meta` (41 economies) and `external_meta` (47), and the precomputed `coupling_default.json.gz` coupling artifact. Together with `manifest.json`, which records every file's SHA-256 fingerprint, these files form the *data pack*; the client checks each file against it before use (§11.2). In the *client* (browser or desktop app), the cross-country system is solved in closed form: the scenario delta obeys the affine fixed point $x = Mx + d$, evaluated by Gauss–Seidel sweeps or directly via $(I - M)^{-1}d$ (Smith doubling), with global commons (US WTI oil, US 10Y, US BAA, US equity) pinned 1:1 onto every RoW bloc. The interactive UI exposes conditioning, counterfactuals, and the stock-flow fiscal block (§6).

terfactual block, and — once a variable is conditioned — the quarter-by-quarter conditioning list;

- the *chart grid*: charts grouped in sections, in the same order for every economy (core, global commons, fiscal, financial, external, other domestic, rest of world, world aggregate; the euro members have no global commons). It opens with all panels. **Minimum set** reduces it to the standard set — policy rate, inflation, output, the fiscal block, one financial rate and the trade balance — and **Show all panels** brings everything back; in the reduced view, **+ N more** and **Add panel. . .** on a section header bring in one section or one panel. A panel can be hidden (the cross in its corner) or dragged to a new place, and **Reset view** restores the default. This is where the user clicks and drags;

- the *scenario bar* along the bottom: one tab per scenario (up to four, **+ Add Scenario**) and the **CSV**, **Excel**, **PNG**, and **PDF** export buttons;

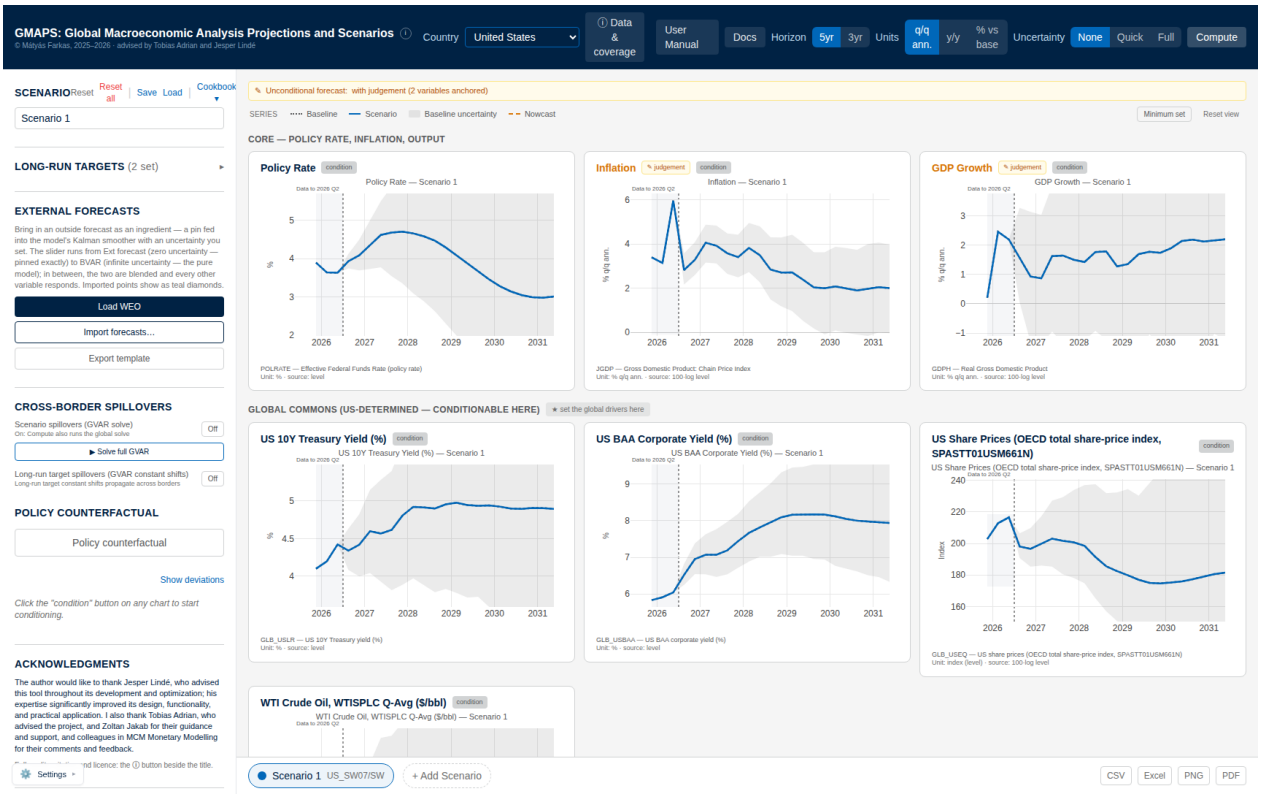


Figure 2: The screen at load (United States, app 4.1.0, data to 2026Q2). The top bar holds the country selector, the *Data & coverage* provenance panel, the User Manual and Docs links, the horizon toggle (5yr/3yr), the display-unit toggles (q/q annualized, y/y, % vs base), the uncertainty setting (None/Quick/Full), and Compute; the Settings drawer sits at the bottom left. The left rail holds the scenario manager, long-run targets, external forecasts (§4.5), cross-border spillovers (§5.5), and the policy counterfactual (§9). Each chart shows history, the dotted baseline, the solid scenario path, and the predictive band, with the history shaded up to the dotted Data to 2026 Q2 line; the charts are grouped in sections (here the core block and the global commons); judgement chips mark anchored variables (§8), and every chart carries either a condition chip (§4.4) or, where it cannot be steered, a derived chip.

- the **Settings** drawer at the bottom left (closed at start): the long-run-target options **Path to target** (**Steady-state**, the default, or **Rigid**) and **Multiple-target solve** (**Joint**, the default, or **Per-variable**) (Section 8.1); the **Experimental features** switch (Section 5.8) and the link to the **Data-release Monitor**, a separate page that shows how each data release moved the current-quarter estimates (Section 11.3); and the **Data pack** line, which names the pack in use and the quarter its data run to — in the desktop application with **Open data folder** and **Reload** for installing a newer pack (Section 11.2).

If a data file fails its check, a red banner at the bottom of the screen names it, and an economy whose data cannot be loaded shows a message instead of charts (Section 11.2). The **Country** dropdown has two groups: **Main AEs** (euro area, Japan, United Kingdom, United States) and **Other countries** (the rest, alphabetically); the grouping is by size, not by model. Thirty-five economies — every

economy except the seventeen euro-area member states — belong to the coupled cross-country system of Section 5, and a scenario built there can reach every other member. The euro-area member states are forecast on their own and receive shocks through the euro-area aggregate (Section 5.6); the **Cross-border spillovers** block says which case applies. Switching country keeps every economy's long-run targets and the pins set on the coupled economies (a euro member keeps its GDP and inflation pins); the counterfactual's rate and spending adjustments and any debt or primary-balance target are cleared (the chosen DSGE model and rule stay).

2.2 Reading one chart

Every chart follows the same grammar, and reading it correctly is most of the tool:

- *History* sits on a lightly shaded background that ends at a dotted vertical line labelled *Data to* and the last data quarter (*Data to 2026 Q2* in the current pack); the first forecast quarter is the first point to its right. The solid line in the shaded area is observed data. If a series' latest quarters have not been published, its history ends in an *amber dashed* tail with open circles — model-filled values (a *nowcast*, Section 11.3), never disguised as data. Hovering one reads *Model-implied nowcast (imputed).* A quarter the publisher skipped inside the history (US consumer prices and unemployment in 2025Q4, for example) carries an open amber circle that reads *Model-implied (no published value).*

- The *dotted grey line* is the **Baseline**: the model's forecast before any scenario, with any long-run targets applied (Section 8). The tool ships default targets that are on at start, so on the United States screen the **GDP Growth** and **Inflation** baselines already carry one; the line above the grid then reads *Unconditional forecast: with judgement* and the number of variables anchored (*model-based* when there are none). The pale grey fan around the baseline is the model's predictive uncertainty, shipped with the estimation (Section 3.5); it is there before anything is computed.

- The *solid coloured line* is the active scenario. Until the user conditions something it lies exactly on the baseline.

- The *header* carries a **condition** button (click to make the variable steerable; it turns blue and reads **CONDITIONED**, the chart gets a blue frame, and a **drag to set path** label appears — **drag to adjust policy** on the policy-rate chart in counterfactual mode). Charts that cannot be steered from this screen carry a grey **derived** label instead: the ratios computed by an accounting identity (**Δ Debt-to-GDP**, trade balance, current account, NFA), the rest-of-world and world read-outs, and, outside the United States, the global commons, which are set on the US screen. **Primary Balance** and **Debt-to-GDP** are derived too but can be conditioned (Section 6.3). An amber **judgement** badge and an amber title mark a variable whose long-run target has moved it off the model's estimate.

- The *footer* gives the variable's definition and unit.

Above the grid, the **Series** key names every line and band on the charts (**Baseline**, **Scenario**, **Baseline uncertainty**, **Scenario uncertainty**, and, when present, **External forecast**, **Counterfactual**, **Nowcast**); clicking an entry hides or shows that series on every chart. The x-axis marks years, and hovering a point gives its quarter and value. The strip along a chart's y-axis zooms it: the mouse wheel zooms, dragging pans, and a double-click restores the automatic range.

The **Units** toggle changes the lens, not the numbers: **q/q ann.** is the quarter-on-quarter annualized rate (the model's native display unit, equation (2) below); **y/y** the change on the same quarter a year earlier; **% vs base** the *difference* between scenario and baseline, centred at zero — a read-only impact view in which a second toggle, **Level** versus **Rate (IRF)**, selects between the cumulative level effect and the quarter-by-quarter response (Section 9.3). Charts that are already rates or ratios (the policy rate, unemployment, the fiscal ratios) are unaffected by the **q/q ann./y/y** choice; in **% vs base** they show the difference in percentage points. A set of index series is always drawn as an index level, because a trade-weighted index quoted q/q-annualized is not an informative number: the oil price and the US equity index (the global commons), every effective exchange rate, the US house-price and non-fuel commodity indices, and China's share-price index. Other economies' house-price and share-price indices are shown as growth rates.

Example 1 (A first scenario in five minutes)**.** *Open the tool; the **Country** dropdown reads **United States**, with **Units** on **q/q ann.** and **Uncertainty** on **None**. (1) On the **Policy Rate** chart, right-click in the forecast region about four quarters out, half a percentage point above the dotted baseline. The **condition** button turns blue and reads **CONDITIONED**, and a dot appears at the cursor. (2) Drag the dot to fine-tune it; release. The solid scenario line passes through the dot — only that quarter is imposed, and the model fills in the most plausible route through the point. (3) Scroll the grid: every chart has updated, not just the rate — growth cools, inflation eases with a lag — because the smoother re-draws the whole forecast consistently with the pin (Section 4). Other countries have not moved; cross-border transmission is off until requested (Section 5.5). (4) Set **Uncertainty** to **Quick** and press **Compute**: after a few seconds a band in the scenario's colour wraps the solid line — the middle two-thirds of plausible outcomes. At the pinned quarter the band closes onto the line, because an imposed value carries no uncertainty. With cross-border transmission off (the default) the rest of the band is the model's own predictive band moved onto the scenario path; with **Scenario spillovers** on, every draw is conditioned on the pin, the partners and the global variables, so the other quarters and variables carry conditional uncertainty as well (Section 4.6). (5) Clean up: in the left panel click **Reset** (clears this country's pins in the active scenario; long-run targets and imported forecasts stay — **Reset all** would also remove the default long-run targets) and set **Uncertainty** back to **None**. Clicking the **condition** button instead conditions* every *forecast quarter at its baseline value — a dot appears on each — so dragging one dot moves that quarter while all the others stay pinned. Right-click a dot to release that quarter, or switch quarters on and off in the conditioning list (**All** toggles them together; Section 4.4).*

Remark 1 (When to press Compute). The central scenario path updates live with every pin; **Compute** is never needed to see it. The button exists for the two calculations that are too heavy to run

on every mouse move: the scenario uncertainty bands (Section 4.6) and, when **Scenario spillovers** is on, the complete cross-country solve (Section 5.5; with the switch off, **Solve full GVAR** runs it). It is greyed out until **Quick**/**Full** uncertainty is selected or a long-run target is set — and because the default targets are on at start, it is already active on the United States screen and in twenty-six other economies. Pressed with **Uncertainty** on **None**, it switches to **Quick** first. **Quick** samples 50 posterior draws, **Full** 200. Bands are a snapshot — change a pin afterwards and the band no longer matches until **Compute** is pressed again.

3 The country models

3.1 Reduced form and data spaces

Each economy ("bloc") c is an n-variable VAR with p lags in reduced form,

$$y_t = c + \sum_{\ell=1}^{p} B_\ell \, y_{t-\ell} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $y_t \in \mathbb{R}^n$ collects the bloc's observables, $c \in \mathbb{R}^n$ is the intercept, $B_\ell \in \mathbb{R}^{n \times n}$ are the autoregressive matrices, and $\Sigma \in \mathbb{R}^{n \times n}$ is the innovation covariance. The forecast horizon is $H = 20$ quarters. The estimator uses $p = 3$ lags in its estimation space; the companion that ships is the equivalent level VAR with $p = 4$ lags (see *Estimation space* below). The deployed dimensions run from $n = 15$ (Singapore) to $n = 40$ (the United States); Appendix B tabulates every bloc. The estimated coefficients are stored as a $(1 + n_{\text{sr}} + np) \times n$ matrix $\hat{\beta}$ whose *first* row is the intercept, followed by n_{sr} seasonal-dummy rows (3 for the 34 economies with quarterly dummies, 0 otherwise; Section 3.3), and whose remaining np rows stack the lag coefficients $[B_1; \ldots; B_p]$ ($p = 4$ in the shipped file) — a convention the reader needs only twice, for the companion construction (7) and the model signature of Section 5.3.

Variables enter the VAR in one of two spaces, recorded per variable in the bloc's specification (`spec.json`) and never changed at run time:

- `log`: the stored series is $\ell_t = 100 \ln Y_t$. Displayed growth is quarter-on-quarter annualized,

$$g_t \;=\; 4\,(\ell_t - \ell_{t-1}) \;=\; 400\,(\ln Y_t - \ln Y_{t-1}), \tag{2}$$

  and year-on-year growth is $\ell_t - \ell_{t-4}$. The factor of 4 on the stored $100 \ln$ space — never 400, which would re-apply the 100 — is the single most load-bearing convention in the tool; every pin, band, and cross-country link below respects it.

- `lin`: the series enters in levels (interest rates in percent, unemployment, ratios to GDP). No growth conversion is applied.

Estimation space. The deployed model (in production since September 2026, re-estimated in October 2026) estimates up to three columns of each bloc in growth rates: real GDP, the bloc's price index (HICP or CPI; for the United States, the GDP deflator) and, where the bloc has a separate one, the GDP deflator. The estimator first-differences them into annualized quarterly growth $4\,\Delta\ell_t$ and gives them a white-noise own-lag prior centre (Section 3.2); every other column stays in its stored space. A VAR with p lags in this mixed space is exactly a level VAR with $p+1$ lags, and the exporter writes that level form (`BVAR_DIFF_GP`=1 in `export_bvar_gvar24_loop.m`). The application, the conditioning and the cross-country coupling therefore all work in the stored spaces above. One consequence: the stored GDP and price levels carry a unit root by construction, so their level forecasts trend, while their growth rates mean-revert.

Display units are chosen per variable for readability: activity and price series are shown as growth and inflation via (2); rates and ratios as levels in percent; and a fixed set of index series as the index level $\exp(\ell_t/100)$: every nominal effective exchange rate, the US trade-weighted dollar, the US non-fuel commodity index, China's real exchange rate and share prices, and the global oil and US equity series (`displayUnits.ts`). House prices, in every economy including the United States, and the other equity indices are shown as growth. Chart titles are the variable names in title case (the euro-area corporate bond yield proxy is titled *EA IG Bond Yield*); the line under each chart gives the variable code and its full name. Conditioning is always performed in the stored 100 ln space. Unemployment-rate bands are floored at zero for the United States, Japan, the United Kingdom and Canada only; policy-rate bands are not, since negative nominal rates occur.

3.2 The hierarchical prior

Estimation follows the hierarchical approach of Giannone et al. (2015), henceforth GLP: the Minnesota prior (Litterman, 1986; Doan et al., 1984) with its tightness treated as a hyperparameter and selected by maximizing its posterior. The prior centers each equation on a univariate own-lag model, with a sum-of-coefficients component (Doan et al., 1984) shrinking toward "a unit increase in every level raises its own forecast one for one." Two hyperparameters are sampled — the overall tightness λ and the sum-of-coefficients weight μ — by a random-walk Metropolis step on the log *posterior* of the hyperparameters: the marginal likelihood of the *completed* data (Section 3.4) plus independent Gamma hyperpriors on λ (mode 0.2, standard deviation 0.4) and μ (mode 1, standard deviation 1), following GLP's hierarchical construction. The chain is seeded by a draw from $N(\hat{\theta}_{\text{csminwel}}, c^2 H)$ around the `csminwel` mode and reflected into the box $[10^{-4}, 5]^2$; the eigenvalues of H are capped so that a proposal's standard deviation is at most a quarter of the box width. The hyperprior is not decoration: it holds λ away from the flat region of the likelihood. There the inverse Hessian H at the mode can be huge, and an unguarded seed draw can land at $\lambda \approx 549$ — so little shrinkage that the companion turns explosive. The reflection and the cap are the guards on the seed draw and on the proposals.

The production runs deliberately depart from two GLP defaults. `MNpsi` $= 0$ freezes the n residual scales ψ at the univariate AR(1) residual variances instead of estimating them, and `sur` $= 0$ drops

the single-unit-root (dummy-initial-observation) prior of Sims (1993) altogether; only the sum-of-coefficients dummy (`noc` = 1) is retained. That is why exactly two hyperparameters remain free, rather than the $n + 3$ a genuine default run would carry.

The estimator is the bundled MATLAB sampler `cgl_bvarV5.m` (all files named here are under `BVAR/MFBVAR/`). It calls `bvarGLPmf.m` (in `GLPreplicationWeb/`) once, to obtain the `csminwel` warm start for (λ, μ) and the AR(1)-based residual scales ψ, and then runs its own data-augmentation Gibbs loop. Its (β, Σ) draws come from `Support/BVAR.m`, and its Metropolis target is `logMLMFVAR.m`, also in `GLPreplicationWeb/`. No external MATLAB packages are required.

Own-lag prior mean: random walk or white noise. The prior mean on variable i's own first lag is δ_i, with $\delta_i = 1$ (random walk, `RW`) the default: for (near-)integrated series — log activity, log prices, nominal yields — "the best forecast of the level is the current level." For a genuinely *stationary* series, a mean-reverting ratio that fluctuates around a constant, the random-walk center is the wrong null: it shrinks toward a unit root and lets the projection drift at long horizons. Such variables carry `prior:"WN"` in the specification and are estimated with $\delta_i = 0$ (white noise): "the best forecast is the unconditional mean."

The mechanism is two lines in the prior construction, and the second is the stronger of the two. First the own-lag prior-mean diagonal is zeroed at the `WN` indices (`diagb(pos)=0` in `Support/BVAR.m`, with `pos` built from the per-variable tags), which moves the prior *centre* from a random walk to white noise. Second, the sum-of-coefficients dummy observation is blanked at those same indices while its regressor block is left intact (`ydnoc(pos,pos)=0`), so the dummy shrinks a `WN` variable's *summed* own-lag coefficients toward zero rather than toward one. This second channel is a dummy observation carrying weight $1/\mu$; at the estimated μ it dominates, and the posterior own-lag sum for a `WN` variable comes out close to zero rather than merely re-centred. Neither line is a hard constraint — the data can still pull a tagged series back toward persistence — but the combination is decisive in practice, which is exactly what the fiscal ratios need. In the deployed panel the `WN` tag is carried by the three fiscal GDP-ratios (Section 6) and by the growth-rate columns of the estimation space (real GDP, the price index and the GDP deflator, Section 3.1); every other variable is `RW` (Appendix B).

Foreign-demand pass-through: a units-consistent difference prior. *Status: not used by the deployed model.* This prior shipped in release 3.3.0 (August 2026) with the level model and was rolled back in 3.4.0. The growth-rate specification in production since September 2026 estimates GDP in growth rates with the standard prior on every star, and does not use it (the estimator prints "NO DVAX" in that mode). The construction is kept here because the option remains in the estimator.

The own-lag prior fixes how a bloc remembers its own past; it says nothing about how it should respond to the world. Left to the default, each foreign "star" coefficient is shrunk toward zero, so a bloc carries essentially no long-run tie to trade-weighted foreign demand — an advanced small open economy would converge on its domestic trend regardless of what its partners do. Theory says otherwise: a country's output should track foreign final demand with an elasticity close to

its *value-added export share* $\mu_i = \text{EXGR_DVA}_i/\text{VALU}_i$, the OECD–WTO TIVA measure of how much domestic value added is embodied in exports (Appendix B). We encode that as a soft, theory-based re-centring of the prior *mean* of the foreign-demand star (ROWDEM) in the GDP equation — and only there. The inflation and foreign-rate stars (ROWINF, ROWFIN) keep the standard zero-centred prior; an inflation-targeting central bank anchors prices domestically, and nominal-rate parity is rejected in the data, so neither earns a theory prior.

The re-centring must respect units, and this is where the construction earns its keep. The star ROWDEM is an *annualized* growth rate, $4\sum_j w_{ij}\,\Delta(100\log Y_j^*)$ (Section 5.1), while domestic GDP is a $100\log$ *level* carrying the random-walk own-lag prior, so under that prior its equation is effectively a quarterly first difference $\Delta(100\log Y_i)$. Matching an annualized rate to a quarterly increment costs a factor of four: a prior mean of $\mu_i/4$, not μ_i, encodes "a one-point rise in annualized foreign growth raises domestic *quarterly* growth by $\mu_i/4$," i.e. domestic growth tracks foreign growth with the value-added elasticity μ_i. The full μ_i on a single lag — the naïve re-centring — is a fourfold over-count that also lets the star's low-frequency movement accumulate into the domestic *level*, and ROWDEM, itself a random walk in the star block, can wander at long horizons. We therefore centre the pair

$$b^{(1)}_{\text{ROWDEM}\to\text{GDP}} = +\tfrac{\mu_i}{4}, \qquad b^{(2)}_{\text{ROWDEM}\to\text{GDP}} = -\tfrac{\mu_i}{4}$$

on the first two lags, so the star enters as a *difference* $\frac{\mu_i}{4}\,\Delta$ROWDEM: the two coefficients sum to zero, the prior injects a zero-mean term into domestic growth, and a sustained foreign boom lifts the domestic level once and then lets it settle rather than compounding without end. Transmission is real but transitory — the co-movement an open economy shows over the cycle, without importing the star's drift. (The reduced form carries no contemporaneous foreign term, so "contemporaneous" and "first lag" map onto lags one and two; the $+\mu_i/4$, $-\mu_i/4$ pattern is the level-space intuition $b_1 = \mu_i, b_2 = -\mu_i$ rescaled to the annualized star.)

Mechanically the prior mean is set at both places the sampler reads it — the posterior (β, Σ) draws and the Metropolis target for the hyperparameters — through a lag-aware `fmean` entry [ROWDEM, GDP, $\pm\mu_i/4$, ℓ] (`Support/BVAR_coint.m` and `cgl_bvarV5_coint.m`); the star's prior variance is tightened by $\kappa = 0.15$ so the posterior stays near the theory value, and ROWDEM is exempted from the sum-of-coefficients dummy so the two re-centrings do not fight. Like every prior here it is soft — the data can override it — and a bloc whose fiscal block is unstable (Japan) is better served letting the data identify the spillover, so it drops the re-centring. The honest limit is that this prior shapes the *pass-through* to domestic GDP, not ROWDEM's own dynamics: the star can still drift under its random-walk prior, but because the net pass-through is $b^{(1)} + b^{(2)} \approx 0$, domestic GDP inherits almost none of that drift — the level system stays anchored while the cross-country co-movement is restored. This is a genuine trade-off, and the alternative — differencing every log level into a growth-domain VAR — tames the star's own path but was measured to push the coupled-solve spectral radius above one; the difference prior keeps the contraction ($\rho(M) \approx 0.97$, §5) while recovering the spillover.

3.3 Seasonal dummies

A subset of the panel is reported *not seasonally adjusted*, so a regular quarter-of-year cycle sits in the level: NSA consumer prices (HICP/CPI for several economies and euro members), unemployment rates, house-price indices, India's government series, and the UK and Korean stock-flow adjustments. Left untreated, a pure autoregression can reproduce such a cycle only through a near-unit root close to the seasonal frequency, which then *explodes* the projection into a spurious sawtooth out of sample. The deterministic remedy is a set of quarterly dummies. For a bloc flagged seasonal, let $q(t) \in \{1, 2, 3, 4\}$ be the calendar quarter of observation t and collect the three non-base indicators

$$D_t = \big[\, \mathbf{1}\{q(t) = 2\},\ \mathbf{1}\{q(t) = 3\},\ \mathbf{1}\{q(t) = 4\} \,\big], \tag{3}$$

with the first quarter as the base. Each equation's regressor vector becomes $x_t = [\,1,\ D_t,\ y'_{t-1}, \ldots, y'_{t-p}\,]$, so the estimated coefficient block acquires three deterministic rows between the constant and the lag matrices,

$$\widehat{\beta} = \big[\, \underbrace{c}_{1};\ \underbrace{\gamma_{Q2};\ \gamma_{Q3};\ \gamma_{Q4}}_{n_{\mathrm{sr}}=3};\ \underbrace{B_1; \ldots; B_p}_{n \text{ lag blocks}} \,\big], \qquad k = 1 + n_{\mathrm{sr}} + n\,p, \tag{4}$$

where n is the number of endogenous variables, p the lag order (3 in the estimator's mixed space; the shipped level companion has $p = 4$, so $k = 1 + n_{\mathrm{sr}} + 4n$ there), and n_{sr} the number of "seasonal rows" (3 when the bloc is flagged, 0 otherwise — the legacy layout, left byte-identical). The dummy coefficients $\gamma_{Q2}, \gamma_{Q3}, \gamma_{Q4}$ carry a *diffuse* prior — the intercept variance V_c, centred at zero — and are zeroed in the sum-of-coefficients dummy observations, so the shrinkage machinery treats them like the constant and never pulls them toward a unit root (`BVAR/MFBVAR/Support/BVAR.m`; the COVID-aware wrapper `cgl_bvarV5.m` passes them through the `qoy` argument and carries the seasonal term in the simulation-smoother artificial path, so the Durbin–Koopman completion of Section 3.4 stays exact).

Which blocs are flagged is data-driven: a panel scan tags a bloc seasonal when at least one series shows calendar-$R^2 \geq 0.40$ with a display amplitude ≥ 2, and every coupon-lumpy bloc of Remark 7 is added because its `SFA` inherits the cash-versus-accrual timing. In the deployed system this is seventeen coupled blocs (CL, CN, CO, CR, EA, HK, ID, IN, IS, JP, KR, MX, NO, PE, TR, UK, ZA) and, since the October 2026 re-estimation, all seventeen euro-area members; the other eighteen coupled blocs keep $n_{\mathrm{sr}} = 0$.

In estimation the three dummies absorb the calendar cycle. **In the projection the deterministic term is seasonally adjusted, and the forecast carries no within-year swing.** `buildStateSpace` precomputes four quarter-specific transition constants $c + \gamma_q$ (with $\gamma_{Q1} \equiv 0$) and `c2AtStep` selects by the forecast step's quarter-of-year phase, but every forecast path in the application sets `seasonalAdjust = true`, which replaces all four by the annual mean $\bar{c} = c + (\gamma_{Q2} + \gamma_{Q3} + \gamma_{Q4})/4$ (`c2ByQ` in `app/src/lib/compute/kalman.ts`; the MATLAB exporter matches with `dfc` $= [.25, .25, .25]$). The projected path is therefore smooth, while the autoregression keeps the benefit of having had the seasonal soaked out of it, so the near-Nyquist root

never appears.

The quarter-specific constants are used only inside the estimator, where the data-augmentation step completes real calendar quarters (Algorithm 1). In the tool every path — forecasts, bands, and the coupled nowcast of the ragged edge — uses the annual mean $\bar{c}$. The phase is still recomputed from the tool's own date grid — the quarter following the last history point, wrapped $1 \ldots 4$ — and checked against the phase the data pack declares for each bloc (`qoyNext` in `manifest.json`, Section 11.2); a disagreement stops the load. Every $n_{\text{sr}} = 0$ path is unchanged to the byte.

3.4 COVID as missing data

The 2020 pandemic quarters are extreme outliers that would otherwise contaminate both the autoregressive coefficients and Σ. Rather than dummying or rescaling them, the estimator treats the pandemic window as *missing* and imputes it inside the estimation, by a data-augmentation Gibbs sampler in the sense of Tanner and Wong (1987): each sweep alternates

(i) a Durbin–Koopman simulation-smoother draw (Durbin and Koopman, 2002) of the missing cells given the coefficients (in the deployed runs a *fixed companion*, built once from the posterior-mode fit, Section 5.2; the textbook version uses the current draw), treating them as `NaN` observations in the companion-form state space of Section 4.1 (Algorithm 1);

(ii) a Metropolis–Hastings update of (λ, μ) on the completed-data marginal likelihood; and

(iii) a draw of (β, Σ) from the completed data.

The surrounding quarters stay in the sample and inform the imputation, so the pandemic history is preserved while the outlier does not drive the dynamics.

Two implementation choices matter. First, the production window is the *fixed* three quarters 2020Q2–Q4 for every coupled bloc; for the seventeen euro members the data layer withholds the four quarters 2020Q1–Q4, including their credit-to-GDP ratio. (A legacy single-bloc variant self-calibrates the window to the GDP trough searched within 2019Q4–2021Q1 — restricting the search matters, since an unrestricted trough lands on the wrong crisis for some economies, e.g. Korea's 2008Q4 — but the deployed GVAR estimator uses the fixed window.) Second, the masking is *macro-only*: over the window the domestic macro columns and the two macro star variables of Section 5.1 are set missing, while the financial columns (rates, yields, equity, house prices, spreads; the `isFinancial` tag in Appendix B) and the financial star stay *observed*. (The stars are masked by position, not by that tag: `spec.json` marks all three stars `isFinancial`, yet `ROWDEM` and `ROWINF` are withheld.) The smoother therefore imputes the pandemic macro holes conditional on financial series that carry genuine signal through 2020. The same device handles every other gap in the panel; each bloc's `estimation_mask.json` lists the cells the sampler drew as missing data (see the initial-condition paragraph below).

Two details of step (1) and step (2) are load-bearing. The artificial path carries the *same* deterministic term as the data — the intercept and, on the seasonal blocs, the applicable quarterly

Algorithm 1 Durbin–Koopman simulation smoother with partial observations (one draw inside the data-augmentation Gibbs sweep; `cgl_bvarV5.m`).

Input: system matrices T, Z, Q of Section 4.1; partial observations $\{y_t^{(o)}\}$ (COVID-masked and ragged cells absent); coefficients (β, Σ) (the fixed companion in the deployed runs).
Output: one draw $\tilde{\alpha}_{0:T}$ from $p(\alpha_{0:T} \mid y^{(o)}, \beta, \Sigma)$, hence imputed values for all missing cells, including those in the first p rows (option `nofill`).

1: ▷ (1) Forward simulation of an unconditional path.
2: Draw $\alpha_0^+ \sim N(M_0, P_0)$, with P_0 zero except on missing cells of the initial block (see below; in the legacy setting $P_0 = 0$ and $\alpha_0^+ = M_0$); **for** $t = 1, \ldots, T$: draw shocks and set $\alpha_t^+ = T\alpha_{t-1}^+ + c_\alpha(t) + \eta_t$, $y_t^+ = Z\alpha_t^+$, where $c_\alpha(t)$ is the intercept plus the applicable quarterly dummy coefficient when $n_{\text{sr}} > 0$.
3: Form synthetic partial observations $y_t^{+(o)}$ on the *same* missing pattern as the data.
4: ▷ (2) Kalman filter (Section 4.2) on the artificial series $y_t^{(o)} - y_t^{+(o)}$ in *deviation* form: the same T, Z, Q, R, but with $c_1 = 0$, $c_\alpha = 0$ and initial state 0, processing only the observed rows at each date.
5: ▷ (3) Backward smoothing recursion (equation (12) of Section 4.2) $\Rightarrow$ smoothed discrepancy $\hat{\alpha}_t$.
6: ▷ (4) Mean correction. $\tilde{\alpha}_t \leftarrow \alpha_t^+ + \hat{\alpha}_t$; read the imputed cells off $\tilde{y}_t = Z\tilde{\alpha}_t$.

dummy — and the correction pass then runs on the centred system. The deterministic parts cancel in the difference $y^{(o)} - y^{+(o)}$, and that cancellation is precisely what makes the mean correction in step (4) exact; running the correction pass with the constants left in place would leave an $(I - SZ)$-weighted deterministic term inside the draw.

The initial condition depends on how gaps are treated. In the legacy setting it is degenerate: $P_0 = 0$, so $\alpha_0^+ = \alpha_0 = M_0$ every sweep, where M_0 holds the first p rows filled once by the interpolation seed (`remNaNs_prev`, 'previous' method), and a missing cell inside that initial block keeps its seed value for the whole run. The deployed estimation (option `nofill`, used since the October 2026 re-estimation, Section 5.2) removes this restriction. A missing cell of the initial block gets a real prior variance, ten times its column's observed variance, and the smoother draws it with the rest: $\tilde{\alpha}_0 = \alpha_0^+ + P_0 A' r_0^*$, with r_0^* the smoothing cumulant of the difference system at $t = 0$. Leading, interior and trailing gaps are therefore all drawn as missing data. The seed value only starts the chain.

3.5 What ships: posterior output and predictive bands

For the deterministic engine the posterior is summarized by the means $(\hat{\beta}, \hat{\Sigma})$. For uncertainty the sampler is run for 1000 sweeps, the first 500 discarded as burn-in and every subsequent sweep retained — $D = 500$ consecutive draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}$, with no thinning, so they carry the chain's autocorrelation and the bands below should be read as the output of a 500-sweep segment rather than of 500 independent draws. They ship per bloc, gzipped, in `companion_draws.json.gz` and are loaded lazily, when the user first presses **Compute**. The exception is the United States: its file,

the largest at about 29 MB compressed, is fetched in the background after the page first draws.

The shipped baseline carries *predictive* bands, not parameter-only bands. For the seventeen euro members they are the Gibbs sampler's own forecast density, which integrates over parameter uncertainty, future innovations and the missing-data imputation, re-centred on the tool's central path. For the 35 coupled economies they are recomputed after estimation with the tool's own engine (`app/scripts/regen_gvar_bands.mjs`): one forward simulation per shipped draw, with that draw's $(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})$ and fresh, seeded innovations, started from the last history rows. In both cases the exported quantiles are

$$\big(q_{05},\, q_{16},\, q_{84},\, q_{95}\big)_{T+h} \;=\; \text{Quantile}\big\{\, \tilde{y}^{(d)}_{T+h} \,\big\}_{d=1}^{D}, \tag{5}$$

of which the charts draw only the 68% band (q_{16}, q_{84}), as the grey fan; the 90% band (q_{05}, q_{95}) ships in the files but is not drawn. No median is exported for the growth-space baseline; the seventeen euro-member `baseline_levels` files additionally carry a q_{50} that the charts do not read. The *central* path is the deterministic forecast from the posterior means, not a quantile of the draws, so an untouched screen shows the stored baseline exactly. Scenario bands — computed in the browser, conditional on the user's pins — are constructed differently and are described with the conditioning engine (Section 4.6).

Every bloc is serialized to one eight-file JSON contract — `spec`, `companion_mean`, `companion_draws` (shipped gzipped), `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations` — plus `estimation_mask` (the cells the sampler drew as missing data), with the 41 fiscal economies adding `fiscal_meta` and the 47 with the external satellite adding `external_meta`; floats are kept at full precision for the coupled blocs and compacted to six significant figures for the euro members. The deployed application reads only this contract and never re-estimates (Section 13).

4 Conditional forecasting

Conditioning a forecast on a chosen path is, in state-space form, the same problem as filtering with missing data: a pinned cell is an *observation* of the future, a free cell is a missing one (Bańbura et al., 2015; Waggoner and Zha, 1999). This section states the state space, the filter and smoother exactly as implemented (`app/src/lib/compute/kalman.ts`), the display construction, and the two kinds of pin — hard and soft — the interface exposes.

4.1 Companion-form state space

Stack the current and lagged observables into the state $\alpha_t = (y_t', y_{t-1}', \ldots, y_{t-p+1}')' \in \mathbb{R}^{np}$. The VAR (1) is then the first-order system

$$\alpha_t = T\,\alpha_{t-1} + c_\alpha + \eta_t, \qquad y_t = Z\,\alpha_t + \varepsilon_t, \tag{6}$$

with

$$T = \begin{pmatrix} B_1 & B_2 & \cdots & B_{p-1} & B_p \\ & I_{n(p-1)} & & & 0 \end{pmatrix}, \qquad c_\alpha = \begin{pmatrix} c \\ 0 \end{pmatrix}, \qquad Z = \big(I_n \mid 0_{n\times n(p-1)}\big), \tag{7}$$

$\eta_t \sim \mathcal{N}(0, Q)$ where Q carries Σ in its top-left $n \times n$ block and zeros elsewhere, and $\varepsilon_t \sim \mathcal{N}(0, R)$ with

$$R = \sigma_\varepsilon^2 I_n, \qquad \sigma_\varepsilon^2 = 10^{-12}. \tag{8}$$

The top block of T is filled from the stored coefficients as $T_{ij} = \hat{\beta}_{j+1+n_{\text{sr}},\,i}$ (the transpose of the lag rows, skipping the intercept row *and* the n_{sr} seasonal-dummy rows of (4)), and c_α from the intercept row plus, on seasonal blocs, the annual mean of the dummy coefficients ($\bar{c}$ of Section 3.3). Dropping the n_{sr} offset would read the quarterly dummies as first-lag coefficients on the 34 economies that carry them. The measurement variance (8) is the *hard-pin* noise: small enough that a pinned value is honoured to machine precision, nonzero for numerical stability. The state is initialized at the last p observed rows, $\alpha_0 = (y_T', \ldots, y_{T-p+1}')'$, with $P_0 = 10^{-7} I_{np}$.

4.2 Filter and smoother with partial observations

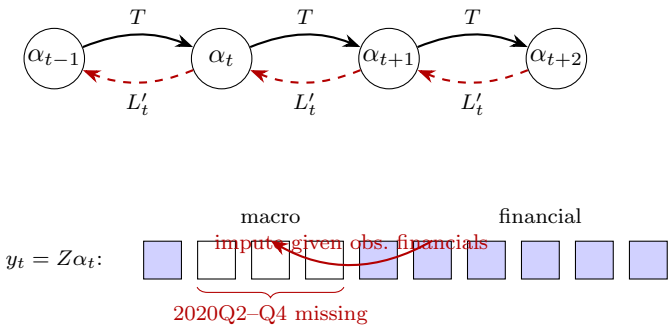


Figure 3: Companion-form state space with the Durbin–Koopman filter/smoother under the COVID-as-missing treatment. States α_t (§4.1) evolve forward through the transition matrix T (solid, Kalman filter, Eq. (9)); the backward smoother propagates $r_{t-1} = (Z_t^{(o)})' F_t^{-1} v_t + L_t' r_t$ (dashed, Eq. (12)). In the observation row $y_t = Z\alpha_t$ (§4.1), over the fixed window 2020Q2–Q4 (three quarters) the *macro* cells (real activity, prices, and the macro stars ROWDEM/ROWINF) are set missing (hollow) while the *financial* cells (rates, yields, equity, house prices, and the foreign-rate star ROWFIN) remain observed (filled). Because the filter processes only the observed rows $Z_t^{(o)}$ of each partially-observed vector, the macro holes are imputed conditional on the observed financials.

Collect the user's pins in a matrix $Y^* \in \mathbb{R}^{H\times n}$ with $Y^*_{t,j}$ the pinned value of variable j at forecast quarter t and `NaN` where free. At each t the filter processes only the observed rows: let Z_t stack the rows of Z for the pinned cells (plus any linear-combination rows, below) and R_t the corresponding

diagonal of measurement variances. For $t = 1, \ldots, H$:

$$a_{t|t-1} = T\, a_{t-1|t-1} + c_\alpha, \qquad P_{t|t-1} = T\, P_{t-1|t-1}\, T' + Q, \tag{9}$$

$$v_t = y_t^* - Z_t\, a_{t|t-1}, \qquad F_t = Z_t\, P_{t|t-1}\, Z_t' + R_t, \tag{10}$$

$$K_t = P_{t|t-1}\, Z_t'\, F_t^{-1}, \qquad a_{t|t} = a_{t|t-1} + K_t\, v_t, \qquad P_{t|t} = P_{t|t-1} - K_t\, Z_t\, P_{t|t-1}, \tag{11}$$

with $a_{t|t} = a_{t|t-1}$, $P_{t|t} = P_{t|t-1}$ at quarters with no pins. F_t^{-1} is computed by LU decomposition with partial pivoting (singularity threshold 10^{-15}), and the covariance updates are symmetrized. The Durbin–Koopman fixed-interval smoother then runs backward with $r_H = 0$:

$$L_t = T\big(I - P_{t|t-1} Z_t' F_t^{-1} Z_t\big), \qquad r_{t-1} = Z_t' F_t^{-1} v_t + L_t'\, r_t, \tag{12}$$

($r_{t-1} = T' r_t$ at pin-free quarters), and the smoothed state is $\hat{\alpha}_{t|H} = a_{t|t-1} + P_{t|t-1}\, r_{t-1}$. The smoothed *variance* is delivered by the companion N-recursion,

$$N_{t-1} = Z_t' F_t^{-1} Z_t + L_t'\, N_t\, L_t, \qquad V_t = P_{t|t-1} - P_{t|t-1}\, N_{t-1}\, P_{t|t-1}, \tag{13}$$

with $N_H = 0$; V_t is the state covariance conditional on *all* pins, so uncertainty about a quarter *preceding* a future pin is correctly reduced by it. The conditional forecast is the observation block of $\hat{\alpha}_{t|H}$; the display shading of the impact view (Section 4.6) reads V_t.

Linear-combination pins. A constraint on a linear combination $h' y_t = v$ — used by the fiscal block to pin the primary balance, $h_{\text{rev}} = +1$, $h_{\text{pexp}} = -1$ (Section 6) — enters the same machinery as one extra observation row $(h'Z,\, v)$ with measurement variance $\sum_k h_k^2\, \sigma_\varepsilon^2$. No separate variable is created.

4.3 The display: exact deltas on the stored baseline

To guarantee that an untouched screen reproduces the shipped baseline exactly, the tool displays

$$\hat{y}_{T+h}^{\text{display}} = \bar{y}_{T+h}^{\text{baseline}} + \big(\hat{y}_{T+h}^{\text{cond}} - \hat{y}_{T+h}^{\text{uncond}}\big), \tag{14}$$

the stored baseline plus the smoother *delta* between a conditional and an unconditional pass (the latter cached and reused). Zero pins give a zero delta and hence the stored baseline to the last digit. For `log` variables the delta is formed in the stored 100 ln level space and converted to the displayed q/q-annualized rate by applying (2) *to the delta itself*, $4(\delta_{T+h} - \delta_{T+h-1})$ with $\delta_{T+h} = \hat{y}_{T+h}^{\text{cond}} - \hat{y}_{T+h}^{\text{uncond}}$ and $\delta_T \equiv 0$ (the history is shared by both passes) — the first difference of the level gap, never $4\delta_{T+h}$, which is the *cumulative* gap and grows without bound. A dragged growth pin is correspondingly cumulated into the level pin, one quarter's worth (deviation/4) per active quarter — the exact inverse.

4.4 Operating hard pins

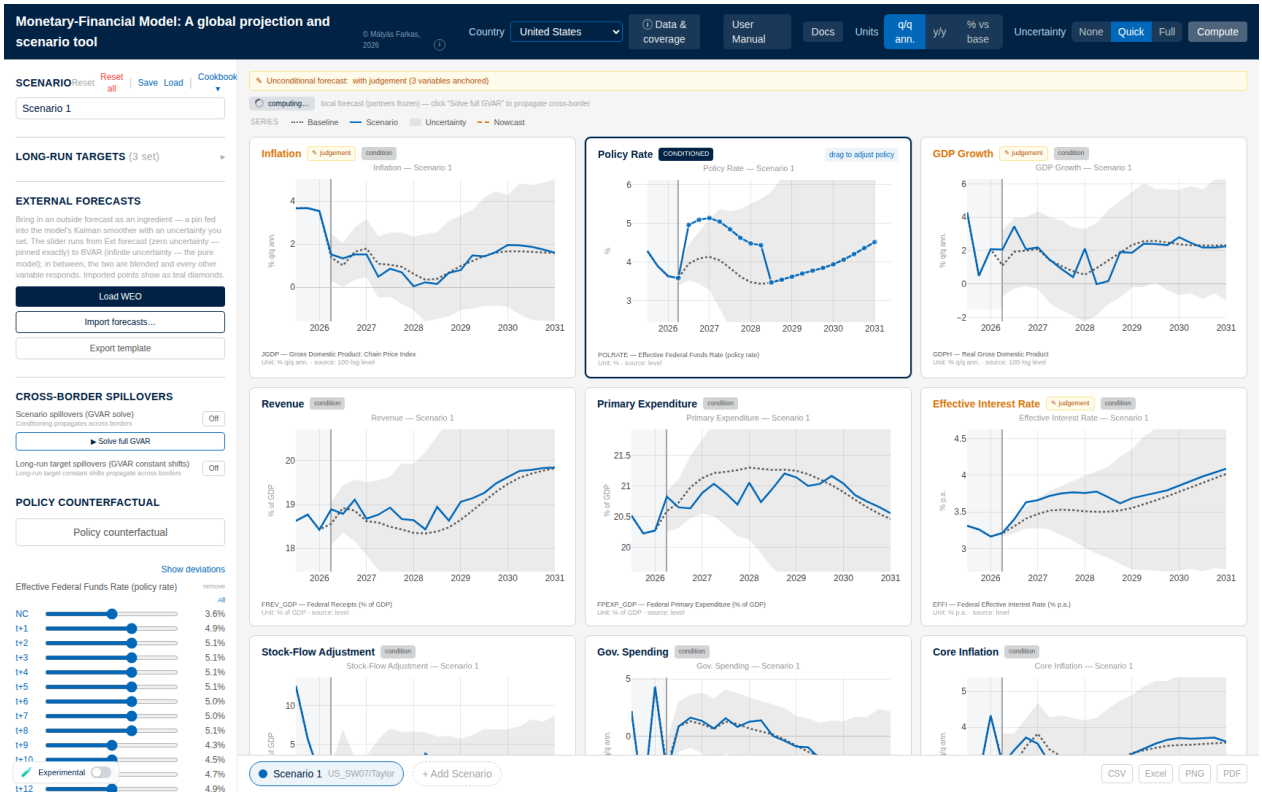


Figure 4: Hard conditioning in action. The policy-rate chart is put in CONDITIONED mode and the path is raised by +100 bp for eight quarters — by dragging the dots on the chart or the per-quarter sliders that appear in the left rail (here $t{+}1, \ldots, t{+}8$). As soon as the pins change, every panel redraws the smoothed scenario (solid; Compute adds the bands) against the frozen baseline (dotted): output growth dips, inflation eases, the effective interest rate on debt climbs, and the fiscal ratios respond — all through the estimated cross-correlations of §4.2; nothing is re-estimated.

A pin is set on the chart (drag on a **CONDITIONED** chart; right-click to condition and plant in one gesture; right-click again to toggle a quarter's pin off) or in the conditioning list, which shows one row per forecast quarter, labelled **NC**, **t+1**, …, **t+19**. **NC** — the nowcast — is the current, not-yet-published quarter (Section 11.3). Each row has an on/off toggle and a value slider; **Show deviations/Show levels** switches between entering distances from baseline and absolute values. Pins are what-you-set-is-what-you-see: a pinned quarter sits exactly at its number, always; the quarters left off are filled by the model. Multiple variables stack, and all pins are imposed *jointly* — the smoother finds the single most plausible forecast through every pin at once, so mutually strained pins show their cost in the unconditioned charts. Up to four scenarios can be held side by side (the bottom bar); **Reset** clears the active scenario for the current country, **Reset all** (after a confirmation) everything everywhere, **Save** writes one country's conditioning to an Excel file, and **Load** reads it back (or a JSON scenario file).

Example 2 (A ready-made scenario, and what identification buys)**.** *The* ***Cookbook*** *(left panel,*

Scenario *header) loads documented, source-calibrated scenarios; each recipe offers* ***Load naive*** *— pin only the headline variable — and* ***Load + restriction*** *— add the short-run pins that shape the shock into its economically meaningful form. The credit-crunch recipe (calibrated to the Federal Reserve's 2020 severely-adverse scenario) illustrates on the United States what the extra pins buy.* Naive*: pin only the corporate bond yield* $+400bp$*. The model reads a corporate-yield spike as generic rates tightening — the whole curve rises (at the 2026Q1 vintage, the policy rate peaks around* $7\frac{1}{2}\%$*, the 10-year near* $8\frac{1}{2}\%$*) and inflation moves* up *on impact: comovement alone cannot tell a credit-supply dislocation from a broad rate shift.* Restricted*: add the documented crunch signature — safe 10-year yield* down *(flight to safety), output down, inflation down, equity down, the policy rate cut. The same* $+400bp$ *spread now produces a deep recession (GDP trough near* -6% *q/q ann.), disinflation, and easing — the policy rate and the safe yield move in the* opposite *direction from the naive read. Loading a recipe* replaces *the active scenario's pins; run the two variants on two scenario tabs to compare.*

4.5 Soft pins: outside forecasts with a trust dial

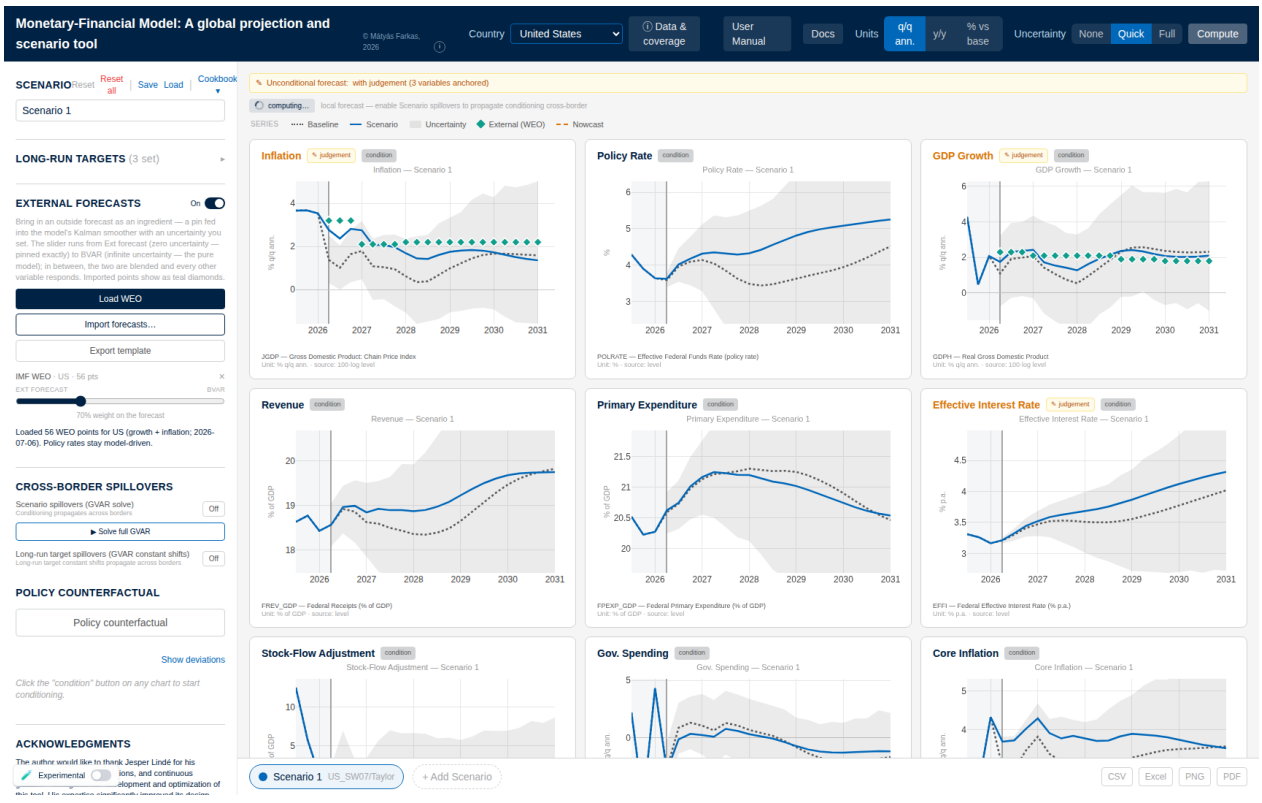


Figure 5: An outside forecast as an ingredient (Load WEO). The teal diamonds are the imported WEO growth and inflation points; the trust dial (left rail, here at 70% weight on the forecast) maps to the measurement-error variance of Eq. (15), so the scenario blends the outside path with the model instead of pinning it; policy rates stay model-driven.

An imported outside forecast — IMF WEO, a consensus survey, desk numbers — enters as a

soft pin: an observation with a measurement variance set by the user's trust in the source, so the model reconciles the outside number with everything else rather than being overridden by it. For a cell of variable j at horizon h held with trust $\tau \in (0,1)$, the measurement variance in (10) is

$$R^{\text{ext}}_{h,j} \;=\; \widehat{\text{Var}}_{h,j}\,\frac{1-\tau}{\tau}, \qquad \widehat{\text{Var}}_{h,j} = \Big(\max\big(\tfrac{q_{84}-q_{16}}{2},\, 10^{-6}\big)\Big)^2, \tag{15}$$

where $q_{84} - q_{16}$ is the model's own shipped predictive band of (5) at that cell, read in the stored level space in which the pin lives (`baseline_levels.json`) — so the dial needs no units: $\tau = \frac{1}{2}$ makes the outside number exactly as informative as the model's own predictive density, $\tau \geq 0.999$ collapses to the hard pin (8), and $\tau \to 0$ is a no-op (τ is clamped to $[0.001, 0.999]$; the default is 0.7). Display-unit values are converted to level-space pins by the conventions of Section 4.3 (growth targets cumulate; any active long-run drift of Section 8 is netted out first, so the *displayed* line lands on the imported number). A cell the user has pinned by hand always wins over an imported one (`externalForecast.ts`).

In the interface (**External forecasts**, left panel): **Load WEO** imports the IMF World Economic Outlook real-GDP-growth and inflation forecast for the viewed economy, each annual figure applied softly to its four quarters. The shipped WEO file (retrieved July 2026) has no numbers for Hong Kong, Peru, the Philippines, Romania and Singapore, and where an economy's price column is a deflator or HICP the WEO CPI figure is applied to it. **Export template** writes an Excel file pre-filled with the model baseline, to be overwritten and re-imported; **Import forecasts…** accepts that file or a small JSON. Each source appears with its trust slider, which runs from **Ext forecast** (left: pinned exactly) to **BVAR** (right: the model alone) in 5% steps. Imported points draw as teal diamonds in the q/q and level views, and the hover shows the trust. Only the source's × removes it: at zero trust the source stays loaded but is ignored. An **On/Off** switch next to the panel title stops conditioning on all imports without deleting them; a new import switches it back on. **Export session**/**Import session** bundle all countries' conditioning, targets, and imports into one JSON for hand-over. On a coupled economy an imported forecast also propagates cross-border through the global solve (Section 5.5), with one performance caveat documented in Section 5.7.

4.6 Scenario uncertainty bands

Pressing **Compute** with **Quick** or **Full** selected (the **Uncertainty** toggle in the top bar; with **None** selected and a long-run target active, **Compute** switches to **Quick**) draws $N_b = 50$ or 200 predictive paths and bands the scenario. The default construction is deliberately simple and fast:

(i) for each band draw, sample a posterior draw $(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})$ from the $D = 500$ shipped draws (with replacement);

(ii) run the simulation smoother of Algorithm 1 on that draw, conditioning in level space on the imposed paths — the global variables on the US solution, the trade-weighted foreign stars on the converged partner levels, and the user's own pins — so the draw carries both parameter and innovation uncertainty yet reproduces every conditioned value exactly;

(iii) report the empirical 16th and 84th percentiles per cell.

Step (ii) runs in full only when the global solve for the current pins drives the displayed line — it has run (**Compute** with the **Scenario spillovers (GVAR solve)** switch on, or **Solve full GVAR**) and either the switch is on or the viewed economy has no pins of its own — and no primary-balance or debt target is set. The band is then genuinely conditional: it collapses onto the line at each hard pin, narrows across variables through the estimated covariance, and elsewhere carries the full parameter-and-innovation predictive width — directly comparable to the baseline fan. In every other case — the default (switch off), a euro member, or a fiscal target — each draw is conditioned on the global variables alone (on nothing for the United States, where they are endogenous) and shifted onto the scenario path of (14). When the band is drawn, a pinned variable's band is closed onto the pin at each pinned quarter, since an imposed value carries no uncertainty; everywhere else this band keeps the model's predictive width and does not take up the information in the pins. On the **% vs base** impact view, in its **Level** mode (not **Rate (IRF)**), the shading is instead ± one smoothed conditional standard deviation from (13) (during a drag the cheaper filtered variance stands in until release): it pinches at hard pins, shrinks partially at soft pins in proportion to (15), and tapers *into* a future pin — the honest conditional-variance picture.

Remark 2 (The default band and the coupled per-draw cloud). The default band conditions each draw through the simulation smoother alone, without re-solving the cross-country system for every draw. A fuller construction — re-solving the entire Gauss–Seidel fixed point once per draw — is available for coupled economies as the **Experimental features** switch in the **Settings** drawer (bottom-left, Section 5.8); like step (ii), it applies only when the coupled solution drives the line. It is the most faithful cloud, but it is slow, and with only a point or two pinned it can collapse too tightly under the estimated prior, a band the user reads as false confidence. Toggling it clears any band on screen, because bands built the two ways must not be read against each other.

Remark 3 (Long-run targets translate bands rigidly). A long-run anchor (Section 8) shifts the central path and every band quantile by the same deterministic amount, instantly and without recomputation — band *widths* are invariant by construction. Only pins change conditional uncertainty; targets move where the forecast is headed, not how well it is known.

5 The global model

Thirty-five of the deployed economies are coupled into a global VAR (GVAR) in the tradition of Pesaran et al. (2004) and Dees et al. (2007): the United States, the euro area as a single aggregate bloc, and thirty-three further economies, spanning roughly ninety percent of world GDP (the roster is in Appendix B). Two design choices organize everything in this section. First, *coupling is imposed at solve time, not at estimation time*: each bloc is an independently estimated BVAR, and cross-country spillovers enter only through the scenario *deviation*, which propagates as a linear fixed point. Second, the United States is *fully endogenous*: it carries its own trade-weighted foreign variables

and iterates inside the fixed point like every other bloc, so shocks abroad feed back into the US — the channel a classical frozen dominant unit cannot represent.

The euro area enters as one aggregate by necessity: a fully disaggregated euro GVAR is non-stationary (the coupling operator's spectral radius exceeds one, $\rho \approx 1.089$, in the disaggregated experiment), while aggregation restores contraction. The seventeen euro-area member states are therefore *standalone* models outside the coupled system, connected to it through a separate euro-area doorway (Section 5.6).

5.1 Star variables and trade weights

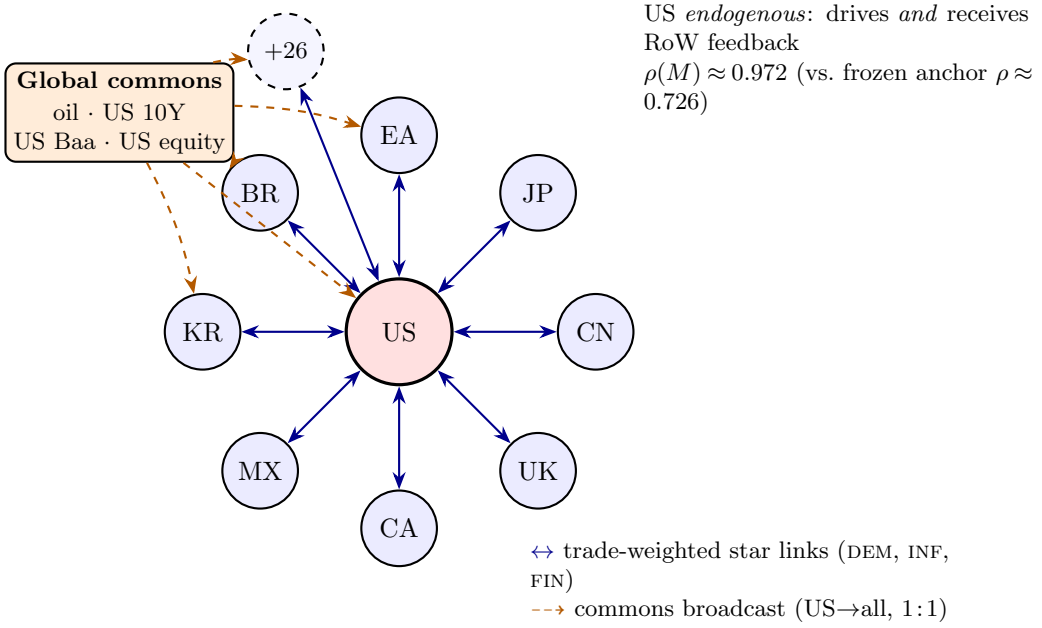


Figure 6: GVAR bloc topology. A central, dominant United States bloc is coupled to a ring of 34 Rest-of-World partner blocs (eight labelled—EA, JP, UK, CA, CN, MX, KR, BR—plus 26 further blocs) through *bidirectional*, trade-weighted "RoW star" links carrying foreign demand (DEM), inflation (INF), and the short rate (FIN). A small global-commons node (oil, the long US rate, US corporate credit, US equity) broadcasts *one-directionally* and identically (1 : 1) to every bloc. The US is now a *fully endogenous* coupled bloc rather than the classical frozen anchor: it feeds the rest of the world *and* receives feedback, raising the coupling operator's spectral radius from $\rho(M) \approx 0.726$ (frozen-US, RoW-only; measured on an earlier estimate) to $\rho(M) \approx 0.972$ at the full 35-bloc endogenous-US configuration (the October 2026 estimate, as stored in the shipped `coupling_default.json.gz`; it moves with each refit), both contractive so the fixed point $x = Mx + d$ is well defined (Sections 5.1 and 5.3).

Every coupled bloc carries three trade-weighted *star* variables summarizing its foreign environment, the weak-exogeneity device of the GVAR literature: foreign demand (`ROWDEM`), foreign inflation (`ROWINF`), and foreign financial conditions (`ROWFIN`). At estimation time the stars are built as levels from the partners' *completed* panels (Section 5.2): with w_{cj} the trade weight of partner j

in receiver c and $\ell_{j,t}$ the partner's stored 100 ln level,

$$\mathrm{ROWDEM}_{c,t} = 4\sum_j \tilde{w}_{cj}\left(\ell^{\mathrm{gdp}}_{j,t} - \ell^{\mathrm{gdp}}_{j,t-1}\right), \qquad \mathrm{ROWFIN}_{c,t} = \sum_j \tilde{w}_{cj}\, r_{j,t}, \tag{16}$$

(`ROWINF` as `ROWDEM` on price levels) — annualized growth for the two macro stars, the factor 4 of (2), and a level average of partner short rates $r_{j,t}$ for the financial star. The weights $\tilde{w}_{cj}$ are renormalized *per component and per quarter* over the partners actually present and eligible, so absent or excluded mass is reabsorbed rather than dropped. Eligibility is read from the data, not from a list: a partner leaves its receivers' `ROWFIN` when its specification resolves no short rate, which on the shipped roster is Singapore alone. A hyperinflation list drops the named bloc from partners' `ROWINF` and `ROWFIN` over the *whole* sample — the test carries no date restriction — while its real demand stays in `ROWDEM`; the list holds Argentina, which the 2026 re-estimation removed from the roster, so on the shipped 35 blocs it currently has no live member. Presence itself is tested jointly at t and $t-1$, before the per-component rules, so a partner's very first in-sample quarter contributes to none of the three stars — including `ROWFIN`, which otherwise needs only the level at t.

A deflate-by-subtraction rule for blocs reporting nominal GDP survives only in the *solve-time* star mapping, where the bloc set is currently empty. The estimator's matching branch was removed outright after China was re-sourced to a real chain-linked volume index, so a new nominal-GDP bloc would need it reinstated there before its star history could be built correctly.

The bilateral weights come from the IMF's International Merchandise Trade Statistics (exports plus imports, 2021–2023 average, euro area as a single counterpart) and ship as one row per coupled bloc — the US row included — each summing to one with a zero diagonal (`weights_24bloc.json`; the file name is historical, the file lists all 35 blocs).

Every bloc in the global model is coupled, and the United States is one of them. There is no distinguished country: the US carries its own `ROWDEM_US`, `ROWINF_US` and `ROWFIN_US` stars and its own weights row, exactly as the other 34 do, so foreign conditions enter the US model as they enter every other. This is what separates the design from the classical GVAR, where the dominant unit is frozen and the rest of the world may respond to it but never the reverse. A shock originating anywhere reaches the US, and the US response feeds back into the world in the same solve.

The topology is read from the data rather than hardcoded: a bloc is a coupled member if and only if its specification carries the three `row`-tagged stars *and* it has a weights row. The engine retains a frozen-anchor fallback for the case where those are absent, which is how pre-2026-06 models are reproduced; it is not the configuration that ships. The economies tried are those in the weights file the app was built with; a data pack can re-weight them but cannot add a new one. An economy that the pack does not ship is simply left out of the coupled set. An economy that the pack ships but that fails to load, or fails its file check, stops the global model: a red banner names the file and the economies it took out, and no cross-country result is shown until the pack is repaired. The tool never shows a coupled answer for a smaller model. The trade-weights file is treated the same way.

The 17 euro-area member economies are a *second* layer and are counted separately. They do not appear in the weights file and are not GVAR members in their own right; they carry euro-area stars (`ROWEA_GDP_xx`, `ROWEA_HICP_xx`) and are coupled to the euro-area *aggregate*, which is itself one of the 35. Their cross-border transmission therefore runs through the EA bloc rather than bilaterally — Section 5.6 sets out what that does and does not let a member scenario say.

Time-varying weights. The trade weights that build the stars are *time-varying*, and this is the state of the deployed models, not an option held in reserve. Rather than freeze one fixed-window row across the whole sample, `buildstars` reads each quarter's stars off that quarter's calendar-year row of an annual bilateral panel, clamped to the file's span, with a partner absent from a given year assigned weight zero and reabsorbed by the same per-component, per-quarter renormalization as above (`export_bvar_gvar24_loop.m`). The weights therefore reach the star *history* on which the blocs are estimated.

The shipped panel is annual IMF IMTS bilateral trade (goods, by partner) for 1995–2024 over 39 blocs, at `app/public/data/bvar/gvar/weights_timevarying.json`. The empirical picture it carries is not the textbook monotone story: the US-to-China trade share rises to a 2017 peak (0.176) and then *falls* after the 2018 tariff episode (to 0.119 by 2024), while the US-to-euro-area share does *not* decline but drifts up to 0.178. Freezing a single recent row would have imposed the end-point of that path on the whole history.

Remark 4 (Where the time variation stops). *Estimation* uses the time-varying panel; the *solve-time* coupling operator of Section 5.3 is still built from a single static weights row, and the browser never reads the annual panel. So a scenario's cross-border propagation today runs on one fixed weight matrix even though the blocs behind it were estimated on a moving one. Promoting the panel to the live solver is a separate, deliberate step. The estimator retains a switch (`GVAR_TV_WEIGHTS`=0) that reverts to the legacy static row, which exists to reproduce pre-2026-07 models and is not the shipped configuration.

5.2 Estimation: an EM fixed point in the completed stars

Before the formal statement, here is the idea in plain terms. To estimate any one country's model you need its *foreign environment* — what its trading partners' demand, inflation, and interest rates are doing. But each partner's path is itself the output of that partner's model, which needs *its* foreign environment, and so on around the world: everything depends on everything, so no country can be estimated on its own and first. The way out is to treat the unknown foreign environment as *missing data* and fill it in by iterating. Start from a rough guess of the foreign environment for every country; estimate each country's model against its guess (filling in the pandemic holes in the same step); then read off from the freshly-estimated countries what the foreign environment *actually* turned out to be, and re-estimate against that. Repeat. Each pass the guesses become more mutually consistent, and the loop stops when the environment each country *assumed* matches the one its partners *produce* — when the guesses have come true. That self-consistent set of country

models is the estimate.

A concrete image helps: fifty country desks share one room, each forecasting its own economy but needing its neighbours' forecasts as inputs. Everyone pencils in a rough guess for the neighbours and produces a first forecast; the forecasts are passed back around the room and each desk redoes its own with the updated inputs; the numbers keep circulating until nobody's change any more. This is exactly the expectation–maximization logic of Dempster et al. (1977) with the "rest of the world" — and the pandemic gaps — as the missing data: an **E-step** that fills in the blanks given the current models, and an **M-step** that re-fits the models given the filled-in blanks, iterated to a fixed point. Two things are worth keeping straight. Nothing is accepted or rejected *across* countries: coherence is the loop's stopping rule, not an acceptance test, and the loop settles on one consistent set of paths rather than a distribution of draws. The genuine Bayesian sampling — the simulation smoother that draws the missing cells and the Metropolis step that updates the shrinkage hyperparameters — lives *inside* each country's single fit (Section 3.4), one level below the loop.

The stars cannot be built directly from observed data: the panels are unbalanced (blocs enter the sample at different dates), the pandemic macro cells are deliberately missing (Section 3.4), and several financial series start late. A star built only on complete quarters would truncate the sample to the common balanced span — exactly the information loss the data-augmentation design avoids. The estimator therefore iterates completion and re-estimation to a fixed point in the stars (in the spirit of Dempster et al., 1977):

- **E-step (completion).** For each bloc, one call to the data-augmentation sampler of Section 3.4 on the panel [domestic | stars] imputes the *interior* missing cells (COVID macro mask included) and the *trailing* ragged edge by the Durbin–Koopman simulation smoother; the posterior-mean completed panel is retained. In the deployed estimation (`GVAR_NOFILL`=1, since October 2026) *every* gap is missing data — leading, interior and trailing — and the sampler draws it. A start-up imputation (interior gaps interpolated, leading and trailing gaps set to the column's first or last observed value) only starts the chain and the hyperparameter fit; every one of those cells is re-drawn. The cells the sampler drew rather than observed are exported per bloc in `estimation_mask.json`, and the app shows them in amber. (The earlier estimations instead back-filled each leading gap with the column's first observed value before the sampler was called, which on late-starting credit and lending-rate columns meant tens of flat quarters treated as data.)

- **M-step (re-estimation).** The same call re-estimates the bloc's coefficients under the GLP prior on the completed data; the posterior means $(\hat{\beta}_c, \hat{\Sigma}_c)$ and the thinned draws are what ships. The hyperparameters (λ, μ) follow the mixed-frequency method of `cgl_bvarV5`: a GLP start-up fit on the full sample (start-up imputation included), then a Metropolis step on (λ, μ) inside the Gibbs chain on the completed data.

- **Outer loop.** All blocs' stars are rebuilt via (16) from the *frozen* start-of-sweep panels, the blocs are re-estimated in parallel (a Jacobi sweep, order-independent), and the loop repeats

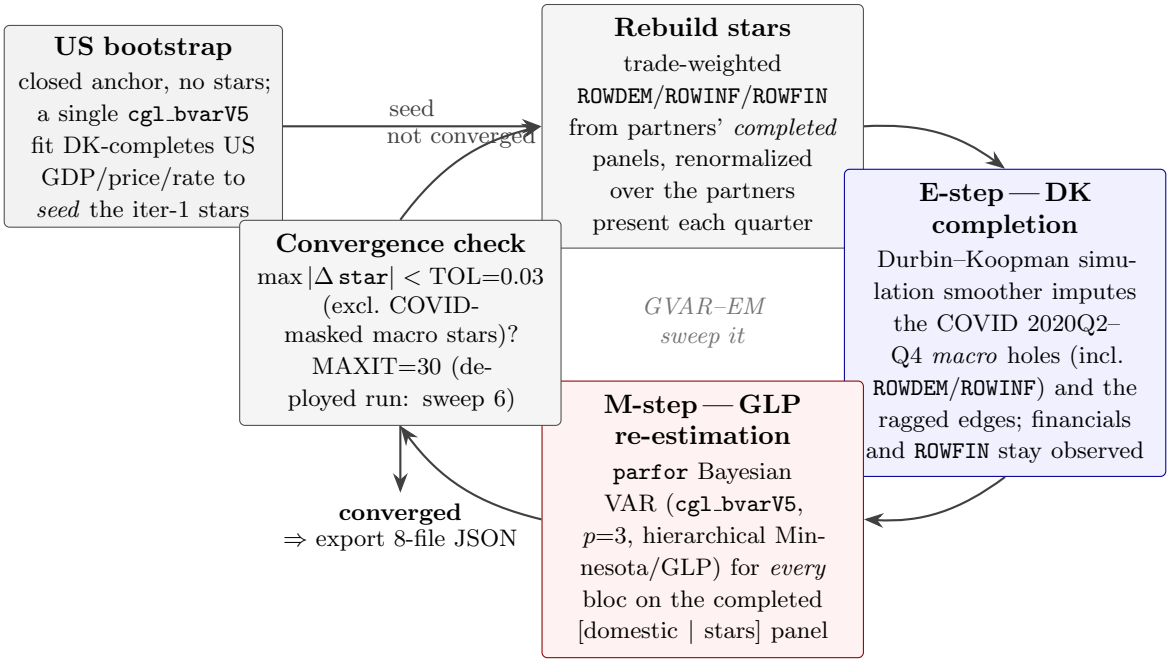


Figure 7: The GVAR-EM estimation loop under pervasive missingness (Giannone et al., 2019; Durbin and Koopman, 2002). A closed US bootstrap (no stars) is estimated once to seed the first trade-weighted foreign variables. Each sweep then (i) rebuilds the ROWDEM/ROWINF/ROWFIN stars from partners' completed panels, renormalized over the partners present each quarter, and (ii) re-estimates every bloc by hierarchical-prior Bayesian VAR (Gibbs, run in `parfor`). Within each bloc's single `cgl_bvarV5` pass the **E-step**—a Durbin–Koopman simulation-smoother completion that imputes the COVID macro holes (the fixed 2020Q2–Q4 window, with financial series and ROWFIN held observed) and the *trailing* ragged edge—and the **M-step**—the posterior re-estimation of the VAR coefficients on the completed [domestic | stars] panel—are carried out jointly by the same sampler. The sweep repeats until the foreign components stop moving, $\max|\Delta\,\texttt{star}| < \text{TOL} = 0.03$ (evaluated after the second sweep and excluding the COVID-masked macro stars, whose cross-iteration churn is imputation noise — a floor of ≈ 0.1 under the original sampler), capped at MAXIT sweeps (30 in the deployed October 2026 run, which converged at sweep 6).

until the stars stop moving. At $m = 1$ only the US panel is a completion, from the bootstrap; every other bloc is seeded with its *raw* transformed data with the COVID macro cells blanked, so the first sweep's stars are essentially the classical data-based construction — which is why the descent's first number is as large as it is. From $m = 2$ on, every panel is the previous sweep's posterior-mean completion. The convergence test is $\max_c \|x_c^{*(m)} - x_c^{*(m-1)}\|_\infty < \text{TOL} = 0.03$ (COVID-masked macro-star cells excluded from the metric — their cross-iteration churn is imputation Monte-Carlo noise, with an observed floor of about 0.06–0.20 under the original sampler), capped at MAXIT sweeps. Until September 2026 the loop never met TOL: runs stopped at MAXIT = 12 on that floor. The October 2026 re-estimation traced the floor to two defects and removed it. (i) On a parallel worker, MATLAB's `rng default` resets to that worker's own random stream, so a bloc's draws depended on which worker ran it; each call now seeds explicitly (`rng(0,'twister')`). (ii) The Metropolis accept/reject step on (λ, μ),

an eigenvector-sign choice in a matrix square root, and the hyperparameter stage's data-dependent use of random numbers made the sampler jump by its Monte-Carlo error for any change in its input, however small. With `GVAR_FIXHYPER`=1 the sweeps hold (λ, μ) at the GLP mode, use a continuous square root and re-seed before sampling, so a sweep is a smooth function of the stars; the shipped posterior then comes from one final full Metropolis pass at the converged stars. The deployed run descended $33.5 \to 1.90 \to 0.59 \to 0.044 \to 0.011$ and converged at sweep 6 (MAXIT = 30). A one-shot, no-stars US completion seeds the first sweep.

Algorithm 2 states the loop (`pipeline/export_bvar_gvar24_loop.m`). Setting MAXIT = 1 collapses the loop to a single sweep — one-shot stars read off the seed panels, no coupling iteration — but it is *not* the standalone-bloc estimator: the sweep still runs the fixed-companion E-step described below. The euro members are produced by a separate single-bloc script (`export_bvar_euro_redesign.m`) that carries two euro-area stars rather than the trade-weighted trio. Since October 2026 it runs in growth rates with the same no-fill rule, seasonal dummies for all 17 members and a fixed companion (`CGL_FIXED_COMPANION`=1). Relative to the classical fixed-weight GVAR — stars built once from observed data on a balanced panel, one OLS pass per country — every departure here serves the unbalanced panel: model-based completion flowing *into* the stars, per-quarter renormalized weights, Bayesian shrinkage with the COVID-as-missing treatment, and the star fixed point itself. The out-of-sample evaluation of Section 12 races exactly this apparatus against its classical ancestor.

Algorithm 2 GVAR-EM estimation loop (`export_bvar_gvar24_loop.m`).

Input: per-bloc observed panels with `NaN` holes; trade weights; eligibility rules; TOL = 0.03, MAXIT (30 in the deployed run); fixed COVID window 2020Q2–Q4.

Output: per-bloc posterior means and draws; converged stars; the eight-file export.

1: ▷ Bootstrap. Complete the US panel with no stars (COVID macro cells masked); its posterior-mean completion seeds the first-sweep stars. Under US_ENDOGENOUS the US then joins the swept set with its own `ROW*_US` stars.
2: **for** $m = 1, \ldots,$ MAXIT:
3: rebuild every receiver's stars (16) from the frozen start-of-sweep completed panels;
4: **parfor each** bloc c: mask the COVID macro cells (domestic macro columns + `ROWDEM`, `ROWINF`; financials and `ROWFIN` stay observed); one `cgl_bvarV5` call ⇒ completed panel + $(\hat{\beta}_c, \hat{\Sigma}_c)$ + draws;
5: commit the new completions; mc $\leftarrow \max_c \|\Delta x_c^*\|_\infty$ (COVID macro-star cells excluded);
6: **if** $m > 1$ **and** mc < TOL: **break**.
7: ▷ Final pass (`GVAR_FIXHYPER`=1): one full Metropolis estimation of every bloc at the converged, frozen stars.
8: Export each bloc to the eight-file JSON contract, plus `estimation_mask.json` under `GVAR_NOFILL`.

Two ways to run the E-step. The imputation inside each M-step (Section 3.4) can be run in either of two ways; the shipped coupled estimate uses only the first, in the outer sweeps and in the final Metropolis pass alike. During the outer sweeps the Durbin–Koopman completion is carried out with

a *fixed companion*: the smoother's state-space matrices are built once from the posterior-mode fit and held across the Gibbs draws (`fixedcomp`= 1 in `estimate_bloc`, `export_bvar_gvar24_loop.m`). With the sampler's random stream seeded explicitly per call, consecutive sweeps then share their random numbers, so a bloc's star moves only for *systematic* reasons and the Jacobi sweep contracts sharply. Before the two fixes described above, runs descended (e.g. $14.7 \to 1.07 \to 0.20 \to 0.11$) to the Monte-Carlo floor and flattened there; the deployed run converges. The textbook alternative rebuilds the companion from the current (β, Σ) draw at every Gibbs sweep — exact Tanner–Wong data augmentation (`cgl_bvarV5.m`; `CGL_FIXED_COMPANION`=0) — but re-randomizing the completions each sweep breaks the common-random-number coupling, leaving a star-churn floor that never meets TOL. The coupled estimator therefore uses the fixed-companion sweeps throughout, and *ships the fixed-companion posterior* from the converged outer loop. An optional exact-data-augmentation *polish* pass at the frozen stars is available (`GVAR_POLISH_EXACT`=1) but is *off by default for the coupled system*: exact augmentation marginally raises companion persistence and lifts the coupled operator's spectral radius above one (§5.3), so it would leave pinned-scenario spillovers non-contractive. The euro members, which have no outer loop, could run exact data augmentation directly; the October 2026 estimation keeps the fixed companion for them as well.

5.3 Solve-time coupling: a linear fixed point

At solve time the coefficients are frozen, and the key observation is that the conditional-forecast operator of Section 4 is *affine in the pinned values for a fixed pin pattern*: the filter and smoother are linear in the data. Each bloc's level forecast can therefore be written

$$x_c \;=\; a_c \;+\; \sum_{v} \sum_{s=0}^{H-1} K_c[v][s]\; \Delta^{*}_{c,v,s}, \tag{17}$$

where a_c is the bloc's forecast with its *own* pins active and its stars and global columns held at baseline, $\Delta^{*}_{c,v,s}$ is the deviation pinned on star (or global) column v at horizon s, and the kernel $K_c[v][s] \in \mathbb{R}^{H \times n}$ is the smoothed response to a unit bump of that single cell. The kernels are computed from one filter–smoother pass per bloc by value-independent delta recursions (`affineBlocKernels`, numerically identical to explicit probing to below 10^{-9} and an order of magnitude faster), and they depend only on the bloc's coefficients and its own-pin *pattern* — never on pin values or on partners — so they are cached and reused across every value change of an interactive drag. A *soft* own pin (Section 5.7) keeps the forecast affine in the pinned values but changes the filter gains through its measurement variance, so its kernels are computed with that variance and cached under it. Since November 2026 the live solve uses (17) twice: to build M, and to rebuild every bloc's path from the solved star and global deviations, with no further smoother pass (route (c) below).

The solver never rebuilds star *levels* from partner levels. It pins each receiver's star column at

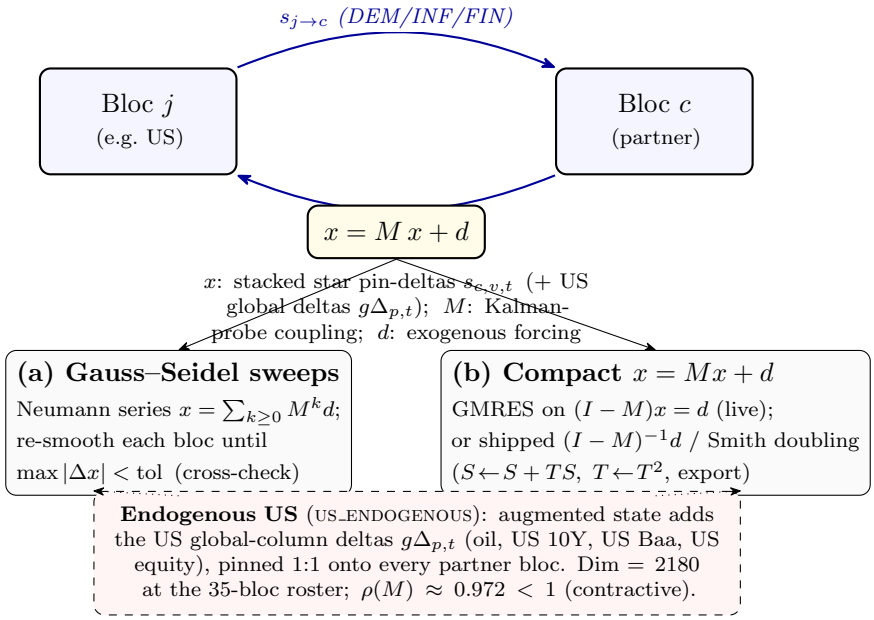


Figure 8: Solve-time cross-country coupling as the affine fixed point $x = M\,x + d$. Two coupled blocs exchange trade-weighted star pin-deltas (DEM/INF/FIN), forming the feedback cycle; the deviation state x stacks these pins (plus the US global-column deltas under an endogenous US). The fixed point admits two equivalent solutions—(a) Gauss–Seidel/Neumann sweeps through the Durbin–Koopman smoother, used as the cross-check, and (b) the compact M-state recursion, solved with the *shipped* $(I-M)^{-1}$ whenever the request's pin structure matches the precomputed artifact's — an $O(\dim^2)$ matvec, and the ordinary case at load and under drift-only forcing — and otherwise by restarted GMRES on $(I-M)x = d$, with every bloc's path rebuilt from its kernels (the network solve, route (c) in the text). It is the $O(\dim^3)$ *inversion*, not its use, that is prohibitive in-browser at dim = 2180; Smith doubling and the export-time precompute sit outside the live path. With the US endogenous the augmented state has dimension 2180 and spectral radius $\rho(M) \approx 0.972 < 1$, so the iteration contracts. The two routes agree to roughly 10^{-5}–10^{-6}, limited by the six-significant-figure rounding applied to M and $(I - M)^{-1}$ when the artifact is exported, not to 10^{-9}.

its own baseline plus the trade-weighted partner *deviation*:

$$\Delta^*_{c,\mathrm{DEM},t} = \sum_j \tilde{w}_{cj} \cdot 4\left[\left(x^{\mathrm{scen}}_{j,t} - x^{\mathrm{scen}}_{j,t-1}\right) - \left(x^{\mathrm{base}}_{j,t} - x^{\mathrm{base}}_{j,t-1}\right)\right], \qquad \Delta^*_{c,\mathrm{FIN},t} = \sum_j \tilde{w}_{cj}\left(r^{\mathrm{scen}}_{j,t} - r^{\mathrm{base}}_{j,t}\right), \tag{18}$$

read off the partners' gdp/price/rate columns (at $t = 0$ the shared last in-sample level cancels in the difference). Stacking all star deviations — and, with the US endogenous, the deviations of the four US-determined global columns (Section 5.4) — into a vector x, the deviation system is the linear fixed point

$$x = M\,x + d, \tag{19}$$

where d collects the exogenous forcing (each bloc's own pins evaluated at zero partner deviation, plus any long-run drift, Section 8.1) and M carries three coupling families: star ← partner star (through the kernels and weights, equations (17)–(18), the US on both sides), the US-global definition rows (a

global column's deviation as the US's own response to its star pins), and star ← global (a partner's star response to its own global pin). At the deployed roster the state dimension is $35\times3\times20+4\times20 = 2180$, and the shipped operator has spectral radius

$$\rho(M) = 0.972 \;<\; 1 \tag{20}$$

(the value stored in the shipped coupling artifact, 0.971875, computed by a telescoped-log-norm power iteration — `spectralRadius` in `gvarSolve.ts`, 600 iterations; a dense eigen-decomposition of the shipped M gives the real dominant eigenvalue 0.9718529, a difference of 2×10^{-5}. The value moves with each re-estimate: 0.973405 in the previous edition of this manual, 0.957724 for the September 2026 model that was live until October, 0.971875 for the October 2026 re-estimation; before 2026-07-17 the field read 0.935044 from an under-converged iteration on a complex dominant pair), so the map is contractive, $(I - M)$ is nonsingular, and the fixed point is unique.

The fixed point is solved four equivalent ways, all implemented in `gvarSolve.ts`:

(a) **Gauss–Seidel sweeps**: re-condition each bloc in turn on the freshest partner deviations — blocs later in a sweep already see partners updated within that same sweep, so this is Gauss–Seidel, *not* the term-by-term Neumann sum $\sum_k M^k d$ (which would be Jacobi, and converges at the slower rate $\rho(M)$). The loop stops when the maximum level change falls below 10^{-4} *or* when a budget of 12 sweeps is exhausted, whichever comes first; the budget binds on the shipped roster, so this route returns an unconverged iterate and reports `converged = false` rather than the exact fixed point;

(b) **closed form** $x = (I-M)^{-1}d$, using a precomputed inverse (below), followed by one conditional-forecast pass per bloc to reconstruct all free variables;

(c) **network solve** (since November 2026), when the precomputed inverse is unusable — the in-browser $O(\dim^3)$ inverse at $\dim = 2180$ would take the better part of an hour and is never built live. Each bloc is a node whose path is the affine map (17) of its input ports (its star and global deviations). Restarted GMRES (Krylov dimension 100, relative residual 10^{-12}) solves $(I - M)x = d$ for the ports, and every bloc's path is then rebuilt from its kernels, $x_c = a_c + \sum K_c\,\Delta^*$, instead of one conditional-forecast pass per bloc. On the shipped pack GMRES needs about 85 matrix–vector products where the inverse-free Neumann iteration $x \leftarrow d + Mx$ used until then needed about 600, and it ends with a residual of about 10^{-13} instead of 10^{-10}. A *soft* imported forecast (Section 5.7) enters through soft kernels (the kernels computed with the import's measurement variance) and a soft own-pin forecast a_c, so it, too, is one exact solve;

(d) **accelerated sweeps**: the sweep of (a) with Anderson acceleration (`solveGVARAnderson`, depth 5), which reaches the same fixed point in far fewer sweeps. Until November 2026 the app used it for a soft import, stopped after 12 sweeps to stay responsive (within 0.04 pp of the converged answer for GDP, inflation and the policy rate of every economy). It now serves

as the reference the network solve is checked against (run to a tolerance of 10^{-10}), and as the soft route of the frozen-anchor fallback topology.

The scenario solve on screen uses (b) when the precomputed inverse fits the pin pattern (nothing pinned), and (c) otherwise, soft imports included. Route (a) runs the per-draw bands (Section 5.8). Offline, it also runs the nowcast of the ragged edge when a data pack is built, at tolerance 10^{-5}; the October 2026 pack needed 115 sweeps. What the validation battery asserts is that the closed form reproduces a tightly-iterated Gauss–Seidel fixed point to 10^{-4} in levels on a six-bloc staging subset, and that the shipped artifact reproduces a live rebuild to 10^{-3}. Those are the guaranteed bounds; the binding limit on route agreement is the six-significant-figure rounding applied to M and $(I - M)^{-1}$ at export, not the arithmetic of either solver.

The convergence flag is a budget indicator, not a divergence alarm: it reports `false` whenever an iterative route stops at its sweep budget with the residual still above tolerance, which on the shipped roster is the *ordinary* case for the plain Gauss–Seidel route, since at $\rho_{\mathrm{GS}} \approx 0.94$ twelve sweeps do not reach 10^{-4}. The scenario solve tells the two cases apart. A result whose ρ is at least one, or whose residual is not finite, is treated as divergent: a console warning says so, and the result is kept out of the solve cache, so it is recomputed on every revisit. GMRES solves $(I - M)x = d$ whenever $I - M$ is invertible, whatever $\rho(M)$, so on the network route ρ is estimated explicitly: 60 Arnoldi steps on M from a fixed start vector, once per pin pattern (cached with M); on the shipped pack the estimate equals the artifact's $\rho(M)$ to four decimals. A result that only ran out of sweeps gets a milder console note and is cached, since solving again returns the same iterate. Either way the path itself is still returned (Section 14.2). The per-draw band solves do not read the flag.

The coupling artifact and the model signature. M depends on the models and the pin *pattern*, not the pin values, so for the default (empty) pattern both M and $(I-M)^{-1}$ are precomputed offline — the inverse by a BLAS solve in MATLAB (`app/scripts/invert_coupling.m`; a NumPy fallback ships alongside) — rounded to six significant figures, gzipped (`coupling_default.json.gz`, 35.6 MB compressed), and decompressed in the browser at load. Hydrating it reduces the first complete solve from a cold build of many minutes to a few seconds; it is mandatory infrastructure, not an optimization. A stale artifact is refused by a *model signature*: per bloc, the dimensions (n_c, p_c, H_c) and three six-digit fingerprints of the posterior mean — the first equation's intercept $\hat{\beta}_{0,0}$, the cell $\hat{\beta}_{n_c p_c,\, n_c - 1}$ (zero-indexed; recall the intercept is row zero), and the leading residual variance $\hat{\Sigma}_{0,0}$. Under the legacy layout that middle cell is the far corner of the last lag row; under the seasonal layout of (4) the same index falls n_{sr} rows earlier, on an interior lag coefficient. The formula is deliberately layout-blind, so that legacy signatures stay byte-identical — a signature needs determinism and collision-resistance, not interpretability. On the shipped models, however, the middle cell is zero in all 35 blocs. It falls on a fourth-lag coefficient of a column that is not differenced, and the shipped level VAR(4) has a nonzero fourth lag only on the differenced columns (real GDP, the price index, the deflator). The stored signature reads `0.000000e+0` there for every bloc. In practice the check therefore rests on the dimensions, the intercept and the residual variance, which

a re-estimation moves. Any re-estimation flips the signature, and the solver falls back to route (c). A pin on any coupled bloc changes the pattern and likewise bypasses the cached inverse; only the changed bloc's kernels are re-probed. Two further checks guard the file. The data pack records its SHA-256 hash, and a coupling file that fails this check stops the global model with an on-screen message; it is never rebuilt silently. The pack also records which trade-weights file the coupling was built from. If that differs from the weights file served with it, the coupling is set aside, a red banner says so, and the coupling is rebuilt in the browser, which takes several minutes.

5.4 Global commons

Four US-determined series are carried by every bloc: the WTI oil price (`GLB_OIL`), the US 10-year yield (`GLB_USLR`), the US Baa corporate yield (`GLB_USBAA`), and US equity (`GLB_USEQ`). They are *endogenous, not exogenous*: ordinary endogenous variables in the US model, and endogenous columns of its own in every other bloc. What ties them together is a *uniqueness restriction* — every bloc's copy must equal the single US-determined path — the global-commons counterpart of the trade-weighted consistency the y^* stars carry, and solved by the same coupled fixed point. The solver imposes it *by deviation*:

$$\widehat{\mathrm{GLB}}_{c,t} = \bar{\mathrm{GLB}}_{c,t} + \left(\mathrm{GLB}^{\mathrm{scen}}_{\mathrm{US},t} - \bar{\mathrm{GLB}}_{\mathrm{US},t}\right), \tag{21}$$

each bloc's own baseline plus the US deviation. Pinning the US *level* instead would inject the blocs' baseline gaps (tens of 100 ln units) as spurious shocks; the deviation convention gives a zero-scenario no-op exactly and an identical common shock under conditioning. With the US endogenous the global deviations are unknowns of (19) — this is precisely the second and third coupling family — so a partner shock that moves the US moves the commons, which feed back to every other bloc, without double counting. On screen the commons are conditionable on the US and read-only elsewhere.

The deviation convention in (21) makes the commons agree across blocs *in deviation*; a second step makes them agree *in level*. Each bloc carries its commons as endogenous columns, so its unconditional baseline extrapolates them on its own path and the baselines would drift apart. At load the tool therefore re-pins every bloc's *baseline* commons to the US baseline path and recomputes that bloc's baseline conditional on them, so $\bar{\mathrm{GLB}}_{c,t} = \bar{\mathrm{GLB}}_{\mathrm{US},t}$ and (21) resolves to $\widehat{\mathrm{GLB}}_{c,t} = \mathrm{GLB}^{\mathrm{scen}}_{\mathrm{US},t}$ exactly. On the shipped 35-bloc system the commons then coincide with the US path to machine precision — about 10^{-12} in 100 ln units in the baseline, and the same under a US oil or rate shock — in both the iterative and the closed-form solves. One consequence to keep in view: the shared commons carry the US baseline's own path, drift included. In the October 2026 model (its baseline before any long-run target) the US ten-year yield stays near its last observed 4.4 percent over the five years. The Baa yield rises from about 6.0 to a peak near 7.5 percent in the third year and eases to about 6.9 by the end. Every bloc inherits that same rate path as its commons baseline. This is the price of one globally consistent set of commons rather than thirty-five mutually

inconsistent ones, not a leak: the drift is US-sourced and identical everywhere by construction.

5.5 Operating the global solve

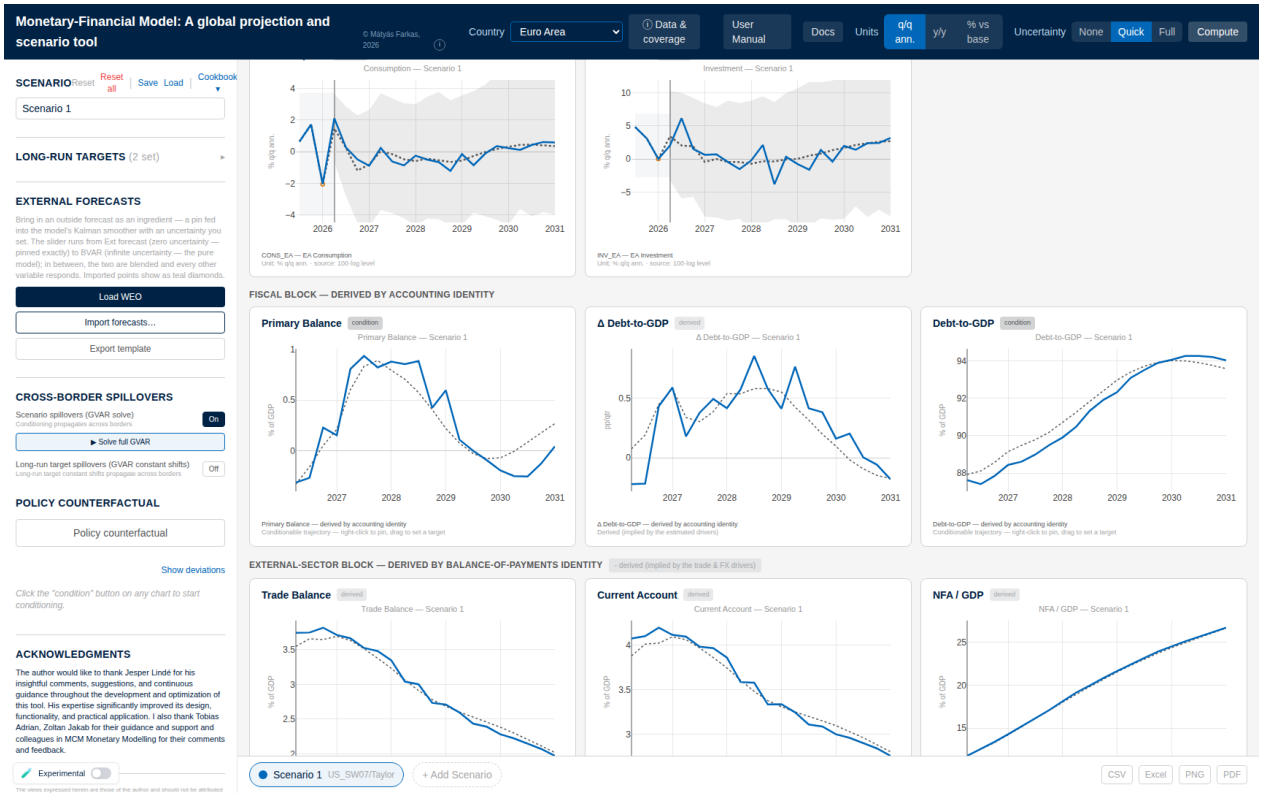


Figure 9: The euro-area view after Solve full GVAR on a US tightening scenario (spillovers On). The conditioning entered in the United States has propagated cross-border: euro-area consumption and investment deviate from baseline (dotted), and the derived satellite blocks — the fiscal block of §6 and the external-sector block of §7 — re-solve on the coupled paths. Switching countries after a full solve always shows the same converged system state.

Every scenario is read in two steps, the domestic solve and the global solve. By default the charts show the *domestic* solve: the economy's own model, conditioned on its own pins. Its foreign (star) variables — foreign demand, foreign inflation, the foreign rate — follow that model's own equations; no other economy is solved, and the global commons stay on the US baseline path outside the United States. This holds whatever is pinned: a second economy, or a global common (oil, the US 10-year yield, the Baa yield, US equity) pinned or imported in the United States, shows its domestic solve until you ask for the global one. The tool never turns cross-border transmission on by itself. The *global* solve is a separate, explicit step:

- **Scenario spillovers (GVAR solve)** (left panel, **Cross-border spillovers**) says whether **Compute** also runs the global solve. With it off, **Compute** stays domestic. With it on, **Compute** runs the complete fixed point (19), and the charts show it once it lands. Switching

it on does not by itself change the charts: before the global solve has run, on and off show the same domestic solve.

- **Solve full GVAR** runs the complete fixed point (19) — all feedback loops, including RoW→US — at once, in a few seconds, and turns **Scenario spillovers** on. A blue **GVAR full solve** badge then shows the sweep count and solve time. The global solve reflects the pins at the moment it runs: edit a pin afterwards and the charts return to the domestic solve of the new pins, and the status line and the **Solve full GVAR** button reappear, until you solve again. No solve runs during a drag. After a global solve, an economy with no pins of its own shows the solved spillover even with the switch off.

The difference between the two solves is the cross-border content of the scenario: what the partners' responses, fed back through the trade-weighted star variables and the global commons, add to the domestic reading. Before quoting a number for a *receiving* country, run the global solve: its domestic solve does not see a foreign shock at all.

One accuracy note on how the full solve is computed. Dragged pins and exact (100%-trust) imported pins are solved exactly — by the shipped inverse or the inverse-free iteration, routes (b) and (c) of Section 5.3 — and the global commons in particular are reproduced to machine precision. A genuinely *soft* imported forecast (trust below one, §7) instead runs the accelerated sweep, route (d), which the app caps at 12 sweeps at tolerance 10^{-4}. On a large shock the cap can leave the *domestic* cross-country spillovers short of the exact fixed point, and the read-out then shows 12 sweeps. The commons still stay identical across economies, because every sweep pins them to the US path. For a headline number from a soft-conditioned scenario, prefer an exact pin.

Example 3 (A global shock reaching a partner)**.** *Load the **Cookbook** energy-supply recipe on the United States (**Load + restriction**): an oil-price spike with its documented supply-shock signature. The recipe pins oil and US equity, two global commons; leave **Scenario spillovers Off**. Switch the **Country** dropdown to the euro area and set **Units** to **% vs base**: the euro-area GDP line is flat at zero — the shock has not been propagated. Return to the US, press **Solve full GVAR**, and switch back: euro-area GDP now shows the imported hit, at the 2026Q1 vintage building to roughly* -0.8 *percent of the baseline level by the end of the horizon. The difference between the flat line and the solved one is exactly the content of the coupled system: a country-only read cannot see a foreign shock at all.*

5.6 Euro-area members: one doorway

The euro members do not set their own interest rate, and the tool respects this: the area-wide instruments — the ECB policy rate, the short rate, the area corporate yield — are *endogenous* in the EA-aggregate bloc and enter each member's model as *weakly exogenous* columns, alongside the member's two rest-of-euro-area stars. The members carry no global commons of their own (no `GLB_*` column): oil, US rates and US equity reach them only through the euro-area aggregate. Shocks reach a member through one doorway, by *deviation transmission*: after a full solve, the EA-aggregate

deviation is mapped identifier-by-identifier onto the member's corresponding columns: policy rate onto policy rate, short rate onto short rate, area corporate yield onto itself, area 10-year yield onto the Bund column the other members carry, and area activity and prices onto the member's rest-of-euro-area aggregates. Each is pinned at the member's own baseline plus the deviation. For the two rest-of-area columns the deviation is that of the area *without* the member, $(\Delta_{\mathrm{EA}} - w_m\,\Delta_m)/(1-w_m)$, with w_m the member's GDP weight, because that is how those columns were built. Germany's own Bund yield is left free, and the member's own estimated dynamics translate the area shock into its GDP, inflation, and yields. A member never updates silently: an amber note above its grid says when a US or EA scenario has not yet been passed through, and **Solve full GVAR** runs the pass. Only a scenario that pins the United States or the euro area is passed to the members. A scenario pinned only on another economy (Japan or China, say) moves the euro-area aggregate in the full solve but does not reach the members.

The reverse direction exists for three kinds of member pin. A pin on a member's GDP or prices adds its GDP-weighted move, $w_m\,\Delta_m$ summed over the pinned members, to the area's GDP or HICP when the coupled system is solved; a pin set on the euro area itself takes precedence. Pinning a member's ECB-rate column mirrors one-for-one into the area policy rate. A member's Bund pin passes one-for-one into the area long yield, and its sovereign-spread scenario passes by the member's GDP weight,

$$\Delta\mathrm{YIELD10}_{\mathrm{EA},t} = \Delta\mathrm{Bund}_t + w_m\,\Delta\mathrm{spread}_{m,t}, \tag{22}$$

the two channels additive, with w_m the member's share in euro-area GDP (real Eurostat current-price weights, shipped in `ea_gdp_weights.json`; a missing weight suppresses the channel rather than inventing one). An Italian spread scenario thus moves the area yield about fifty times more than an Estonian one.

5.7 Outside forecasts across borders

A hard external forecast (trust at 100%) is indistinguishable from a hand pin and rides the same path. A *soft* one changes the effective filter gains through (15), so the kernels of (17) for that bloc are computed with the per-cell variances (and cached under them), and its own-pin forecast a_c is the soft conditional forecast. The cross-border solve is then the network solve (route (c) of Section 5.3) and reaches the exact soft fixed point. Until November 2026 a soft import went to the accelerated sweep (route (d)), capped at 12 sweeps, which could stop short of the fixed point by a few hundredths of a percentage point; a difference between a soft-conditioned cross-border path and a hard-pinned one is now the trust dial alone. An imported forecast, hard or soft, propagates cross-border only on **Solve full GVAR** (or **Compute** with **Scenario spillovers** on), never live while typing; the viewed economy still updates instantly.

5.8 Per-draw coupled bands

With **Experimental features** on (in **Settings**), on a coupled economy whose cross-border path is on screen and with no fiscal target set, **Compute** re-solves the whole coupled system once per posterior draw (N_b of the D shipped draws, selected by deterministic stratification; parameter uncertainty only, no innovation shocks; only the viewed bloc's model and pins are rebuilt per draw, partners held at their means) and reports the empirical 16th/84th percentiles of the resulting conditional cloud. This band pinches naturally at every pin — and can collapse too tightly when only a point or two is pinned, which is why it is opt-in (Remark 2). Each draw runs the full Gauss–Seidel budget (tolerance 10^{-4}, 12 sweeps), so **Full** on a coupled bloc is the slowest computation in the tool.

5.9 Identification: what is assumed where

The separable estimate-then-couple design rests on stated assumptions. *Weak exogeneity of the stars*: each bloc is estimated conditional on its trade-weighted foreign aggregates, valid when no single partner dominates the weights — the granularity condition of Chudik and Pesaran (2016). The condition is assumed, not checked. On the shipped weights it is weakest for close neighbours of a large partner: the United States holds about two-thirds of Canada's and Mexico's weights, and the euro area about two-thirds of Czechia's, Poland's, Hungary's and Romania's. The EM loop relaxes the one-shot version of this assumption by iterating the conditioning information to mutual consistency, and the endogenous US removes the frozen-dominant-unit asymmetry altogether; what keeps the relaxed system well posed is the contraction (20). *Euro-area monetary identification*: policy is set for the area as a whole — endogenous in the aggregate, predetermined for each member — so member-level monetary counterfactuals are conditional scenarios on the area instrument, not independent national experiments. *The scenario responses are conditional projections*: a multi-period pin bundles the direct conditioning with the coupled feedback and is not a structural impulse response (Section 12 keeps this distinction explicit).

6 Public finances

Forty-one economies carry a fiscal block: the six majors (US, euro area, Japan, UK, Canada, China), the seventeen euro-area member states, and eighteen further economies (Appendix B). The design follows the stock-flow-consistent division of labour shared by the CBO's budget mechanics and the IMF's financial-programming and debt-sustainability frameworks (Escolano, 2010; International Monetary Fund, 2013): the BVAR forecasts a small set of fiscal *flow drivers* jointly with the macroeconomy, while the debt stock and the headline balances are *derived* by the budget identity. Debt is never a free VAR variable — the identity has a near-unit autoregressive root $(1+i)/(1+g)$, and small coefficient errors compound into divergent paths; imposing the identity by construction is the discipline the frameworks demand (`app/src/lib/compute/fiscalAccounting.ts`).

6.1 Estimated drivers

Four drivers enter the VAR as ordinary `lin` variables and inherit the full estimation machinery of Sections 3–4:

- `FREV_GDP`, `FPEXP_GDP`: general-government revenue and primary (non-interest) expenditure, percent of GDP, annual ratios;
- `EFFI`: the *effective* interest rate — net interest over lagged gross debt, percent per annum. The stock reprices only as maturing securities refinance, so this rate lags market yields and is estimated separately from them;
- `SFA`: the stock-flow adjustment, percent of GDP — the deficit–debt discrepancy (intragovernmental holdings, valuation effects, other below-the-line operations), constructed on history as the *exact* residual that closes the recursion below, with the pandemic quarters imputed rather than observed.

The three GDP-ratios carry the stationary `WN` own-lag prior of Section 3.2 — they are mean-reverting shares, and the random-walk center would let the projected ratio drift at the long end; under `WN` a ratio's own lags are pulled toward zero rather than toward one, so the forecast is not held at its end-of-sample value. The ratio still moves with the rest of the system, whose growth and price columns carry unit roots, so it need not return to its sample mean: in the October 2026 pack the US revenue ratio (sample mean 17.9, last value 18.7 percent of GDP) ends the five-year forecast at 18.9 percent. `EFFI` keeps the `RW` prior: a stock-average rate is genuinely persistent. Each fiscal bloc ships a `fiscal_meta.json` recording the column indices the recursion reads (`idx` = {`gdp`, `deflator`, `revenue`, `primaryExp`, `effRate`, `sfa`}), the seed debt ratio d_0 with its quarter (`d0_quarter`; where d_0 was rolled forward, also the published `d0_observed`), and the structural constant `sfaDefault` (the long-run SFA mean; in the October 2026 pack US 1.75, EA 0.19, JP −0.94, UK 0.23, CA 3.11, CN 3.25 percent of GDP).

6.2 The debt recursion

Let d_t denote debt-to-GDP in percent, i_t^{eff} the effective rate in percent per annum, and g_t^n the *quarterly* nominal growth rate in decimal, read off the level-space forecast as

$$g_t^n \;=\; \tfrac{1}{100}\Big[\big(\ell_t^{\text{gdp}} - \ell_{t-1}^{\text{gdp}}\big) + \big(\ell_t^{\text{defl}} - \ell_{t-1}^{\text{defl}}\big)\Big], \tag{23}$$

the sum of the real-GDP and deflator log differences in the stored 100 ln space, divided by 100. The price column is the GDP deflator `DEFL_<CC>` in 38 of the 41 economies, the GDP price index `JGDP` for the US, and the CPI for China and New Zealand (a bloc whose GDP series were nominal would omit the deflator term via the `gdpIsNominal` switch; no shipped bloc currently uses it). With the primary balance $pb_t = rev_t - pexp_t$ (percent of GDP, annual, positive = surplus), the debt stock

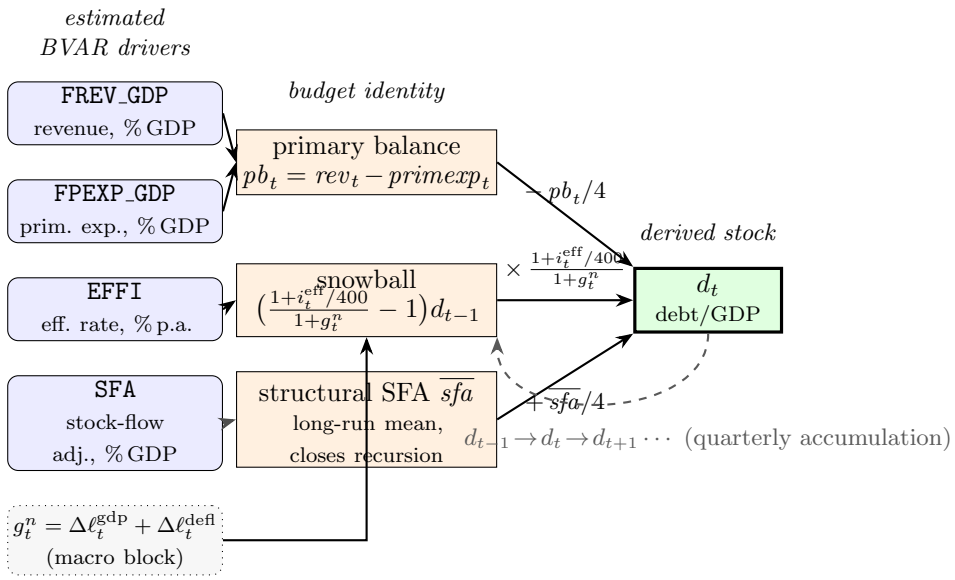


Figure 10: Fiscal stock-flow schematic. The four estimated BVAR drivers (`FREV_GDP`, `FPEXP_GDP`, `EFFI`, `SFA`; Section 6.1) feed the budget identity: revenue and primary expenditure form the primary balance pb_t (§6.1); the effective rate `EFFI` and nominal growth g_t^n (23) form the interest–growth *snowball* (25); and the stock-flow adjustment enters as the *structural constant* $\overline{sfa}$ = `sfaDefault` that closes the recursion. The estimated `SFA` column is charted and conditionable but does *not* enter the debt arithmetic (Remark 5) — hence the greyed, dashed arrow. These flows accumulate into the derived debt-to-GDP stock d_t by the quarterized roll-forward $d_t = d_{t-1}\,(1+i_t^{\text{eff}}/400)/(1+g_t^n) - pb_t/4 + \overline{sfa}/4$ (24), with the lagged stock d_{t-1} fed back each quarter (dashed). Debt is *derived*, never estimated (`computeFiscalBlock`, Section 6.2).

rolls forward one quarter as

$$d_t \;=\; d_{t-1}\,\frac{1+i_t^{\text{eff}}/400}{1+g_t^n} \;-\; \frac{pb_t}{4} \;+\; \frac{\overline{sfa}}{4}, \qquad d_{-1} \equiv d_0, \tag{24}$$

exactly as coded: the annual rate quarterized by 400, the quarterly decimal growth entering directly, and one quarter's worth of the annual flow ratios accruing per step. The quarterly convention is pinned by a back-cast test that reconstructs the observed debt ratio from realized drivers. Two derived flows complete the identity: the interest bill $int_t = i_t^{\text{eff}}\, d_{t-1}/100$ (annual, percent of GDP) and the deficit $def_t = int_t - pb_t$. Subtracting d_{t-1} from (24) isolates the automatic *snowball* from the discretionary balance,

$$\Delta d_t = \underbrace{\left(\frac{1+i_t^{\text{eff}}/400}{1+g_t^n} - 1\right) d_{t-1}}_{\text{snowball (interest–growth)}} \;-\; \frac{pb_t}{4} \;+\; \frac{\overline{sfa}}{4} \;\approx\; \frac{i_t^{\text{eff}}/400 - g_t^n}{1+g_t^n}\, d_{t-1} \;-\; \frac{pb_t}{4} \;+\; \frac{\overline{sfa}}{4}, \tag{25}$$

the quarterized $(i-g)$ form: debt rises mechanically whenever the quarterized effective rate exceeds quarterly nominal growth.

Remark 5 (The SFA in the roll-forward is the structural constant). Equation (24) deliberately uses

the constant $\overline{sfa}$ = `sfaDefault`, *not* the forecast `SFA` column, even though the column is estimated and displayed with its own chart, dynamics, and bands. The forecast column responds to everything else in the VAR — including long-run anchors' cross-responses — and can be dragged to spurious extremes that would distort the derived debt path; the long-run mean is the economically correct closure for a projection. The `SFA` *chart* shows the model's forecast of the discrepancy; the debt *arithmetic* uses the stable constant. Conditioning the SFA chart is still meaningful — it moves the rest of the system through the estimated covariances — but it does not enter (24).

The recursion (23)–(25) is `computeFiscalBlock`, the only fiscal recursion wired to the charts: it produces the baseline, scenario, and counterfactual debt paths and, run per posterior draw, the fiscal bands. It is *passive*: lagged debt enters every term, so a single forward sweep is exact, and there is no debt feedback onto the primary balance or the rate — the displayed balance is exactly $rev_t - pexp_t$ and the displayed rate exactly the BVAR forecast. Three derived charts follow the four driver charts in the **Fiscal block** section, each described as "derived by accounting identity": the **Primary Balance**, the per-quarter change **Δ Debt-to-GDP** (a pure identity output with a **derived** badge, not conditionable), and **Debt-to-GDP**. The section badge reads **drivers estimated · debt derived by identity**, or **derived variables implicitly conditioned** once any fiscal chart is conditioned. The remaining identity outputs (interest, deficit, overall balance, snowball) are computed internally but not charted.

Remark 6 (Timing of the debt ratio). The denominator convention follows GFSM 2014 and the Eurostat quarterly Maastricht series: d_t is the *end-of-quarter* gross debt stock divided by *contemporaneous rolling-annual* nominal GDP, $d_t = D_t / \sum_{s=t-3}^{t} Y_s$ — the four-quarter sum *ending in* quarter t, not lagged GDP. All fiscal ratios (`FREV_GDP`, `FPEXP_GDP`, debt/GDP) use the same contemporaneous four-quarter denominator. The seed d_0 is the debt ratio at the last history quarter (2026Q2 in the October 2026 pack). Where published debt ends earlier — 38 of the 41 economies, mostly at 2026Q1 — the last published ratio is rolled forward to that quarter with this same identity on the completed history (`app/scripts/roll_d0_forward.mjs`). d_0 is then a model estimate wherever those flows are nowcast, and `fiscal_meta.json` keeps the published value as `d0_observed`. The one deliberately *lagged* object is the effective rate, $\texttt{EFFI}_t = 400\, int_t / D_{t-1}$: interest accrues on the pre-existing stock, the financial-programming convention. In the roll-forward (24) the denominator's growth enters through the factor $(1 + i_t^{\text{eff}}/400)/(1 + g_t^n)$ with g_t^n the quarterly nominal-GDP growth of the same quarter; in sample, any wedge between quarterly growth and the rolling-sum growth is absorbed into the estimated `SFA` residual by construction, so the identity holds to machine precision on history.

Remark 7 (Coupon-lumpy blocs: trailing four-quarter interest). Seven blocs record interest payments on an annual or semiannual coupon calendar — CL, CO, CR, ID, MX, TR, ZA (Israel was the eighth before it left the roster) — identified by a systematic within-year pattern in the raw interest flows (calendar-$R^2 > 0.5$ with coefficient of variation > 0.25, or near-perfect quarterly alternation). The lumpy coupon would otherwise inject a mechanical sawtooth into $\texttt{EFFI}_t = 400\, int_t / D_{t-1}$. Since

the October 2026 rebuild these blocs measure the effective rate over the last four quarters,

$$\texttt{EFFI}_t = 100\,\frac{\sum_{j=0}^{3} int_{t-j}}{D_{t-4}},$$

a genuine, one-sided measure that runs to the latest published quarter and needs no filled values (`effi_trailing4` in `pipeline/sources/derived.py`). Thailand, whose `EFFI` was rebuilt in 2026 from actual interest payments, uses the same definition. Primary expenditure keeps the *raw* interest — total minus interest, $te - int$, stays true non-interest spending — and the exact debt-identity residual `SFA` absorbs the genuine cash-versus-accrual timing, which is why the coupon blocs also carry the seasonal dummies of Section 3.3. Earlier estimations instead replaced the four payments within each complete calendar year by their common mean (`even_distribute_years` in `pipeline/build_fiscal22_columns.py`), which removed the `EFFI` sawtooth (mean absolute quarterly change from ~ 5 to $\lesssim 0.25$ points for CL, MX, ZA) and moved the near-Nyquist companion root off -1 (e.g. ZA from -0.97 to -0.63); the trailing sum smooths the coupon in the same way without redistributing within a year.

6.3 Targeting a debt or balance path

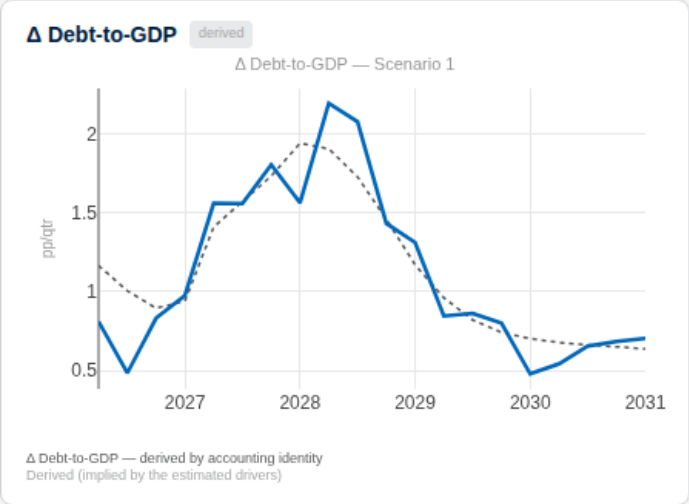


Figure 11: The derived debt block under the tightening scenario: the quarterly change in debt-to-GDP (solid) rises above baseline (dotted) as the snowball term absorbs the higher effective rate, then reverts. This panel is a read-only identity output (**derived** badge); the **Debt-to-GDP** and **Primary Balance** panels are conditionable: right-click pins a point, dragging sets a target path, and the smoother back-solves the primary balance that delivers it (§6.3).

The recursion also runs backward. Given a target debt path $\{d_t^{\text{tgt}}\}$, the primary balance required at each quarter is the exact inverse of (24),

$$pb_t \;=\; 4\left[\frac{d_{t-1}^{\text{tgt}} + int_t/4}{1+g_t^n} \;-\; d_t^{\text{tgt}}\right] \;+\; \overline{sfa}, \tag{26}$$

with the lag taken along the target path and $int_t = i_t^{\text{eff}}\, d_{t-1}^{\text{tgt}}/100$ (`requiredPrimaryBalance`). The tool evaluates the effective rate and growth on the unconditional forecast, computes (26), and pins the combination $rev_t - pexp_t = pb_t$ as a linear-combination constraint (Section 4.2); the smoother finds the most plausible revenue/expenditure mix.

One step is easy to overlook and is not optional. The inversion is performed on the path the user actually *sees*, which on the twenty-seven blocs carrying default long-run anchors is the *drifted* forecast of Section 8, not the raw unconditional one: the effective rate and nominal growth in (26) are read off $\hat{y} + \delta$, and the pinned combination is pre-compensated by the drift wedge,

$$rev_t - pexp_t \;=\; pb_t \;-\; \left(\delta_t^{\text{rev}} - \delta_t^{\text{pexp}}\right), \tag{27}$$

because the smoother solves on the undrifted forecast while the chart adds δ to both columns afterwards. Without the compensation a balance target advertised as exact would land beside itself by $\delta^{\text{rev}} - \delta^{\text{pexp}}$, and a debt target would be inverted against the wrong interest–growth gap — a real error, since an anchor such as the UK effective rate or the US growth-and-deflator pair sits directly on the recursion's own inputs. With no anchor active $\delta \equiv 0$ and (27) is the plain pin. For a balance target the compensation makes the hit exact. A debt target still carries the one-pass approximation of Example 4, and on the current growth-rate models that miss is larger, not smaller, when a drift is active — a known limitation awaiting a second refinement pass.

Example 4 (Drawing a debt path)**.** *On a fiscal economy, right-click the* ***Debt-to-GDP*** *chart where the path should pass — on this chart the dragged value is the absolute target,* 100 *meaning* 100 *percent of GDP — and shape the trajectory; right-click a quarter to pin or release it. A* ***Primary Balance*** *target set the same way is hit* exactly *(it is one linear constraint). A* ***Debt-to-GDP*** *target is hit* approximately*: the inversion* (26) *conditions on the model's unconditional rate and growth, which themselves respond a little to the implied consolidation, and the tool does not iterate on that feedback. Expect a miss of a point or two of GDP without long-run targets, and more when targets are active (the default on 27 economies): in the 2026Q1 pack the US debt path ended about 5 points off its target at the five-year horizon. For an exact decimal the instrument is the balance target; for a coherent consolidation scenario the debt target is the natural handle. Note that conditioning* any *fiscal variable implicitly conditions the derived row — a note next to the block header says so. A primary-balance or debt target is solved on the economy's own model only: while one is active, the charts show the single-economy scenario even with* ***Scenario spillovers (GVAR solve)*** *on, and trade partners do not respond to the target.*

Remark 8 (Fiscal bands are identity-consistent, and re-centred)*.* Each band draw pushes its own simulated drivers through (24), so the debt band widens exactly as fast as the underlying budget-and-growth uncertainty implies. Because the roll-forward compounds, the per-draw debt distribution is right-skewed: its median sits below the debt path computed at the mean drivers, which would leave the central line hugging the band's upper edge. At render the band is therefore re-centred so that its *median* sits on the displayed path, which preserves both the width and the right-skew; a shift of

the midpoint would instead force the fan symmetric and mask the asymmetry. This is the fan-chart convention "central projection ± uncertainty"; at actively pinned quarters the band collapses onto the line. The same treatment applies to the external position below.

7 The external accounts

Forty-seven economies carry an external block — forty-one of them chartable in the tool (Brazil, Chile, Estonia, Ireland, Peru and Romania ship it but do not chart it: their projected current account drifts 12–22 points, or their trade data are distorted or in mismatched units; Costa Rica, Japan, Korea, Türkiye and Taiwan have none) — deriving the trade balance, the current account, and the net foreign asset (NFA) position from the estimated trade, price, and exchange-rate variables by the balance-of-payments identity (`app/src/lib/compute/externalAccounting.ts`). All three charts are read-only (**derived** badges): to move them, condition the drivers — exports, imports, the exchange rate, growth — and the external row follows.

With b_t the NFA-to-GDP ratio in percent, seeded at the latest published net international investment position ($b_{-1} \equiv nfa_0$: US −66.75, EA +11.0, CN +20.35 percent of GDP at the current vintage), the per-quarter identity as coded is

$$ca_t = \overline{ca} \;+\; \big(nx_t - nx_{\text{seam}}\big), \qquad tb_t = ca_t - \frac{i^{\text{NFA}}\, b_{t-1}}{100}, \tag{28}$$

$$val_t = -\,\phi^{\text{FX}}\, b_{t-1}\, \frac{\Delta \ell_t^{\text{FX}}}{100}, \qquad b_t = \frac{b_{t-1}}{1+g_t^n} \;+\; \frac{ca_t}{4} \;+\; val_t, \tag{29}$$

where $nx_t = 100\,(X_t - M_t)/Y_t$ is the model's real net-export ratio, built from the exponentiated export, import and GDP levels (for the US and the euro area a real-trade-to-real-GDP ratio; for China, whose trade is in US dollars, the real GDP index is reflated with the CPI and a calibrated dollar scale, and the monthly-average trade flows are multiplied by three), $\Delta \ell_t^{\text{FX}}$ is the quarterly change of the 100 ln effective exchange rate (positive = home appreciation, which shrinks the home-currency value of net foreign assets — hence the sign), and g_t^n is quarterly nominal growth built as in (23) but with the external block's own price column (`idxIds.deflator` in `external_meta.json`): the consumer price index (CPI or HICP) in 46 of the 47 economies and `JGDP` for the US, not the GDP deflator the fiscal block uses. The stock update divides the inherited ratio by $(1 + g_t^n)$ — the automatic denominator effect — adds one quarter of the annual current account, and adds the valuation term; the return on the stock enters through the income term, computed on the undiluted lagged ratio (the second-order cross-terms this drops are noted in the code header).

The current account is *anchored*, and this matters more than it looks. $\overline{ca}$ is the published current-account ratio at the seam, sourced per bloc and shipped in `external_meta.json` (all 47 blocs carry one: AT +2.63, AU −3.07, BR −2.29, CA −0.14, CH +5.04, US and EA likewise). The model supplies only the *deviation* of its real net-export ratio from that same seam. The reason is that nx_t is a real-volume ratio and not the nominal balance: for an economy whose relative prices sit away from the chain-link reference year, its *level* can be wrong by several points of GDP and

can even carry the wrong sign, while its *change* is exactly the trade-balance response the estimated block is entitled to claim. Anchoring keeps the level honest and lets the model move it. A bloc shipping no $\overline{ca}$ would fall back to the raw absolute ratio; none currently does.

Remark 9 (The income and valuation terms are switched off). Two inputs of (28) and (29) are modelling assumptions rather than estimates: the return on net foreign assets i^{NFA} and the share of the position exposed to the exchange rate ϕ^{FX}. **Both ship at zero in all 47 external blocs**, pending a data review. Consequently the reported trade balance and current account coincide — whatever primary income the economy actually earns on its position is folded into the sourced $\overline{ca}$ rather than modelled separately — and the *entire* valuation term of the NFA recursion is identically zero on every path the tool draws, so the position accumulates the anchored current account alone and exchange-rate revaluation does not move it. The equations are stated here in full because the machinery is present and these two constants are the only thing standing between it and a live income and valuation channel — but no result currently shown depends on either.

8 Long-run anchors

The long-horizon behaviour of the forecasts is inherited from the estimated VAR, whose level dynamics carry roots on or near the unit circle; the far end of a projection can therefore drift away from any value the user regards as a credible steady state — a two-percent inflation objective, a potential growth rate, a stable revenue ratio. The **Long-run targets** panel lets the user pin the terminal forecast of a chosen variable to a number, exactly and instantly. The mechanism is a deterministic counterfactual on the companion drift in the spirit of Villani (2009), implemented in `app/src/lib/compute/driftSetter.ts`.

The shipped consensus set. The tool ships a reviewed default anchor set (`default_anchors.json`; July-2026 transition-default revision: 34 anchors across 27 blocs) tuned under the now-default steady-state transition. Because that path phases the adjustment in rather than shifting the whole path, the set is anchored to *official* long-run values wherever the model shows a genuine terminal gap — central-bank inflation targets throughout (euro-area members at the ECB's two per cent, national targets elsewhere), medium-term potential growth where the raw path is clearly off, and three fiscal anchors (the UK effective rate, Canada's primary expenditure and China's revenue ratio). This is a real gain of the transition shape: fourteen of these official-target anchors (Canada, Chile, China, Costa Rica, Czechia, Finland, Greece, Mexico, Norway, New Zealand, Poland, Portugal, Sweden, Thailand — mostly inflation, to two-to-four per cent) are rejected as harmful under the rigid shift, which would jump their near term; twelve of the fourteen are clean under the transition, and the two that are not — Costa Rica and New Zealand inflation — fail on terminal cross-drag rather than on the near term, a shape-independent property. Anchors are admitted only if the sensitivity clears the guard of §8.1, and the shipped set is *curated* with the smell test (`anchor_smell.mjs`) and locked by a panel-wide non-explosiveness regression. Two guards still bite — both shape-independent, being

properties of the estimated steady state rather than the path: a target that lands within the redundancy threshold of the model's own terminal is dropped, and one that would displace an unanchored headline variable by more than a point at the terminal is dropped or co-anchored. Re-run at the vintage of that review the battery still flagged eight of the thirty-four — five inside the redundancy threshold (Australia, the euro area, Indonesia, India, Korea) and three on cross-drag (Costa Rica and New Zealand inflation, Mexico growth) — which are retained deliberately, for cross-country consistency of the published targets; the script is a review instrument, not a build gate. The euro-area primary-expenditure anchor stays removed for the second reason: at the targets considered it drags the unanchored euro-area growth and inflation terminals by more than a point. Its effect on the euro-area policy-rate terminal is a separate, human diagnostic — the cross-drag gate scans growth, inflation and the effective rate only, so no policy-rate displacement can trip it. A handful of high-inflation emerging markets — Colombia, Hungary, Türkiye, Malaysia (and Israel, before it left the roster) — are shipped model-driven and flagged: their estimates link inflation and growth so tightly that pinning one drags the other several points at the steady state, which no univariate anchor and no choice of path can repair; these await re-sourcing or a joint re-estimation. Japan ships no default anchors (its sliders work and start at the model estimate). The anchor set was not re-tuned for the October 2026 re-estimation: the same 34 anchors ship on the new model, and its review battery has not been re-run. One consequence was found and accepted: Costa Rica's raw inflation forecast is now far below its target, so its inflation anchor needs a large shift, and the overlay it adds to every column of the bloc reaches about 155 points on its own copy of world oil under the rigid shift (about 80 under the default transition); the safety bound of the drift-setter test was raised from 150 to 175 for it (user decision, October 2026).

8.1 The drift setter

Write the deterministic part of the companion recursion (6) as $Y_{T+h} = T^h Y_T + S_h\, c_\alpha$ with

$$S_h = \sum_{k=0}^{h-1} T^k, \tag{30}$$

the partial sum of companion powers, built once per bloc (H matrix products, cached). A shift δc of the drift moves the forecast by exactly $\Delta Y_{T+h} = S_h\, \delta c$ — an algebraic identity, verified against the iterated recursion to 10^{-11}. We deliberately avoid the closed-form steady state $(I - \sum_\ell B_\ell)^{-1} c$: in the shipped models real GDP, the price index and the GDP deflator are estimated in growth rates, so their level columns carry an exact unit root and $I - \sum_\ell B_\ell$ is singular (near-singular for the other persistent columns), whereas the finite-horizon operator (30) is well posed. A target on variable j shifts only j's own equation, $\delta c = e_j\, \delta c_j$, and the coherent system-wide response is the j-th column, $\Delta Y_{T+h} = S_h[:, j]\, \delta c_j$. The scalar is chosen so the *displayed* terminal forecast lands on

the target $\psi_j^\star$:

$$\delta c_j = \frac{\psi_j^\star - \psi_j^0}{\text{sens}_j}, \qquad \text{sens}_j = \begin{cases} 4\big(S_H[j,j] - S_{H-1}[j,j]\big), & \texttt{log} \text{ variable}, \\ S_H[j,j], & \texttt{lin} \text{ variable}, \end{cases} \tag{31}$$

with ψ_j^0 the current terminal display forecast — the slider's default position, so an unmoved slider is a byte-exact no-op. A variable whose own sensitivity is below the guard $|\text{sens}_j| < 0.3$ has its slider disabled ("near-singular response"): forcing a target there would require an explosive drift. The terminal is the last quarter of the twenty-quarter forecast (2031Q2 in the October 2026 pack), whichever display horizon is chosen. Under the default transition, the anchored growth and inflation columns have closed only about 45 to 80 percent of the gap (median 60) by the end of the **3yr** view.

Two paths to the target: steady-state transition (default) or rigid shift. Equation (31) fixes *how far* the terminal must move; a second choice — the **Path to target** toggle in the **Settings** drawer (lower left; buttons **Steady-state** and **Rigid**) — fixes *how the adjustment is distributed across the horizon.* The constant-drift response $\Delta Y_{T+h} = S_h[:, j]\, \delta c_j$ places the *entire* effect from the first forecast quarter: because $S_1 = I$, a growth variable's displayed deviation at $h = 1$ is $4\,\delta c_j = \text{sens}_j\, \delta c_j / T^{H-1}[j,j]$, the terminal gap divided by $T^{H-1}[j,j]$, applied at once. On the shipped anchors that divisor runs from about 0.4 to 1.6, so the first-quarter jump can be smaller than the terminal gap or more than twice it (on US real growth, anchored alone at 2.2 per cent, -1.89 points at $h = 1$ against a terminal gap of -1.88). Thereafter the deviation is $4\, T^{h-1}[j,j]\, \delta c_j$, which follows the column's own estimated response and need not be monotone. This **rigid shift** is exact at the terminal and band-coherent, but it *jumps* the near term at the origin and, for a large gap, can manufacture a near-term contraction or deflation. The shipped default is instead a **steady-state transition**: the same terminal δc_j is phased in along the anchored variable's own estimated convergence profile

$$w_j(h) = \frac{S_h[j,j]}{S_H[j,j]}, \qquad w_j(1) = \tfrac{1}{S_H[j,j]}, \quad w_j(H) = 1, \tag{32}$$

where $w_j(1)$ is about 0.03–0.13 on the unit-root growth and inflation columns the anchors mostly target (0.04–0.09 for most) — small — but not on a fast-reverting ratio, where it is 0.10, 0.30 and 0.81 for the three fiscal anchors the tool ships (China's revenue ratio, the UK effective rate, Canada's primary expenditure; there the $S_H[j,j]$ normaliser cancels and the own-column path coincides with the rigid shift). Where $w_j(1) \approx 0$ the near term stays data-consistent (no origin jump) and the forecast *glides* to the target by the end of the horizon at the model's own mean-reversion speed. This is the reactive, no-re-estimation counterpart of a steady-state (mean-adjusted) prior in the spirit of Villani (2009): rather than translating the mean path by a constant, it lets the estimated dynamics carry the forecast from the observed edge to the long-run target. It is finite-horizon by construction: w_j is a ratio of two partial sums of companion powers, well defined for any T — the

unit-root case included — and it never forms the singular $(I-\sum_\ell B_\ell)^{-1}c$ that motivated (30). (In a scalar AR(1) it collapses to the familiar $w(h)=(1-\rho^h)/(1-\rho^H)$; the matrix case does not simplify, because $(I-T)^{-1}$ does not factor out of a single diagonal entry.)

The terminal is identical to the rigid shift in *display* space, so anchored targets are hit under either path. The level paths are not identical: for a `log` variable the transition integrates a phased-in growth deviation, so its cumulated terminal level is appreciably smaller — US real GDP −4.25 against −6.79 in 100 ln points with real growth alone anchored at 2.2 per cent (−3.92 against −6.18 with the shipped US pair, growth and deflator, solved jointly). Levels are what the fiscal recursion (24) consumes, so the debt ratio does differ across the toggle even where the displayed growth and inflation terminals coincide; ratio (`lin`) columns are unaffected, their level and display deviations being the same object. Each horizon is still a rigid translation, so band widths remain invariant (Remark 3). Cross-variable drag phases in at the *source* anchor's rate applied to the terminal cross-response, which has a consequence worth stating: a source with a large $w_j(1)$ moves the other columns already at the first forecast quarter, where the rigid shift — whose $S_1 = I$ kills every off-diagonal at $h = 1$ — leaves them untouched.

Several targets: per-variable or joint. With two or more targets the cross-responses interfere — a drift on equation j moves every other anchored terminal through $S_H[k, j]$. The **Settings** drawer exposes a **Multiple-target solve** toggle (**Joint** / **Per-variable**):

- **Per-variable**: each δc_k from its own diagonal sensitivity (31). Stable and bounded, but approximate — it ignores the cross-responses, so with several targets each terminal lands slightly off its stated value.
- **Joint**: solve the $K \times K$ display-unit system $M^{\mathrm{lr}}\,\delta c = \psi^\star - \psi^0$, whose diagonal is exactly (31), by Gaussian elimination with partial pivoting. Every anchored terminal then lands exactly — but a near-collinear target set (say two anchored interest rates the dynamics cannot separate in the long run) makes the system ill-conditioned: the exact solution buys terminal precision with huge offsetting drifts (measured up to $|\delta c| \approx 459$ where the diagonal needed ≈ 11) that wreck the path elsewhere. A singular system falls back to per-variable numerically; an ill-conditioned but solvable one is returned unchanged. The code marks it (joint drift above twice the per-variable maximum), but the interface does not show the mark: if an anchored path turns erratic, switch to **Per-variable**.

Joint is the shipped default: the steady-state transition removed the front-loading that once made per-variable the safer choice, so the exact solve ships and per-variable stays one click away in the Settings drawer for a near-collinear anchor set (an erratic anchored path is the sign; the tool shows no warning). The near-term implausibility that used to accompany a large gap was the signature of the *rigid shift*, not of the joint solve. At the 2026Q1 origin, anchoring the US deflator — model terminal 3.91 against the 2.0 target — displays a −1.98 point disinflation shock at the very first forecast quarter under the rigid shift, pushing the shown series from 3.78 down to 1.80; the steady-state transition of (32) removes it, easing the same series from 3.78 to 3.69 in the first quarter and

onto 2.00 at the terminal, a first-quarter move of under a tenth of a point (-0.09). The two toggles are orthogonal: the solve mode (per-variable / joint) sets *how far* each terminal moves; the path (transition / rigid) sets *how the move is phased in*.

Bands, fiscal forwarding, and visibility. The response ΔY is added identically to the central path and to every shipped band quantile — a rigid translation, widths provably invariant (Remark 3) — and, in level space, forwarded into the fiscal recursion (24), so the debt path responds coherently: a real-growth target owns the direction of the debt ratio through the $(i - g)$ snowball, which is why its row carries the tag **sets debt path**. Judgment is flagged: the affected chart title turns amber with a **judgement** badge stating target and estimate, and the grid-level status line counts the anchored variables. Every control starts at the model estimate, and clearing all targets restores the raw model byte-exactly.

Consensus preset. Blocs with publicly defensible long-run values carry an **Apply consensus anchors** preset — deliberately minimal (34 anchors across 27 economies at the July-2026 transition revision): at most potential growth and the inflation objective per economy, plus three fiscal anchors (the UK effective rate, Canada's primary expenditure, China's revenue ratio); never policy rates, unemployment, or asset prices, whose collinear anchors would ill-condition the joint system. Each anchor was reviewed under the transition path on the July 2026 model; eight were kept despite a flag, and the set has not been re-checked on the October 2026 model (see the paragraph above). Where a preset ships it is on by default (the button then reads **Consensus applied**) and fully reversible (**Clear**); the model-driven-and-flagged economies of the preceding paragraph ship none and stay entirely model-driven.

On a coupled bloc the target is marginal. The drift setter operates on one bloc's companion. On a GVAR economy it shifts that bloc's own long run; partners are not re-solved unless the second spillover switch, **Long-run target spillovers (GVAR constant shifts)**, is on. That channel is instant — the drift deviation enters the coupled system only through the forcing term d of (19), leaving M and its precomputed inverse untouched, and the receiving country's banner turns amber with **+ spillover from ...** naming the origin — judgment stays visible across borders.

One term is held out by construction: the *viewed* bloc's own δc is excluded from d. What its chart shows is therefore its single-bloc response plus the inbound spillover from the others, not the coupled fixed point of its own anchor; its outbound-and-back feedback is treated as second order against the direct effect. The exclusion is also what makes the viewed-only-anchored case an exact no-op for the spillover term, so switching the channel on cannot move a chart whose bloc is the only one anchored.

8.2 The in-prior long-run prior

The long-run prior was evaluated and is not used. No deployed model carries it. It is recorded here because the experiment shaped the design, and because a reader who finds the script should know why it does not run.

The "prior for the long run" (PLR) of Giannone et al. (2019) augments the prior of Section 3.2 with dummy observations shrinking a set of long-run linear combinations Hy toward non-cointegration. It is a proper Bayesian prior: it changes the posterior and requires re-estimation. A US head-to-head (run on an earlier, pre-expansion $n = 36$ specification; the deployed bloc is $n = 40$) found that the PLR acts mainly on the *bands*, tightening the GDP-growth 68% band by roughly 7 percent at unit tightness while leaving the central long-run path essentially unchanged. Since the question it was brought in to answer — how to pin a long run — is answered by the drift setter of Section 8.1, and since the cost is a full re-estimation of every bloc, it was not adopted.

What remains is one standalone script, `pipeline/export_bvar_us_plr.m`, which reproduces that US experiment. The deployed estimator (`export_bvar_gvar24_loop.m`, `cgl_bvarV5.m`) contains no PLR term of any kind. To pin a long run, use the drift setter; there is no second, in-prior route in this tool.

Remark 10 (A retired mechanism). An earlier release pinned long runs by entropic tilting (Robertson et al., 2005) — reweighting the posterior draws so the weighted terminal moment hits the target. It degenerated in practice: as the target moved away from the draw cloud the effective sample size $1/\sum_d w_d^2$ collapsed toward one (a joint 13-constraint tilt reached ESS $\to 1$ at $D = 500$), leaving the anchored band resting on a handful of draws precisely where the user most wanted a firm answer. The drift setter replaced it wholesale — deterministic, exact at any target, instant — and the tilt code has since been removed from the app; only MATLAB comparison scripts remain in `pipeline/_verify/`.

9 Policy counterfactuals

A scenario asks what happens if the data come in differently; a policy counterfactual asks what a *deliberate policy change* would cause. The distinction is identification: the smoother's response to a pinned rate path is the estimated historical co-movement, which cannot tell a policy surprise from a response to something else. The counterfactual layer therefore borrows structure from a DSGE model of the user's choice, drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016), and overlays its identified responses on the scenario.

9.1 Impulse responses and shock inversion

The MMB catalogue in the tool lists 169 models; pre-computed impulse responses ship for 155 of them, each under up to nine interest-rate rules (Taylor, Smets–Wouters, CEE, Gerdesmeier–Roffia, Levin–Wieland–Williams, two Orphanides–Wieland variants, Coenen, Christiano–Motto–Rostagno). A model–rule pair is one JSON file of responses to a monetary (and, where the model has

a fiscal sector, a government-spending) shock, with the rule embedded in the model's equilibrium when the responses were generated — the tool never evaluates a policy rule at run time. Rules without computed responses for the selected model are greyed out.

Let $h_r(s)$ denote the interest-rate response at horizon s to a unit monetary shock. Given a user rate adjustment Δr_t on a set of *constrained* quarters, the tool solves for the shock sequence by forward substitution on the lower-triangular Toeplitz system — a new shock only where the user pinned, zero elsewhere:

$$\varepsilon_t = \begin{cases} \dfrac{\Delta r_t - \sum_{j<t} h_r(t-j)\,\varepsilon_j}{h_r(0)}, & t \text{ constrained}, \\ 0, & \text{otherwise}, \end{cases} \tag{33}$$

so at unconstrained quarters the rate follows the decaying response of the earlier shocks — the model's rule, not a snap-back. The implied inflation and output responses are the convolutions $\Delta\pi_t = \sum_{j\le t} h_\pi(t-j)\,\varepsilon_j$ and $\Delta y_t = \sum_{j\le t} h_y(t-j)\,\varepsilon_j$; a parallel inversion runs on the fiscal shock for the spending sliders.

To use it, press **Policy counterfactual** in the left panel (the button then reads **Exit counterfactual mode**). Under **DSGE Model**, **Browse...** searches the catalogue and the **Policy rule** list picks the rule. Every new scenario starts on `US_SW07` with the Taylor rule, whatever economy is on screen, and each scenario tab keeps its own choice. **Rule-based | Optimal (B&M)** chooses how the rate path is set (Section 9.4). A hand-set path is entered with the twelve **Interest rate adjustment (pp)** sliders (Q1–Q12, ±2 pp in steps of 0.05, with **Undo** and **Reset**), or by dragging the **Policy Rate** chart while counterfactual mode is on (not in **Optimal (B&M)** mode, where the sliders show the optimum read-only). Every quarter touched becomes a *constrained* quarter of (33). Switching country clears the rate and spending adjustments; the model and rule stay.

9.2 Re-conditioning and the response modes

The counterfactual is not an additive overlay of every variable: the DSGE deltas are imposed on the *key* variables and the BVAR fills in the rest. The **Macro response** selector offers two modes:

- **DSGE/Taylor-rule (IRF-consistent)** (default): the rate is pinned at scenario plus Δr (monetary) plus Δr^{fiscal}, inflation and output at scenario plus the DSGE deltas; every variable the user has *not* conditioned is re-conditioned through the smoother, so its response is consistent with both the identified transmission and the estimated covariances, while the other variables the user *did* condition are held at their scenario path — at every quarter, not only the quarters they pinned. (The rate, inflation and output always carry the DSGE deltas, even when the user conditioned them.) That is what makes a zero adjustment reproduce the scenario exactly, and it also means a conditioned variable shows a flat zero policy response, which is an artefact of the experiment and not an economic result. One convention subtlety is handled per variable: the inflation response used is the MMB series `inflationq`,

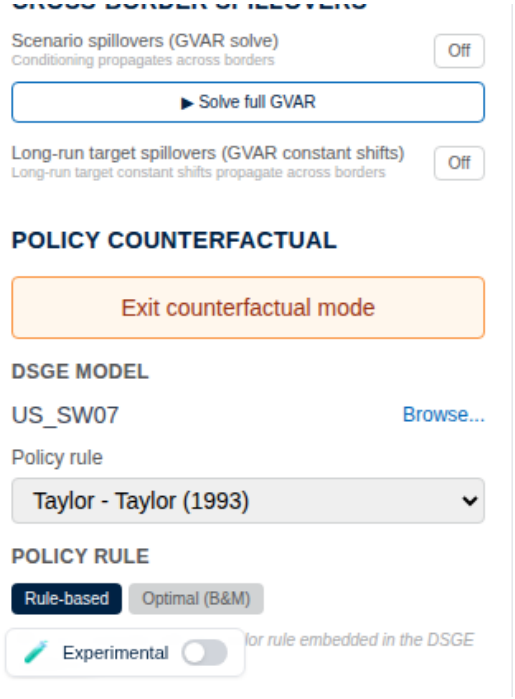


Figure 12: The counterfactual control rail: entering counterfactual mode selects the DSGE model (here US_SW07) and the policy rule — a named rule (Taylor 1993) or the optimal rule of §9.4 — whose impulse responses are layered on the BVAR scenario as described in §9.1.

the annualized quarterly rate, cumulated into the 100 ln price level ($\sum_{s\le t}\Delta\pi_s/4$) so that the implied price level moves exactly as the model's own; the output-gap response a percent *level* deviation (applied directly, not cumulated), and a rate response an identity.

- **BVAR-endogenous (let adjust)**: only the rate path is pinned; everything else responds through the estimated co-movements, exactly as in a scenario. The two modes answer different questions — "what does this theory say the move would do" versus "what usually accompanies such a path" — and their disagreement is informative. If the selected pair has no computed responses the tool falls back to this mode and says so in an amber notice.

In both modes the displayed counterfactual is the scenario plus the smoother delta between the counterfactual pass and a reference pass with zero adjustment, converted to display units by the conventions of Section 4.3; a zero adjustment reproduces the scenario exactly. The counterfactual is computed on the viewed economy's own model alone: it is not passed on to other economies, and on a GVAR bloc whose charts show the coupled scenario it is still built on that bloc's single-country (partners-frozen) scenario.

9.3 Reading the result

The counterfactual of the active scenario draws as a dashed rust line on every chart (on the policy rate, with diamond markers); the scenario line stays put. The second scenario tab uses the same

rust, drawn solid, so tell the two apart by the dash. To isolate the policy effect, set **Units** to **% vs base**; in that view the **Level | Rate (IRF)** toggle selects between the cumulative effect on the *level* of each variable (the "% chg." reporting convention) and the quarter-by-quarter response in native units. For a variable whose DSGE response is a genuine *rate*, the latter reproduces the MMB response exactly: inflation is injected as the cumulated level $\sum_{s\le t}\Delta\pi_s/4$, so differencing and scaling by four inverts the cumulation, and the policy rate passes through as an identity. That inflation response is the MMB series `inflationq`, the annualized quarter-on-quarter rate — the Inflation panel's unit, and the series the drivers layer of Section 10 uses — not the year-on-year `inflation`, which in the shipped responses is the four-quarter average of `inflationq`. (Until 11 October 2026 the counterfactual read `inflation`: its impact response was a quarter of the quarterly one and its peak came later.) The output gap is the exception. It is a percent *level* and is injected directly, so it is the **Level** view that reproduces the MMB output-gap IRF, while **Rate (IRF)** shows the implied q/q-annualized growth deviation 4Δ of it — four times the MMB number at impact, and of the opposite sign thereafter whenever the gap is closing.

Two further readings of this view are easy to get wrong. Each line is a deviation from the *unconditional* baseline, so the counterfactual line still carries the scenario's own conditioning: the policy effect is the vertical *gap* between the scenario line and the counterfactual line, and the two coincide only when the scenario conditions on nothing. And in the IRF-consistent mode a variable the user has conditioned — other than the rate, inflation and output — shows no policy response of its own, because the counterfactual pass holds it at its scenario path (Section 9.2). Uncertainty bands, when computed, belong to the scenario; the counterfactual is a single model-implied path.

Remark 11 (The rule choice is a first-order magnitude decision). The same one-quarter rate move can have an order-of-magnitude different effect across rules. In the shipped responses of the default US model (`US_SW07`, Smets and Wouters, 2007), the peak inflation response to a unit monetary shock under the model's own Smets–Wouters rule is about 30 times the response under the Taylor rule, the output response about 6 times. Treat pass-through magnitudes as model-and-rule dependent, never as a fact; flipping the model while the scenario stays put (each scenario tab carries its own model and rule) is the cheapest robustness check in the tool.

9.4 Optimal policy

Instead of hand-crafting the path, select **Optimal (B&M)**. The tool then sizes a policy move that is optimal *given the scenario on screen*, following Barnichon and Mesters (2023), and keeps it up to date: every change to the scenario, the model, the rule or a slider recomputes it. The scenario supplies, for each quarter $h = 0, \dots, H-1$ of the display horizon, its inflation π^s_h (the Inflation panel, % q/q annualized, long-run targets included), its output gap x^s_h (defined below) and its policy rate i^s_h. The loss is

$$\mathcal{L} = \sum_{h=0}^{H-1} \Big[\lambda_\pi \big(\pi^s_h + \Delta\pi_h - \pi^\star\big)^2 + \lambda_y \big(x^s_h + \Delta x_h\big)^2 + \lambda_{\Delta i} \big(i_h - i_{h-1}\big)^2 \Big], \qquad i_h = i^s_h + \Delta i_h, \tag{34}$$

with i_{-1} the last observed policy rate. The first two terms are the Barnichon–Mesters loss. The third, rate smoothing, is *not* part of their approach; it is an optional slider (default 0). The changes $\Delta\pi$, Δx and Δi come from the chosen DSGE's responses to a monetary-policy surprise under the chosen rule — the sufficient statistics: the MMB `inflationq`, `outputgap` and `interest` responses from impact, scaled so that the rate rises by one point on impact. The macro response is always taken from the DSGE (the IRF-consistent mode of Section 9.2, whatever the **Macro response** menu says): a BVAR's response to a rate pin is a conditional forecast, not the response to an identified policy shock, so it cannot serve as the sufficient statistic.

One move today (the default). The policy object is a single change δ in the policy rate in the current quarter; afterwards the rate follows the chosen rule, so $\Delta i_h = \delta\,\tilde{r}_h$, $\Delta\pi_h = \delta\,\tilde{p}_h$ and $\Delta x_h = \delta\,\tilde{x}_h$, with $\tilde{r}$, $\tilde{p}$, $\tilde{x}$ the responses per point of impact rise. The loss is then a parabola in δ and the optimum is a ratio of two sums,

$$\delta^\star = -\,\frac{\sum_h \left[\lambda_\pi\,\tilde{p}_h(\pi^s_h - \pi^\star) + \lambda_y\,\tilde{x}_h\,x^s_h + \lambda_{\Delta i}\,\Delta\tilde{r}_h\,\Delta i^s_h\right]}{\sum_h \left[\lambda_\pi\,\tilde{p}_h^2 + \lambda_y\,\tilde{x}_h^2 + \lambda_{\Delta i}(\Delta\tilde{r}_h)^2\right]}, \tag{35}$$

well posed in every model, with nothing to invert (Δ is the first difference, $\Delta i^s_h = i^s_h - i^s_{h-1}$). This is the concept of the Federal Reserve Bank of San Francisco's Policy Calibration Tool (Barnichon and Singh, 2026): a forecast under the expected policy, one set of effects of a rate change, a weight between the two objectives, and the move that best serves them — here with a model's responses in place of one empirical estimate, for every economy GMAPS covers. The counterfactual pins the current quarter only; the rest of the drawn rate path is the rule's own propagation of that one surprise.

Three uncertainty ranges. Under **Uncertainty ranges** three switches each show one source of uncertainty around the optimal move, and each holds the other two fixed: **Outlook uncertainty** (the outlook varies; model and rule fixed), **Model uncertainty** (the model varies; scenario and rule fixed) and **Transmission uncertainty** (the model varies; the picked move fixed). They can be combined freely; each band has its own colour, and the read-out names each by colour and chart. The rule is fixed in all three.

Model uncertainty. Tick **Model uncertainty** to see how much the answer depends on the model. Every model of the Macro Model Data Base that ships the chosen rule and passes an admissibility screen gets its own $\delta^\star$ from equation (35) on the same scenario. The read-out gives the median and the 10–90% range of these moves and the share of models in which the chosen model's move lowers the loss; the Policy Rate chart shades the 10–90% band of the models' rate paths in the counterfactual's orange (they are alternative versions of that dashed line), with the median as a dotted line. The set is either all admissible models or those in the chosen model's catalog category (the 69 estimated US models, the 20 estimated euro-area models, and so on). A model is admissible when a surprise tightening raises the rate on impact and lowers inflation and the output gap on

average over the first two years, when its rule is in force (FI_AINO16, whose policy is exogenous, is excluded) and when no numerical gate failed in the 2026-10-13 cross-check of the 69 estimated US models against an independent run (US_CET15 hard-codes its own smoothing; US_CCF12 has a unit root). Models outside that cross-check are taken as shipped, and the read-out says how many. Under the Smets–Wouters rule 130 models are admissible, 62 of them estimated US models. The range is the honest display of model uncertainty: it does not choose a model, and the set itself is a choice — in the trial behind this design (`docs/reest-2026/bm_trial`), dropping five strongly transmitting models moved the robust answer by 1.5 points.

Transmission uncertainty: pick a move, then see its outcomes. Model uncertainty answers one question: how much do the models disagree about the *move*, taking the scenario's inflation and output as given. A second question follows once a move is chosen: how sure is one about what it *does*? The tool keeps the two apart (since 2026-10-26; each has had its own switch since 2026-11-04). **Policy path** picks the move the counterfactual carries: **Chosen model's move**, the $\delta^\star$ of the model and rule in the drop-down (the default), or **Median move across models**, the median of the admissible models' own moves — one number, which the chosen model then carries with its rule, so the dashed line is the chosen model's view of the median move and the read-out states both moves. (The dotted median *path* on the Policy Rate chart is the quarter-by-quarter median of the models' paths; it is no model's own path and could only be imposed with a surprise in every quarter, the full-path problem, so the pick is the median move, not that path.) With **Transmission uncertainty** on, the picked move is applied in every admissible model, and the Inflation and GDP charts shade the 10–90% band of the outcomes the models give in violet, with their median dotted, each model's path converted to the panel's display unit first. The read-out adds the share of models in which the picked move lowers the loss and where inflation and the gap lie after one and two years. The band is not drawn on a chart whose variable is pinned in the scenario: a pin fixes the scenario path, not the response, but a band around a path the user asserted reads as forecast uncertainty, so only the chosen model's dashed line is drawn there. The two bands show different things and are told apart by colour and label (since 2026-10-27): the orange band on the Policy Rate chart is the models' own moves, the violet bands on the Inflation and GDP charts are what one move does across models, and the read-out names each band by its colour and chart ("Model uncertainty, orange band (Policy Rate chart): the moves the 130 models would make …", "Transmission uncertainty, violet bands (Inflation and GDP charts): the picked move in each of the 130 models …"). Both are pointwise percentiles, not an envelope of whole paths. The median move is a central value, not a guarantee: the trial behind this design found no formal robust rule that survives the handful of strongly transmitting models, which is why the control says "median" and not "robust".

Outlook uncertainty (2026-11-04). The third switch holds the chosen model and rule fixed and lets the outlook vary. After **Compute** (with **Quick** or **Full** uncertainty), each draw d of the scenario's uncertainty band replaces the scenario's inflation, output gap and policy rate in equation (35), and the chosen model's move $\delta^\star_d$ is recomputed: the move that would be best if the

outlook turned out like that draw. Because $\delta^\star$ is a ratio of two sums and linear in the scenario paths, each draw costs one pass over the horizon. The Policy Rate chart shades, in magenta, the 10–90% band of the draws' optimal rate paths (each draw's own rate plus its move carried by the rule), with the median dotted; the read-out gives the median and 10–90% range of the moves themselves and the share of draws in which the picked move lowers the loss. The path band therefore also carries the uncertainty of the scenario's own rate path; the range of the moves isolates the decision. The draws carry both parameter and future-shock uncertainty, so this range is wider than, and only related to, the forecast uncertainty Barnichon and Mesters simulate (their Section 5). The draws are those of the last **Compute**: changing the loss weights, $\pi^\star$ or the gap inputs updates the band at once, while a new pin needs a new **Compute**, as the scenario bands do. The range is wide, and with **Quick**'s 50 draws its median is noisy: on the US baseline (US_SW07, SW rule; 2026-11-04) four **Quick** runs gave medians of −2.6, −1.4, +0.2 and −1.4 pp against −1.33 pp on the displayed line, and two **Full** runs (200 draws) −1.5 and −1.8 pp, with the 10–90% range of the moves about −9 to +5 pp — wider than the range across the 130 models (−5.3 to +1.0 pp). Use **Full** for numbers to quote.

Rules without a one-move experiment. In optimal mode the Taylor (1993), OW08 and OW13 rules are greyed out and the solver refuses them. Under Taylor and OW08 the rate carries no inertia, so a surprise leaves the rate for one quarter only and the computed move becomes implausibly large (Remark 11); OW13 is a difference rule (coefficient one on last quarter's rate), so a surprise moves the rate level permanently — the opposite extreme. Switching to optimal mode on one of them moves the rule to SW where the model has it. All three remain available for the rule-based counterfactual.

The output gap. The BVAR has no potential output, so the gap is measured against a potential path that starts at the last data point and grows at a set rate: $x^s_h = (y^s_h - y_T) - \frac{g^\star}{4}(h+1) + x_0$, with y the scenario's 100 ln real GDP, y_T its last observation, $g^\star$ the **Potential growth** (% a year) and x_0 the **Economy today vs potential** (%, default 0). A straight line needs a starting point and a slope, and a growth rate alone cannot say where the line starts, which is why both are asked: the first input says how far GDP is above or below potential in the last data quarter, the second how fast potential grows from there. The two sit together under the heading **Potential output — the line the output gap is measured against**, and the read-out restates them in one sentence ("the output gap is GDP against a potential level that starts x_0% from today's GDP and grows $g^\star$% a year") together with the quarter in which the scenario's gap is largest. On the GDP chart the potential growth is drawn as a thin dash-dot line from the last data point; the starting level cannot be shown on a growth chart, so the read-out carries it. The potential growth defaults to the bloc's enabled long-run GDP target (Section 8) and, where none is set, to the growth of the shipped history; the read-out says which is in use. Nothing is fitted on a window. The gap today is an assessment the user makes; at the default 0 the economy is taken to be at potential in the last data quarter and the policy answers only the growth shortfall or excess the scenario shows relative to potential. (Until 2026-10-15 the gap was GDP against an ordinary least-squares trend over the

shipped forty quarters, with the pandemic inside; for the United States that trend put the inflation scenario's gap at -5.6% at its trough, which decided cut versus hike on its own.) The DSGE's `outputgap` response — the response the GDP panel's counterfactual draws — is applied to this gap one for one. There is no flexible-price output underneath the level: the only flexible-price element is that response. In most of the shipped models it is simply the output response, because a policy surprise leaves their flexible-price output unchanged: in 107 of the 130 models admissible under the Smets–Wouters rule the two shipped responses coincide, in 20 they differ (nine of the 62 estimated US models; US_SW07 is not among them) and three ship no output series. When the chosen model is one of the 20, the read-out says that a change in the policy rate also moves its potential output.

Full path (illustrative). The 2026-10-11 design sets the rate in every quarter of the horizon: a sequence of surprises ε moves the variables by the lower-triangular Toeplitz operators of the responses, and the stacked least-squares problem

$$\varepsilon^{\star} = \left(\tilde{\mathcal{H}}'\tilde{\mathcal{H}}\right)^{-1}\tilde{\mathcal{H}}'\,\tilde{b}, \qquad \tilde{\mathcal{H}} = \begin{pmatrix} \sqrt{\lambda_{\pi}}\,\mathcal{H}_{\pi} \\ \sqrt{\lambda_{y}}\,\mathcal{H}_{x} \\ \sqrt{\lambda_{\Delta i}}\,D\,\mathcal{H}_{i} \end{pmatrix}, \tag{36}$$

is solved with a negligible ridge ($\tilde{b}$ stacks the target gaps, D is the first-difference operator). It is kept as **Full path (illustrative)**. It hits the targets exactly when one weight dominates, but the problem is badly conditioned (condition number about 1,300 at twenty quarters on US_SW07), the paths swing from quarter to quarter and are several times larger than any policy shock in the model's sample, and it is not what Barnichon and Mesters, or the San Francisco Fed tool, compute. Read it as an illustration of what a full plan would need, not as advice.

The controls. One slider sets the weights ($\lambda_{\pi} = 5w$, $\lambda_{y} = 5(1-w)$, w from 0 at the left end, **Full employment**, to 1 at the right end, **Price stability**); $\pi^{\star}$ is the inflation target (% q/q annualized). Its default is the bloc's current official inflation target as published by its central bank (the data pack's `inflation_targets.json`: the target, its band, the price measure targeted, the institution and the source page; the midpoint when only a range is stated); the caption under the slider says so, or that the value was set here, or that the bloc has no numerical target and the tool's default of 2 is used (an exchange-rate anchor, or a definition of price stability without a number). A value set here is kept until **Use the official target** is clicked. **Economy today vs potential** and **Potential growth** define the gap; **Rate smoothing** is $\lambda_{\Delta i}$ (0 to 5); **Policy move** chooses one move today or the full path; **Policy path** picks the chosen model's move or the median move across models (which switches **Model uncertainty** on); the three **Uncertainty ranges** switches (**Outlook**, **Model**, **Transmission uncertainty**) and the model set select the bands. The rate sliders show the resulting path read-only. The read-out gives today's move (and, under the median pick, the chosen model's own move beside it), the horizon, the potential level and growth with their source, the quarter of the largest gap, the loss before and after, the cross-model median and range of

the moves, the share of models in which the picked move lowers the loss with the one- and two-year outcome ranges, and warns when the path takes the rate below zero (no lower bound is imposed) or moves it by more than five points.

What to expect. In the 2026-10-13 trial on the United States (equal weights, a twenty-quarter horizon, the earlier trend gap), the US_SW07 responses under the Smets–Wouters rule gave one move of −0.66 points in the baseline, −1.33 with inflation two points above the baseline throughout (the trend gap's trough drove the cut), +3.10 in a four-quarter boom and −4.43 in a bust. Across the 62 admissible estimated US models the inflation case ranged from −11.5 (10%) to +1.5 (90%) with a median of −4.1, and SW07's move lowered the loss in 71% of them. With all weight on inflation the same scenario calls for a hike. The answer is only as good as the model's transmission and the gap assessment: compare rules, models and the gap inputs before drawing conclusions.

A fiscal lever completes the block: twelve **Fiscal adjustment — gov spending (pp)** sliders (±3 pp per quarter) set a path for government spending on top of the scenario. In the IRF-consistent mode that path is inverted into the model's government-spending shock, as for the rate; 83 of the 155 models with responses ship such a shock. Where the model has none, or in the BVAR-endogenous mode, the sliders only pin the bloc's government-spending column, where it has one. For projecting the public finances themselves, the fiscal block of Section 6 is the instrument.

10 Scenario drivers: reading a path through a model

The counterfactual layer of Section 9 starts from a deliberate policy move and asks what it would cause. The drivers layer runs the same borrowed structure in the opposite direction. It takes a path the user has *already* built — the displayed scenario, or an imported outside-forecast combination (Section 4.5) — and asks which sequence of a chosen DSGE model's own structural shocks would have produced it over the first twelve forecast quarters. The answer is drawn as a stacked contribution chart in the same units as the main panel, with an explicit grey residual for the part the named shocks cannot reproduce.

The claim is conditional on the model, and on nothing else. The reading is: *through model M's lens, this is the mix of M's shocks that reproduces the path.* It is not a statement about what caused the scenario, and it is not model-free shock labelling; a different model on the same path gives a different mix, and the tool makes no attempt to adjudicate between them. What is validated is same-model recovery: a path is generated from a model's own known shock sequence, decomposed under that same model, and the recovered contribution paths are compared with the truth. Those checks speak to the estimator, not to the model and not to the world. The feature is experimental and says so on screen — the **Drivers** button carries an amber `exp` badge — and the licence conditions of Sections 10.5 and 10.6 are enforced by the interface rather than left to the user's discretion.

10.1 The question and the object

The decomposition answers the question a policy meeting asks of a projection: *which shocks would have produced this path?* It is the forward-looking counterpart of the historical shock decomposition that central-bank models report over the sample, and it is performed the way a structural model is used to interpret an outside forecast — filter the path through the model and read off the shocks — except that the object filtered is a scenario the user drew, not a data vintage.

Formally, let $H_d = 12$ be the decomposition horizon in quarters, q the number of model observables watched, and K the number of shocks (or shock clusters, Section 10.7) the user has named. The input is the scenario's *level deviation* from the baseline, z, expressed in the model's observable space by the conventions of Section 10.4. Because the long-run drift of Section 8 translates baseline and scenario rigidly, it cancels in the deviation by construction, and a zero scenario gives a zero decomposition.

10.2 The stacked system and the estimator

Write $R_{k,o}(s)$ for model M's response of observable o to a unit of structural shock k at horizon s, taken from the same pre-computed MMB response files that feed the counterfactual layer (Section 9.1). A shock arriving in quarter τ moves observable o in quarter $t \geq \tau$ by $R_{k,o}(t-\tau)$, so the path implied by a whole shock sequence is the convolution

$$z_o(t) \;=\; \sum_{k=1}^{K} \sum_{\tau=0}^{t} R_{k,o}(t-\tau)\, \varepsilon_k(\tau) \;+\; u_o(t), \qquad o = 1, \ldots, q, \quad t = 0, \ldots, H_d - 1, \tag{37}$$

with $\varepsilon_k(\tau)$ the size of shock k in quarter τ and $u_o(t)$ the part of the path the named shocks leave unexplained. Stacking observables and quarters gives the linear system

$$z = \Theta\, \varepsilon + u, \qquad \Theta[\, oH_d + t,\; kH_d + \tau\,] = R_{k,o}(t-\tau)\, \mathbf{1}\{t \geq \tau\}, \tag{38}$$

where $z \in \mathbb{R}^{qH_d}$ holds the target path (observable-major, quarter within observable), $\varepsilon \in \mathbb{R}^{KH_d}$ the unknown shock sequence (shock-major, arrival quarter within shock), and Θ is the $qH_d \times KH_d$ stacked response matrix whose every (o,k) block is lower-triangular Toeplitz. Only the first H_d horizons of each response are used; the shipped response arrays carry twenty-one entries, of which index 0 is the pre-impact zero, so $R_{k,o}(s)$ is read at offset $s+1$ — the same convention the validation benches use.

The system is square only by coincidence. With three observables and three shocks it has 36 rows and 36 columns; with the full seven-observable US configuration, 84 and 84; with two shocks and three observables, 36 rows and 24 columns. What is reported is the *minimum-norm least-squares* solution: among all ε minimizing the weighted fit error, the one of smallest Euclidean norm. Let $\nu_j = \|\Theta_{\cdot j}\|_2$ be the response energy of column j, $V = \mathrm{diag}(\nu_1, \ldots, \nu_{KH_d})$, and W the diagonal row

weighting of Section 10.5 (unity on the three common observables). The estimator is

$$\hat{\varepsilon} = V^{-1}\tilde{\varepsilon}^{\star}, \qquad \tilde{\varepsilon}^{\star} = \arg\min \|\tilde{\varepsilon}\|_2 \quad \text{over} \quad \arg\min_{\tilde{\varepsilon}} \left\| W\left(\Theta V^{-1}\tilde{\varepsilon} - z\right)\right\|_2^2, \tag{39}$$

computed by forming the Gram matrix of $W\Theta V^{-1}$, adding a ridge $\alpha = 10^{-8}$ times its mean diagonal entry, factorizing once, and then running two sweeps of iterative refinement against that same factorization. Every refinement iterate stays in the row space of Θ, so the sequence converges to the minimum-norm solution and the ridge acts only as a numerical stabilizer, not as a prior on the shocks. Columns whose norm falls below 10^{-12} keep a scale of one, so a shock with no response stays inert instead of being amplified.

The reported contributions are computed from the *raw* responses and the un-normalized shocks,

$$c_{k,o}(t) = \sum_{\tau \le t} R_{k,o}(t-\tau)\,\hat{\varepsilon}_k(\tau), \qquad u_o(t) = z_o(t) - \sum_{k=1}^{K} c_{k,o}(t), \tag{40}$$

so neither the column scaling nor the row weighting touches what is drawn: $\Theta V^{-1}\tilde{\varepsilon} \equiv \Theta\hat{\varepsilon}$ identically, and the bars plus the residual reproduce the scenario line to eight decimal places in the regression tests.

Whether a residual can exist at all is a property of the configuration, not of the scenario. The drawer computes the rank of Θ by a pivoted Cholesky on the smaller-side Gram matrix with relative pivot tolerance 10^{-12}, and reports the regime: a banner at the foot of **Customize** names it with the rank and the row count, and the default view repeats only the residual-able case, in amber. When the rank reaches the number of rows qH_d the system is *exact-fit*: the named shocks span the observable paths, the residual is identically zero by construction, and it is evidence of nothing. Only when observables are watched beyond what the named shocks can span — the *residual-able* regime, flagged in amber with the rank and the row count — does the grey band carry information. On the shipped menu the two regimes are about twenty orders of magnitude apart in conditioning, so the rank threshold is not a delicate choice: full-rank lenses have Θ condition numbers of order 10^2, while the rank-deficient Canadian lens sits near 3×10^{17}.

10.3 Equal response energy across shocks

The column scaling V^{-1} in (39) is not cosmetic. A minimum-norm criterion applied to the raw columns penalizes each shock by its own size in shock units, which means it quietly prefers whichever shock moves the observables most per unit of ε. Model shocks are not comparable in that metric. Measured on the shipped Smets–Wouters lens (Smets and Wouters, 2007) at three observables and twelve quarters, the stacked column norms run from 1.28 for the monetary shock and 2.12 for the government-spending shock up to 11.58 for the wage markup and 13.84 for the price markup: an order of magnitude between the smallest and the largest. Under the raw criterion the monetary shock is priced out of its own attribution, because explaining a given path with it costs far more in $\|\varepsilon\|$ than explaining the same path with a markup shock.

Normalizing every column to unit response energy replaces that implicit ranking with an explicit neutral one: the criterion is indifferent between two shocks that move the watched observables identically. Where the configuration is identified the scaling is a pure reparameterization and changes nothing — the licence fixtures still recover their generating shock sequence to 10^{-6} — and where it is not identified it removes the bias. On the same three-observable Smets–Wouters configuration, mean off-class leakage falls from 32 percent under the raw criterion to 5 percent under the normalized one. The signature of the choice is easiest to see in the extreme case: given two exactly collinear shocks whose response norms differ two to one, the shipped solver splits the contribution equally, whereas the raw criterion would split it four to one, entirely on the strength of the units the model happens to write its shocks in.

10.4 From the panel to the model's observables

The scenario lives in the BVAR's smoother level space of Section 3.1 — 100 ln for log variables, native units for linear ones — and the model lives in its own observable space. Three observables are common to every MMB model in the menu and are always watched: output, quarterly inflation, and the policy rate. They are built from the active bloc's panel columns by the conversions of the upper block of Table 1. Two factors of four deserve naming, because getting either wrong silently rescales an entire class of shocks: the model's `inflationq` is an annualized quarterly rate, so the price-level deviation is differenced and multiplied by four, while the model's `interest` is already annualized and is taken as an identity, with *no* division by four. These are not assumptions. The shipped Smets–Wouters file carries both the MMB common block and the model's own native observables, and a regression test reads the ratios off it, checking on every shock and every horizon with a non-trivial response that `inflationq/pinfobs` and `interest/robs` both equal 4 to better than 10^{-9}.

Table 1: The observable mapping on the United States bloc under the Smets–Wouters lens. Deviations are scenario minus baseline in the stored level space; all differences use a zero pre-sample value, $\text{dev}_{-1} = 0$. The upper block is the three common observables, watched under every model; the lower block is the four native observables the user may add. Spec column names are the US bloc's.

Observable	Meaning	Built from the panel	Drawn as
`output`	Output	`GDPH` level deviation, identity	4Δ, % q/q ann.
`inflationq`	Inflation	4Δ of the `JGDP` level deviation	identity, % q/q ann.
`interest`	Policy rate	`POLRATE` deviation, identity	identity, %
`dc`	Consumption (Δ)	Δ of the (`CH` + `JC` − `JGDP`) deviation	×4, % q/q ann.
`dinve`	Investment (Δ)	Δ of the (0.2384 `FRH` + 0.7616 `FNH`) deviation	×4, % q/q ann.
`dw`	Real wage (Δ)	Δ of the (`LXBC` − `JGDP`) deviation	×4, % q/q ann.
`labobs`	Hours, proxied by employment	`LANAGRA` deviation, level	4Δ, % q/q ann.

Every one of these maps is *linear* — a gain, or a scaled first difference — which is what makes

the chart legible: contributions and residual pass through the display operator additively, so the stacked bars and the grey residual sum to the scenario line exactly rather than approximately.

The four native observables of Table 1 require the spec columns `CH`, `JC`, `JGDP`, `FRH`, `FNH`, `LXBC` and `LANAGRA`, and of the deployed blocs only the United States carries all seven. Everywhere else the drawer runs at three observables. Two rows only approximate the model's observable, and their chips carry a warning mark with the reason: `labobs` uses employment where the model wants hours per capita, and `dinve` is the linear, share-weighted form of total fixed investment. The Christiano–Motto–Rostagno lens exposes a parallel set of four — consumption, investment, hours and the real wage — but its consumption and investment rows still use the single columns (Δ`CH` and Δ`FNH`) and its hours row `LANAGRA`; those three chips are marked as proxies too. Its observation equations were measured from the shipped response file rather than assumed, and the anchor is that its `gdp_obs` equals the first difference of the common `output` column for every non-trend shock, which fixes consumption, investment and the real wage in the same percent quarterly growth units through the trend loading they share with it, and its `hours_obs` in the percent level units of `labobs`. Three of its observables are excluded as duplicates of an already-watched column, for the reason given in Section 10.5. It exposes no financial observables in this pass: credit, net worth and the external-finance premium are not mapped.

10.5 Two rules: the naming budget and the collinearity flag

The decomposition is only as informative as the configuration allows, and two rules keep the interface from offering readings the estimator cannot support. Both come out of the validation bench `pipeline/bench_simple_decomp.m`, which generates paths from a model's own shocks and measures how well the shipped solver recovers them.

The first is the **naming budget**: at q watched observables the user may name at most q shocks. The bench's identified reference point is the Smets–Wouters lens at its native seven observables with all seven usable shocks, where Θ has rank 84 out of 84 rows, noiseless recovery of the generating sequence is accurate to 3.6×10^{-15}, and recovery succeeds in 100 percent of instances at one percent noise. The rule is enforced twice over: the engine refuses a decomposition with more shocks than observables, and the interface disables every un-named shock chip the moment the budget is reached. Adding a native observable raises the budget by one — the heading of the observable chips says so — which is why the four addable observables of Table 1 carry the seven-shock US configuration.

Not every observable a model exposes may be added, and the exclusions are the substance of the rule rather than an exception to it. The Smets–Wouters observables `dy`, `pinfobs` and `robs`, and their Christiano–Motto–Rostagno counterparts, are each an invertible linear transform of an observable already watched: a differenced or rescaled copy of output, inflation or the policy rate. Watching one would raise the naming budget by one while buying no new rank at all, and the identification the budget rule promises would be fictitious. The exclusion is pinned against the data rather than against a curated list: a regression test requires each of the four legitimate additions to buy exactly $H_d = 12$ new rank and each of the three excluded transforms to buy exactly zero,

alone and all together.

The second rule is the **collinearity flag**. Two shocks whose stacked responses point in nearly the same direction cannot be told apart by any estimator, whatever the rank test says. For every model the build step computes the cosine between each pair of unit-normalized stacked three-observable, twelve-quarter response paths, and ships every pair with $|\cos| > 0.95$ in the index. Naming both members of a flagged pair is permitted but warned about, with the measured cosine quoted to two decimals; the one-click **Use all** selection instead walks the model's shock list in order and skips any shock that forms a flagged pair with one already chosen. Among the small lenses this bites in four places: the Christiano–Motto–Rostagno lens flags three pairs among its financial and investment shocks (the largest at 0.998), the Smets–Wouters 2003 euro-area lens one pair at 0.963, the Hong Kong lens two (one at 1.000), and the Canadian lens one at 0.975. The Canadian lens is the instructive case: with three usable shocks, two of them near-duplicates, its response columns span only 24 of the 36 stacked rows however many are named, so the greedy selection names two and the decomposition runs in the residual-able regime. The two FRB/US lenses are a different order of problem entirely — 109 and 117 flagged pairs — and are handled by grouping rather than by skipping (Section 10.7). The discipline to take from the flag is not to avoid such shocks but to read them jointly: the sum over a flagged pair is identified, the split within it is not.

Remark 12 (The measurement-error slider is inert at budget-max). Added observables can be held with measurement error, on a strict-to-permissive slider (me from 0 to 4, default 1) that sets their row weight to $w = 1/\sqrt{1 + me}$ (so $w \approx 0.71$ by default) while the three common observables stay exact. The control does nothing when the budget is exhausted, and is disabled there with a tooltip that says so. The reason is arithmetic: at $K = q$ the stacked system is square, and scaling both Θ and z by the same positive diagonal W leaves the solution of (39) exactly unchanged. Row weights matter only when observables outnumber named shocks, which is precisely when the model is being asked to reconcile more information than it has shocks to reconcile it with. Un-name a shock to give the slider a role.

10.6 The class layer and where it is licensed

Individual shock names are not how a projection is discussed. The **Classes** view therefore aggregates contributions into Monetary, Demand and Supply — and, where a model has one, Financial — by a fixed class map shipped with the tool. Aggregation is exact in the sense that class bars are sums of shock bars, and by linearity of the display operator they still reconcile with the scenario line. Whether those sums are *recoverable* is a separate question, and it is the question the second bench, `pipeline/bench_class_recovery.m`, was written to settle. It generates paths from known class-active shock sequences over twenty seeds and applies two pre-committed criteria: the recovered class-total path must correlate at least 0.95 with the true class total in at least 90 percent of active-class instances, and mean off-class leakage must not exceed 25 percent. Both must pass.

Under the shipped solver, exactly one configuration passes: the Smets–Wouters lens with all seven observables watched and all seven of its class-map shocks named, which recovers class totals

in 100 percent of instances with 0 percent off-class leakage and a perfectly diagonal attribution matrix. Nothing else does. The same lens at three observables recovers in only 50 percent of instances — it fails the recovery gate even though its leakage, at 5 percent, is excellent. The Christiano–Motto–Rostagno lens at seven observables reaches 78 percent against the 90 percent bar. The FRB/US lens at three observables, the only configuration in which it can be run, reaches 37 percent with 47 percent leakage and fails both gates. The **Classes** toggle is greyed everywhere outside the single licensed configuration, with a tooltip stating that class totals are not reliably separable at that combination and pointing at the results file.

Two consequences of that scope are worth stating plainly. The first is that the licence is a joint property of three coordinates — the model, the set of watched observables, and the set of named shocks — and not of the model alone, nor of the model and the observable set (Remark 13). The second is what the licence does and does not assert: it says the estimator recovers class totals when the truth is generated by that same model at that configuration. It says nothing about whether the Smets–Wouters classes are the right classes, and nothing about the world.

Remark 13 (Un-naming one class member revokes the class view). The licence predicate takes three coordinates — the model, the set of watched observables, and the set of *named shocks* — because the bench held all three fixed: it ran the Smets–Wouters lens at its seven-observable set with all seven class-map shocks active. Drop one member under **Customize**, say the wage markup, and **Classes** greys immediately, even though the model and the observable set are untouched. This is not pedantry. The dropped shock's energy does not vanish; it is redistributed across the classes that remain, which is exactly the quantity the off-class-leakage criterion measures and exactly the quantity the bench never measured for that six-shock split. An earlier build gated on model and observables only, and would have kept the licensed labels and the "100 percent recovery, 0 percent leakage" tooltip on a configuration no bench had seen.

Two of the remaining lenses group their shocks without claiming a class split. The Christiano–Motto–Rostagno lens shows its own class map as four *families* — shown as "Monetary shocks", "Demand shocks", "Supply shocks" and "Financial shocks" — purely as an organising grouping of its nine shocks, and the caveat under the chart prints the number the bench refused them on: 78 percent recovery with 3 percent leakage at seven observables (United States only), 45 percent with 32 percent leakage at the common three, which is what every other bloc sees. They are not a licensed class split and should not be quoted as one. Every other model in the menu, including the euro-area, Canadian, Hong Kong, Brazilian and UK lenses, has no class map at all and falls through to one series per named shock under the model's own mnemonics, with the monetary and fiscal shocks given plain-language labels.

10.7 The FRB/US lenses: empirical clusters

The two FRB/US census lenses in the menu each carry 54 shocks: the policy and fiscal shocks, plus 52 equation residuals. They are handled differently from every other model, for two independent reasons. The class view is refused for them by the bench at the only configuration they can be

run in, as reported above. And the individual residuals are not separable in the three common observables in the first place: 109 of the 1,431 shock pairs exceed the 0.95 collinearity threshold in one variant, 117 in the other. Naming three of them at random would produce a chart with no identification behind it.

The drawer therefore groups by the structure the response space actually has. The flagged pairs define a graph on the 54 shocks; its connected components are the *clusters*, labelled by the equation name of their largest member and ordered by size. One variant has 23 clusters, the other 21; the largest gathers seventeen or eighteen financial, housing and fiscal residuals, and about three-quarters of the clusters are singletons. Naming a cluster names one column of Θ — the first principal component of its members' unit-normalized response paths, computed by power iteration and signed to agree with the largest member — so the naming budget counts clusters, not members: three clusters at three observables, whatever their membership.

A full-rank Θ is not evidence that a naming is usable, and here the gap is wide. The build step scans every possible cluster triple through the shipped solver over twenty sparse-truth seeds and records which recover their generating sequence to 10^{-6}: 123 of 1,771 triples pass in one variant, 122 of 1,330 in the other. Most of the rest pass the rank test and fail the recovery test. The shipped default is chosen deterministically among the passers — maximize the number of shocks covered, prefer a triple containing the monetary cluster, then take the first in cluster order — and it covers 19 of the 54 shocks with a worst recovery error of 2.5×10^{-9} in one variant, 12 of 54 at 1.3×10^{-7} in the other. Leave the default naming and the drawer says so in an amber notice, because nothing backs the alternative. The obvious naive choice, the three largest clusters by membership, fails the recovery criterion by several orders of magnitude.

Remark 14 (Cluster members are chained, not mutually parallel). Collinearity is not transitive. A cluster is a connected *component* of the $|\cos| > 0.95$ graph, so two of its members are linked through a chain of collinear neighbours and may themselves sit well below the threshold against each other. It is therefore wrong to read a cluster as a bundle of near-identical shocks, and wrong to describe its composite column as a representative member: the composite is a genuine first principal component of a set that is merely connected, and its economic interpretation rests on the equations that happen to be linked, not on a shared mechanism. The drawer states this in the chip tooltip, in the cluster-membership footnote, and in the caveat under the chart. Read a cluster as "whatever in the model moves the watched observables in roughly this direction", and open the membership list before naming it in a briefing.

10.8 Operating the drawer

The **Drivers** button sits in the scenario-action row above the chart grid, and appears only when there is something to decompose — some displayed, isolated or coupled scenario deviation, or an active outside forecast, above a numerical tolerance. It opens a fixed panel on the right of the window, closed with the × or with ESC, and nothing is solved while it is closed. The controls, in the order they appear:

- The **lens**: the model in force, any shocks dropped from it because they do not move the common observables (`em` for Smets–Wouters, `e_term` for Christiano–Motto–Rostagno), and a **Change model** link opening a picker grouped into own-country models, the two US reference models (listed there only when they are not the bloc's own), and the rest. The choice is remembered per bloc. Until one is made, every bloc opens on the Smets–Wouters lens with **Use all** in force: seven observables and seven shocks on the United States; elsewhere the common three observables and the first three shocks (monetary, government spending and technology), with **Classes** greyed.

- **Classes** | **Groups**: aggregate into classes, or show one series per named shock or cluster. **Classes** is disabled unless the configuration is licensed (Section 10.6).

- The **chart view**: **Forecast** draws contribution bars anchored on the baseline path with the scenario line over them — the familiar decomposition picture — while **Deviation from baseline** drops the baseline and draws the contributions around zero. The view follows the main panel's **Units** setting (**Deviation from baseline** while **Units** is **% vs base**) until the user picks one, and then stays there until **Reset** (Remark 15).

- **Customize** opens the rest: **Use all**, which watches every mappable observable and then names shocks up to the budget skipping collinear duplicates; the input source, either the displayed scenario or an imported outside-forecast combination, and for the scenario a **Domestic** | **Global** toggle when both an isolated and a coupled path exist; the observable chips, three padlocked and the addable ones annotated with the budget they buy; the measurement-error slider (Remark 12); the cluster chips on the FRB/US lenses; and an **Experimental: raw shocks** fold, which on the FRB/US lenses replaces cluster columns with individual shocks and on every other lens is where the named shock chips themselves live.

- Below the chart, a **shares** strip reports each series' share of the total, and an expander lists cluster membership where it applies. The share is the Euclidean norm of that series' contribution path for the observable on display, measured in the model's observable units, divided by the sum of all such norms including the residual's. It is a norm share, not a variance decomposition: it always sums to one hundred and can never show a negative or offsetting contribution as negative.

The chart shows one observable at a time, chosen from a dropdown, but the solve always uses every watched observable jointly — switching the displayed observable does not re-solve anything. The horizon is twelve quarters whatever the display horizon of the main grid. Besides the residual-able and collinear-pair notices, amber notices flag named units whose response columns are rank-deficient (their split is not identified), a **Path** choice that decomposes a path other than the one displayed, and a cluster naming other than the validated default. If a model's response file fails to load, the drawer says so and offers **Retry**. **Reset** returns the drawer to its default — the Smets–Wouters

lens with **Use all**, **Classes**, the scenario as input and the chart view following **Units** — and makes Smets–Wouters the remembered model for that bloc.

Example 5 (The zero-click reading on a US scenario)**.** We decompose a scenario we have already built on the United States, without touching a single control, and read the licensed class split. *Build any US scenario — the credit-crunch recipe of Example 2 will do — and press* ***Drivers****. The drawer opens on the Smets–Wouters lens with all seven observables watched and all seven usable shocks named, which is the one configuration the class bench licenses. The view is* ***Classes****: three stacked series in blue, amber and teal for Monetary, Demand and Supply, a grey residual, the dotted baseline and the scenario line on top, with output on the vertical axis in percent quarter-on-quarter annualized. The caveat under the chart, and the banner at the foot of* ***Customize****, report rank* 84 *of* 84*: the system is exact-fit, so the grey residual is flat at zero and is telling us nothing (Section 10.2). No collinearity or rank warning fires, because the Smets–Wouters lens has no flagged pair. Change the observable dropdown from output to inflation and the same seven-shock solve is re-plotted against a different row block of* (38)*; nothing is recomputed. The strongest sentence this chart supports has the form "through the Smets–Wouters lens, this path is mostly such-and-such a class with such-and-such an offset in the second year". Which classes those are is read off the chart; the qualifier in front of them is not optional.*

Example 6 (Two ways to leave the licence)**.** We deliberately step outside the licensed configuration, twice, to see what the drawer does about it. *Start from the state of Example 5. (1) Open* ***Customize****, then the* ***Experimental: raw shocks*** *fold, which is where a lens without clusters keeps its shock chips, and un-name the wage markup shock. The chart redraws with six shocks — and the* ***Classes*** *toggle greys, with a tooltip saying that class totals are not reliably separable at this model, observable and shock-set combination. The view falls back to* ***Groups****: the three class series become six shock series, labelled with textbook names and the model's mnemonics ("Price markup (epinf)" and so on). Six shocks cannot span seven observables, so an amber notice also reports the residual-able regime (rank* 72 *of* 84*), and the measurement-error slider now has an effect. Nothing about the model or the observables changed; what changed is that the wage markup's energy is now redistributed over the classes that remain, and no bench ever measured that split (Remark 13). (2) Re-name the shock, then use* ***Change model*** *to select the FRB/US census lens. The observables collapse to the common three, because the native mapping exists only for the two US models that expose one; the naming budget falls to three; and the chips are now* 23 *collinearity clusters rather than* 54 *shocks. The shipped default naming is in force, covering* 19 *of the model's* 54 *shocks, and the caveat under the chart changes: it no longer cites* `bench_simple_decomp.m`*, which never ran an FRB/US lens, and instead states that what backs this lens is a build-time check that the default naming recovers a known shock sequence to* 10^{-6}*, and that other namings are unvalidated. Name a different cluster and an amber notice appears saying exactly that. The same scenario, read through two lenses, gives two different answers with two different warranties; that is the feature working as intended, not a defect.*

Remark 15 (The delta view and the panel need not share units). The drawer's chart view follows the main grid on opening: with **Units** set to **% vs base** it opens in **Deviation from baseline**. The two are not the same object. The drawer's deviation bars are always the *rate* deviation of the panel series, while the main grid's **% vs base** carries its own **Level | Rate (IRF)** sub-toggle and defaults to the cumulative level effect (Section 9.3). Auto-following can therefore put the contribution chart in different units from the panel that prompted the question. The axis title always states the unit in force; read it before comparing a number in the drawer with a number on the grid.

10.9 What the decomposition does not establish

It is worth saying plainly where this stops, in the same voice as what it does. The feature is experimental. Its claim is model-conditional: it reports which mix of a chosen model's own shocks reproduces a path, never what caused the path, and it is not model-free shock labelling. Its validation is same-model recovery, so every number quoted above measures the estimator on paths generated by the model itself, not the model's fidelity to any economy. Different lenses give different answers, and they are not equally validated — the simple-decomposition bench ran eight of the models in the menu and none of the FRB/US lenses (the class bench ran the 54-shock FRB/US model only at three observables, where it fails), and one model in the menu, the two-shock 2008 FRB/US lens (`US_FRB08`), carries neither a bench row nor a build-time recovery scan. Class aggregation is licensed at exactly one point in the configuration space and greyed elsewhere; the families of the Christiano–Motto–Rostagno lens are an organising device, not a licensed split; and the FRB/US clusters are an empirical grouping of response directions, not an economic taxonomy. Within a flagged collinear pair the split is not identified. In the exact-fit regime the residual is zero by construction. The shares strip is a norm share, not a variance decomposition. The seven-observable mode, and with it the only licensed class view, exists on the United States bloc alone.

None of this makes the reading useless; it makes it a reading. Used the way the counterfactual layer is used — flip the lens, keep the scenario, and see what survives — it answers a question the reduced form cannot answer at all, which is what a scenario would have to be made of if a particular theory of the economy were true.

11 Data, vintages, and updates

11.1 Sources

No single provider supplies current quarterly data for the full roster, so the panels are assembled economy by economy from published sources. Each economy has a machine-readable source registry (`pipeline/sources/registry/<CC>.json`) that names, for every variable, the provider, the series key and the transformation. The fetchers apply strict quarterly rules — a monthly series is averaged only over complete quarters, a fiscal ratio only over full four-quarter windows — and the result is recorded, series by series, in the coverage manifest (`coverage_manifest.json` under `app/public/data/`), from which the roster table of Appendix B is generated.

Table 2: Repository layout. Data are fetched by the source registries in `pipeline/sources/`; estimation in `pipeline/` produces a per-bloc JSON export; the export, with its `manifest.json`, becomes the data pack in `app/public/data/`; the front end and its compute kernels read that pack and never re-estimate.

Top-level	Key contents
`pipeline/` (MATLAB + Python)	`sources/` (one source registry per economy, the fetchers, and the derived columns — no fills); `export_bvar_gvar24_loop.m` (GVAR–EM estimator); `export_bvar_euro_redesign.m` (euro members); `port_gvar_exports.py`; `fix_{fiscal,external}_meta_indices.mjs`.
`app/` (Next.js + compute)	`src/lib/compute/` (Durbin–Koopman, GVAR solve, fiscal accounting); `src/lib/data/dataManifest.ts` (pack loading and verification); `scripts/` (`precompute_gvar_coupling.mjs`, `nowcast_complete.mjs`, `pack-build.mjs`, `pack-verify.mjs`, `auto_update.mjs`); `public/data/` (the data pack: `manifest.json` and `bvar/<CC>/`).
`docs/` (manual + runbooks)	`GMAPS_User_Manual.tex`; `UPDATE_MANUAL.md` (updating data, model and app); `REESTIMATING_MODELS.md`, `ADDING_A_COUNTRY.md`, `ARCHITECTURE_FOR_AI.md`; `reest-2026/` (decisions, review and run scripts of the October 2026 re-estimation).

- **Eurostat and the ECB** supply the EU and EFTA economies: chain-linked volumes, HICP, unemployment, interest rates, house prices; euro members carry the ECB policy and lending rates.
- **The BIS** supplies credit, residential property prices and, for thirty economies, the broad nominal effective exchange rate (the IMF index is used for Australia, Chile, Costa Rica and New Zealand, Eurostat's for the euro members, and a real effective rate for China). Every exchange rate is an index of the domestic currency: a rise is an appreciation.
- **FRED** supplies the US bloc with exact series identifiers, the four global commons, and a number of series for other economies.
- **The OECD, the IMF and national sources** (statistics offices, central banks and finance ministries) supply the remaining national accounts, prices, rates and fiscal series.
- **Derived series** — the stars, the effective interest rate, the stock-flow adjustment, the rest-of-euro-area aggregates — are computed from these, not pulled (Sections 5.1 and 6; `pipeline/sources/derived.py`).

No fills. A quarter that a source has not published stays missing. In estimation the sampler draws it as missing data (Sections 3.4 and 5.2); in the application it is drawn in amber as a model estimate. Earlier vintages filled short tails with temporally disaggregated IMF WEO projections and back-filled late-starting series; the October 2026 rebuild removed every such fill. A few series remain proxies or built from annual data: China's GDP before 2011 (annual growth disaggregated

to quarters) and the German non-financial corporate yield used as the euro-area corporate rate, both labelled in **Data & coverage**; China's fiscal block (IMF annual revenue and expenditure, each year's value spread evenly over its four quarters, through 2024), which the dialog does not yet label; and a small number of credit and investment concepts.

The **Data & coverage** dialog (top bar) lists, for the economy on screen, every variable's source, coverage span, last observed quarter, and caveats — the on-screen rendering of the coverage manifest. A last observed quarter shown in amber means the series stops before the panel quarter and its remaining quarters are nowcast. In the charts that nowcast tail is drawn as an amber dashed segment with open circles, and a release skipped inside the history as an open amber marker. Cells that estimation drew as missing (the COVID quarters and other gaps, listed in `estimation_mask.json`) are not marked: the charts show the published value there, or a gap where nothing was published.

11.2 Data packs: the data and the model are separate from the tool

Everything the application shows — histories, forecasts, estimated models, posterior draws, trade weights, the coupling artifact — is read from a *data pack*: the folder `app/public/data/` plus one `manifest.json`. The manifest records the pack's identity (`packId`), its format version and the oldest tool version it needs (`minToolVersion`); the panel quarter; for every economy its last historical quarter, estimation end and seasonal phase; and for every file its path, size and SHA-256 fingerprint. The GVAR weights and the coupling artifact are listed with the fingerprint of the weights the coupling was built from, so a coupling paired with the wrong weights is caught.

The tool reads the manifest before anything else. It refuses a pack it cannot use — an incompatible format, a pack that needs a newer tool — with one full-screen message, and it checks each file's size and fingerprint when it reads it, refusing any that differ. The dates come from the data: each history file records the quarter of its first row (`firstQuarter`), so the last quarter is always first quarter plus rows minus one, and the tool sets its calendar from the manifest at start-up. No date is compiled into the program.

This separation is what lets the data and the model change without a new version of the tool. On the web the pack is published with the site. The desktop application carries a built-in pack and can read a newer one from its data folder (**Settings → Data pack → Open data folder**): copy the pack there — or extract an update pack's zip — so that `manifest.json` sits directly in the folder, then press **Reload**. An *update pack* carries only the files that changed and is accepted only if every file it reuses matches the built-in pack; deleting or renaming `manifest.json` in the data folder and pressing **Reload** returns the tool to the built-in data. **Settings → Data pack** and the **About** dialog show the pack in use: its `packId` and the quarter its data run through. A pack is never changed in place: any change gets a new `packId`. The live pack at this writing is `2026Q2-reest-20261108-consistent` (data through 2026Q2; app 4.2.0 on the websites). The commands and checks that produce and verify packs are in `docs/UPDATE_MANUAL.md`.

A problem with the pack as a whole stops the tool with a full-screen message. A single file that fails its check later is named in a banner, and what depends on it is not shown: the economy's charts

give way to an error panel, and the global model refuses to load from a partial set of economies rather than solve without one.

Three limits remain. The list of economies is still compiled into the tool, so a pack can update the economies it knows but cannot add one. The tool contains a build-time copy of the trade weights and of six economies' specifications (`gvarSolve.ts`), used as start-up defaults before the pack has loaded. And the manifest itself is not signed: every file is checked against it, but nothing checks the manifest.

11.3 Vintage policy: estimate once a year, shift quarterly

The data packs come from two kinds of update.

Model update (once a year, after the previous calendar year is published for every economy). Every economy is rebuilt from its sources to the latest published quarter, and the whole system is re-estimated on each economy's full, unbalanced history (Section 5.2). There is no common estimation endpoint: a series that ends early is simply missing at the end, and the sampler draws it. The current pack is the October 2026 re-estimation, with data through 2026Q2 wherever published and $\rho(M) = 0.972$.

Data shift (within the year). When a new quarter is published, the panel advances by one quarter with the estimated model unchanged: the US quarter is taken observed from FRED, other published values are pinned at their releases, and the rest is nowcast. The forecast re-anchors from the unchanged companion, and the result is a new pack. Because the coupled system links economies row by row, the panel origin is *uniform*: every coupled bloc's history ends at the same quarter (2026Q2 at this writing). Blocs whose sources lag reach that quarter through the amber, flagged nowcast described next — never through fabricated observations.

The not-yet-published quarters of the remaining series are *nowcast* by the coupled system itself: the panel is truncated to the last quarter at which every cell is observed, every observed cell in the ragged window is pinned at its published value, and one coupled solve (Section 5.3) delivers each missing cell's conditional expectation given everything already observed anywhere in the panel — the star channel carries the leaders' releases into the laggards' estimates. Observed cells are never altered; every imputed cell carries a provenance flag and draws amber (`nowcastCompletion.ts`). These amber cells lie inside the history, at or before the panel quarter. The label **NC** in the conditioning list is something else: the first forecast quarter, the one after the panel quarter (2026Q3 at this writing). It is a model forecast like **t+1**, not an amber cell.

The **Data-release impact monitor** (**Settings** → **Data-release Monitor**) compares the live panel with an earlier vintage. Choose the vintage to compare against, an economy and a variable: a graph shows the base and the live path quarter by quarter, with the revision at the origin quarter, and a table ranks the economies by the revision of GDP growth, inflation and the policy rate. A comparison against a vintage made with an earlier model is labelled "model + data", with a banner, since it mixes both changes; a second banner warns when the base origin is older than the live one, so that the single-quarter columns compare different quarters. At this writing all three bases (the

2025Q4 re-estimation, 2025Q4 as shipped, and the 2026Q1 pre-release) predate the October 2026 re-estimation, so every comparison is "model + data". Source restatements of history are accepted as current best knowledge and logged with the data (`refresh_revisions.json`); the page does not list them.

12 Evaluation

The system's forecasting performance was evaluated in a separate repository (`em-bayesian-gvar-eval`) under a protocol designed before the runs; this section summarizes the design and the results. The results come from that repository's July 2026 run, on the model as it was then — before the September 2026 move of real GDP and prices to growth rates and before the October 2026 re-estimation without fills — with 200 draws per origin. They have not been re-run on the current model: read them as evidence on the method, not as the accuracy of the forecasts now on screen. Two questions are asked: does the estimation apparatus — Bayesian shrinkage, COVID-as-missing, the EM star fixed point, the endogenous US — earn its keep out of sample, and what does the endogenous US buy that a frozen anchor cannot deliver.

12.1 Design

Models. Nine forecast densities are scored: the proposed EM-Bayesian GVAR in its two variants (`emgvar_endo`, the full endogenous-US loop, and `emgvar_frozen`, the frozen-anchor ablation), six benchmarks estimated on the same truncated panels — a driftless random walk in the scored space, an AR(p) per series (p by AIC, iterated multi-step; Marcellino et al., 2006), a single-country BVAR with the stars dropped (the value-of-coupling ablation), the textbook fixed-weight OLS GVAR of Pesaran et al. (2004) and Dees et al. (2007) with no data augmentation (the headline classical competitor), a single-pass Bayesian GVAR (the value-of-EM ablation), and a Bayesian dynamic factor model — plus a sample-mean climatology sibling of the random walk.

Scheme. Pseudo-real-time recursive expanding window on the final vintage: quarterly origins 2010Q1–2024Q4 (60 raw origins; the four 2020 origins are dropped from the headline tables, leaving 56), horizons $h \in \{1, 2, 4, 8\}$, three targets per bloc (real-GDP growth and inflation in the 4Δ scored space, the short rate in levels), six headline blocs (US, EA, JP, UK, CA, CN). Look-ahead is excluded by construction: every panel is truncated to the origin before any transformation or star is built; trade weights are rolling, as-of-origin three-year averages (the shipped 2021–2023 weights are forbidden in the sweep); and the COVID mask is re-applied in-sample at every origin. The Bayesian models (both proposed variants, the one-pass Bayesian GVAR and the single-country BVAR) need a longer warm-up and are scored at 39 origins at $h = 1$ (35 at $h = 8$), against 56 (52) for the other models. Predictive densities use 200 retained draws per origin with a stationarity backstop that discards explosive posterior draws.

Scoring. Point accuracy by relative RMSE against the random walk. Density accuracy by the continuous ranked probability score (CRPS), the primary metric, computed from the draw ensemble by the standard energy-form estimator

$$\widehat{\mathrm{CRPS}} = \frac{1}{m}\sum_{k=1}^{m}|X_k - y| - \frac{1}{2m^2}\sum_{k=1}^{m}\sum_{l=1}^{m}|X_k - X_l| \tag{41}$$

(Gneiting and Raftery, 2007), which degrades gracefully at modest draw counts; calibration by PIT histograms and interval coverage (Berkowitz, 2001). Because most model pairs are *nested*, predictive-ability inference uses the Giacomini and White (2006) conditional test as the headline (valid under nesting and correct for the expanding scheme), Clark and West (2007) for nested point comparisons, and Diebold and Mariano (1995) with the Harvey et al. (1997) correction only for non-nested pairs; multiplicity is controlled by Romano–Wolf step-down (Romano and Wolf, 2005), and the per-cell *model confidence set* (MCS) of Hansen et al. (2011) reports which models are statistically indistinguishable from the best.

12.2 Results

The numbers below are the nine-model headline run of July 2026 (`results/tables_v3.{txt,json}` in the evaluation repository, kept byte-identical in this repository under `docs/eval_results/`; 30,744 scored cells); it is the canonical run for this section, and any later partial sweep in that repository is a diagnostic, not a restatement.

On *point* accuracy the univariate AR(p), the factor model, and the sample-mean climatology lead — a familiar macro-forecasting result — and the EM-GVAR does not win the horse race outright. Its best point showing is on inflation, where at $h = 1$ the whole Bayesian family lands on top of itself — relative RMSE 0.80 for both proposed variants, 0.80 for the one-pass Bayesian GVAR, 0.80 for the single-country BVAR — against 0.64 for the AR and 0.75 for the factor model, which are ahead of all of them. The evidence that matters is in the density comparison and the model confidence set: the EM-Bayesian GVAR sits *inside* the 90% MCS for **real GDP growth at every horizon**, for **inflation at every horizon**, and for the **short rate at** $h \leq 2$ — statistically indistinguishable from the best model in ten of the twelve (target, horizon) cells. It leaves the set only for the short rate at $h = 4$ and $h = 8$, the two cells where every persistent multivariate model is beaten by the random walk on a near-unit-root series.

Two structural findings follow, and the first runs against the convenient reading. The endogenous-US loop does *not* pay for itself in forecast accuracy: the frozen anchor is weakly better throughout, carrying the lower relative RMSE in eleven of the twelve cells and surviving in the MCS at the short rate at $h = 4$ where the endogenous variant is eliminated, and being eliminated later than it at $h = 8$. Everywhere else the two are in the set together. The endogenous US is therefore justified by what it makes computable — the rest-of-world-to-US feedback of Section 12.3, which is structurally absent under a frozen anchor — and not by out-of-sample accuracy. Second, and central to the design's claim, the apparatus *decisively* beats its classical ancestor: the textbook OLS GVAR is

eliminated from five of the twelve MCS cells — growth at $h \in \{1, 4, 8\}$ and inflation at $h \geq 4$ — against two for the endogenous variant and one for the frozen, and it carries badly miscalibrated densities, with one-quarter-ahead 90%-interval coverage of 0.39 on growth and 0.36 on inflation against the 0.90 target.

Two caveats are part of the record. The draw-based densities of the proposed variants are themselves over-confident at short horizons (realized 90% coverage 0.74 at $h = 1$ pooled across targets, rising to 0.82 by $h = 8$): the forecast draws propagate parameter and shock uncertainty but not the smoother uncertainty of the COVID-imputed states. The CRPS and MCS compare score *differentials* and are robust to this; the under-coverage is a calibration caveat, not a ranking artifact. And the log predictive score is numerically unreliable on under-covered clouds (a parametric tail fit explodes on a handful of origins), so ranking rests on CRPS and the MCS only.

12.3 The endogenous-US contrast

What the solver computes is a *conditional-mean scenario response* — pin a path, read the coupled deltas via (19) — not a generalized impulse response or a variance decomposition; the evaluation keeps that distinction explicit. The realized contrast propagates a US +100bp policy-rate path through both topologies: under the frozen anchor the world-to-US direction is structurally absent — not small, absent — while the endogenous configuration returns a non-trivial RoW→US feedback, with sizeable, sign-heterogeneous gaps between the two topologies across partner blocs. These are conditional-projection deltas (the rate is pinned along the whole path, bundling the direct conditioning with the coupled feedback), so their *existence and direction* is the documented result; calibrated spillover *multipliers* await the structural extension (generalized IRFs and connectedness indices on the coupled system, Pesaran and Shin, 1998; Diebold and Yilmaz, 2012), which is constructible from the shipped exports but not part of the tool.

13 Implementation and reproducibility

13.1 Client-side computation

All matrix algebra, filtering, and simulation run in the browser in TypeScript on `Float64Array` storage (IEEE 754 double precision): LU decomposition with partial pivoting for F_t^{-1} (singularity threshold 10^{-15}), Cholesky factorization for shock draws (positive-definiteness threshold 10^{-14}, with a diagonal fallback), and row-major layouts throughout. Table 3 collects the numerical constants a replicator needs.

Table 3: Numerical constants of the deployed engine.

Constant	Value	Role
σ_ε^2	10^{-12}	hard-pin variance (8); soft-pin threshold
P_0	$10^{-7}I$	initial state covariance
LU pivot	10^{-15}	singularity threshold in F_t^{-1}
Cholesky	10^{-14}	positive-definiteness threshold
D	500	shipped posterior draws per bloc
N_b	50 / 200	band draws, **Quick** / **Full**
band quantiles	q_{16}, q_{84}	scenario bands (shipped baseline adds q_{05}, q_{95})
trust default / clamp	0.7 / $[0.001, 0.999]$	soft-pin dial (15); ≥ 0.999 is hard
GS tolerance / sweeps	10^{-4} / 12	iterative solve (per-draw bands)
GMRES tolerance / restart	10^{-12} rel. / 100	network solve (live; hard and soft pins)
ρ estimate	60 Arnoldi steps	divergence guard, once per pin pattern
$\rho(M)$ / dim	0.972 / 2180	shipped coupling artifact
artifact precision	full / 6 sig. fig.	GVAR blocs / members, coupling
sens guard	0.3	long-run slider disable (31)
B&M ridge	$10^{-12}\times$ mean diagonal	optimal-policy solve (36)
EM TOL / MAXIT	0.03 / 30	estimation star fixed point (converged at 6)

Interactivity is engineered so the heavy work never blocks the mouse: no coupled solve fires during a drag (a signal bus outside the rendering state carries the drag flag) and none on release — a pin edit shows the domestic solve, and the global solve runs only on request; the pinned dot is sourced from the click itself through an exact display-to-storage inverse, so it snaps to the cursor instantly while the line and bands reconcile; and each chart panel is memoized so a conditioning edit repaints one panel of the roughly thirty-five, not all of them. The full contract is asserted in the test suite.

13.2 Repository and replication

One repository holds the three concerns: `pipeline/` (MATLAB and Python: data assembly and estimation), `app/` (the front end and its compute kernels), `docs/` (this manual and the operational runbooks). The estimator writes a per-bloc JSON export; the export, with its `manifest.json`, becomes the data pack in `app/public/data/` (Section 11.2), and the deployed application reads only that pack. Dependencies: a recent MATLAB (the Parallel Computing Toolbox is optional and only runs the blocs in parallel; the bundled `BVAR/MFBVAR` code needs no further packages; MATLAB also inverts the coupling in stage 4), Python 3 for the data pulls (a FRED API key via `FRED_API_KEY` is recommended), and Node 22 for the application.

The path from raw data to a deployable build has six stages. The operator's runbook is `docs/UPDATE_MANUAL.md` (section 7), the estimation detail is in `REESTIMATING_MODELS.md`, and the exact scripts of the October 2026 re-estimation are in `docs/reest-2026/runs/`. One script, `app/scripts/model_update_chain.sh`, runs stages 3 to 5 up to the full verification of the pack

(not the tests):

1. **Build the data**: assemble every economy's panel, genuine data only, from its source registry in `pipeline/sources/`.
2. **Estimate**: run `export_bvar_gvar24_loop.m` — the GVAR-EM loop of Algorithm 2 — and the euro-member script; each exports the eight-file contract and the `estimation_mask` per bloc.
3. **Port**: copy the GVAR exports into the application at full precision, port the euro members with `port_gvar_exports.py` (six significant figures), and re-resolve the fiscal and external column indices by variable id.
4. **Date, stamp an interim pack, then regenerate the coupling**: date the histories (`stamp_calendar.mjs`, then the coverage manifest), then stamp an interim pack (`pack-build.mjs`), because the steps that follow read the panel dating from its manifest. Then `precompute_gvar_coupling.mjs` assembles M, inverts $(I - M)$ offline in MATLAB, and writes the gzipped artifact of Section 5.3; $\rho(M)$ must be below one.
5. **Complete and verify the pack**: nowcast the ragged edge, roll the debt starting points forward, regenerate baselines and bands, then stamp the pack under a new `packId` (`pack-build.mjs`) and verify every file (`pack-verify.mjs --full`); then the type check and the test suite (`npm test`; the multi-second solver canaries run under `npm run test:heavy`).
6. **Deploy**: pushing a change under `app/` to `main` of both remotes triggers the builds of both websites (GitHub Pages through a GitHub Action, and Netlify); a push that changes nothing there builds nothing. Each build re-verifies the whole pack before it publishes. The desktop application is built separately and can take a new pack without being rebuilt (Section 11.2).

The model signature (Section 5.3) changes when a coupled bloc is re-estimated, so a coupling artifact left from an earlier model is not used and the solver builds the coupling itself. The signature reads only a few cells per bloc, so the real safeguard is the procedure above: regenerate the coupling after every export. The regenerated per-economy tables of Appendix B enforce the same discipline on this document.

14 Final thoughts

The design bets on a separation of concerns. The baseline is computed once, offline, and shipped — the browser never estimates. The scenario layer imposes user paths without asking where the deviation comes from, in the tradition of Waggoner and Zha (1999); the counterfactual layer adds structural identification exactly where it is needed — the policy transmission — and lets the estimated covariances carry the rest, because no single DSGE model matches the empirical co-movement of thirty variables. The accounting blocks derive what must never be estimated freely. And every

judgment device — pins, trust dials, anchors, counterfactuals — is built to be an exact no-op at its neutral setting. The one device switched on at start-up is the shipped set of 34 long-run anchors on 27 blocs; clear them and the screen is the model, to the last digit.

14.1 Limitations

The dynamics are linear and time-invariant: coefficients are fixed at their posterior means for the deterministic engine, with no time-varying parameters, stochastic volatility, or regime switching — restrictive in periods of structural change. Innovations are Gaussian; crisis-period fat tails are not accommodated. Counterfactuals are point-identified under one model–rule pair at a time (the user can flip pairs, but the tool does not average over them), and the Lucas critique applies to the re-conditioned variables whose responses are governed by the estimated, policy-invariant-by-assumption covariances. The coefficients are frozen at the estimation vintage within the year: data revisions are accepted as knowledge and the origin advances, but the parameters see new data only at the annual re-estimation. Scenario responses — including the cross-country ones — are conditional projections, not causal impulse responses (Section 12.3).

14.2 Known issues

Specific, identified defects of the current deployment, documented in the interest of honesty for a live research tool:

1. **Divergence is detected, not prevented.** The coupled solve does not refuse to run if a future roster pushed $\rho(M) \geq 1$: the convergence flag trips, a console warning fires, and the result is not cached — but a (diverged) path would still be displayed. A hard spectral-radius backstop is the durable fix.

2. **Fiscal and external metadata use column positions.** The fiscal and external blocks read their inputs by *positional* indices stored in `fiscal_meta.json` and `external_meta.json`. A re-export that changes a bloc's variable order moves them: in October 2026, adding a deflator column left the external block's exchange-rate index on the wrong series in 35 blocs (no numerical effect, as that channel is switched off). Both are now re-resolved by variable id (the fiscal indices from the fixed fiscal ids, the external ones from ids stored in `external_meta.json`), by the porter and by repair scripts for exports copied raw, and a data test in the suite checks that every index points at the right family of series.

3. **Mid-drag previews use the filtered variance.** The impact-view shading requests the smoothed variance (13) except while a drag is in flight, when the cheaper filtered variance is shown; the run-up to a future pin is momentarily overstated until release.

4. **Some blocs carry an explosive own-companion baseline.** The stored GDP and price levels carry a unit root by construction (Section 3.1), so roots at one are expected. In the October 2026 estimates, 16 of the 52 posterior-mean companions have a largest root above 1.0005,

most of them only marginally (below 1.005); the largest are Costa Rica (1.034), Malaysia (1.021), Sweden (1.018) and Japan (1.017). The *unconditional* baseline the browser recomputes from such a companion drifts away from the shipped `baseline_levels.json` at long horizons (the gap is small inside the twenty-quarter window and grows beyond it). Two things contain it. First, every per-bloc scenario is displayed as a *deviation*: the explosive component is common to the baseline and the scenario path and cancels in scen − base, so the elasticities the user reads are correct. Second, the GDP-weighted world read-out is aggregated the same way — world scenario growth is world baseline + $\sum_c \omega_c \, \Delta(\text{scen}_c - \text{base}_c)$, built from the solver's *delta* rather than its recomputed level — so it shares one baseline with the world baseline line and is immune to the divergence (a null scenario returns exactly the baseline). What remains affected is only the *absolute* internal baseline of those blocs at horizons past the display. The durable fix is a stationarity safeguard on those blocs' own dynamics, such as a per-draw contraction truncation.

5. **A data shift updates only the previous quarter.** When a source revises its history, a within-year shift brings only the quarter before the new one to the latest vintage; earlier quarters keep the older vintage until the next rebuild, so the growth rate across that seam can mix vintages. (After the US annual revision of September 2026, US 2025Q4 growth showed 1.6% instead of 0.2% until the October 2026 rebuild.)

14.3 Directions

Natural extensions, in rough order of value: a hard contraction guard on the coupled solve; integrating the smoother uncertainty of the imputed states into the predictive draws (the documented source of the short-horizon under-coverage, Section 12.2); structural spillover objects — generalized IRFs and connectedness — on the coupled system; scenario priors for identification inside the BVAR (Farkas and Jakab, 2026); and mixed-frequency estimation to bring monthly indicators into the nowcast.

A Notation

Table 4: Notation. Symbols are local to their sections; the two uses of d (the GVAR forcing vector of Section 5, the debt ratio d_t of Section 6) never co-occur.

Symbol	Meaning
Estimation (Section 3)	
y_t, n, p	bloc observables ($n \times 1$), variable count, lag order ($p = 3$ in estimation, $p = 4$ in the shipped level companion)
c, B_ℓ, Σ	intercept, lag matrices, innovation covariance of (1)
n_{sr}	seasonal-dummy rows of $\hat{\beta}$ (3 on the 34 seasonal blocs, else 0)
$\hat{\beta}$	stored $(1 + n_{\text{sr}} + np) \times n$ coefficients; row 0 the intercept

Symbol	Meaning
ℓ_t	stored level of a `log` variable, $100 \ln Y_t$
g_t	displayed q/q annualized growth, $4(\ell_t - \ell_{t-1})$
δ_i	own-lag prior mean: 1 (`RW`) or 0 (`WN`)
λ, μ	GLP tightness and sum-of-coefficients hyperparameters
D	shipped posterior draws (500)
State space and smoother (Section 4)	
$\alpha_t, T, c_\alpha, Z, Q, R$	companion state and system matrices (7)
$a_{t\|t-1}, P_{t\|t-1}$	predicted state mean and covariance
v_t, F_t, K_t	innovation, its variance, the Kalman gain
r_t, L_t, N_t, V_t	smoother recursions (12)–(13)
Y^*, H	pin matrix (NaN = free) and forecast horizon (20)
τ, R^{ext}	trust and soft-pin variance (15)
N_b	band draws (50 quick / 200 full)
Global model (Section 5)	
$w_{cj}, \tilde{w}_{cj}$	trade weight of partner j in bloc c; renormalized
x^*_c	bloc c's star columns (DEM, INF, FIN)
$K_c[v][s]$	affine response kernel (17)
x, M, d	deviation state, coupling operator, forcing of $x = Mx + d$
$\rho(M)$	spectral radius (shipped: 0.972)
Accounting blocks (Sections 6–7)	
d_t, d_0	debt-to-GDP (%) and its seed
i^{eff}_t, g^n_t	effective rate (% p.a.) and quarterly nominal growth (decimal)
$pb_t, \overline{sfa}$	primary balance; structural stock-flow constant
b_t, nfa_0	NFA-to-GDP (%) and its seed
$i^{\text{NFA}}, \phi^{\text{FX}}$	NFA return (% p.a.) and FX-exposed share; both ship at 0
Anchors and counterfactuals (Sections 8–9)	
$S_h, \delta c_j, \text{sens}_j$	drift operator, drift shift, terminal sensitivity (31)
$\psi^\star_j, \psi^0_j$	target and current terminal display forecast
h_r, h_π, h_y	MMB impulse responses; ε_t the inverted shocks
$\lambda_\pi, \lambda_y, \lambda_{\Delta i}$	loss weights of (34) (inflation, output gap, rate smoothing)
Scenario drivers (Section 10)	
H_d	decomposition horizon (12 quarters)
q, K	watched model observables; named shocks or clusters
o, t	observable index and forecast quarter, $t = 0, \ldots, H_d - 1$
k, τ	shock index and arrival quarter, $\tau = 0, \ldots, H_d - 1$
$R_{k,o}(s)$	response of observable o to shock k at horizon s
Θ	stacked response matrix, $qH_d \times KH_d$ (38)
z, ε, u	observable-space deviation path, shock sequence, residual
ν_j, V	column response norms and their diagonal, $\tilde{\Theta} = \Theta V^{-1}$
W	diagonal row weights, $1/\sqrt{1 + me}$ on added observables

$c_{k,o}(t)$	contribution of shock k to observable o at quarter t (40)

B The deployed economies

The tables of this appendix are *generated* from the deployed data files and regenerated whenever the data change, so they cannot drift from the shipped models. The coverage matrix below (Table 5) is built by `build_coverage_table.py` from the coverage manifest; the roster (Table 6) and the per-economy variable tables that follow are built by `generate_manual_tables.py` — the roster from the same manifest and the per-bloc metadata, the variable tables from each bloc's `spec.json`.

The single grid in Table 5 answers, at a glance, which of the nine variable groups each of the 52 deployed blocs carries: a green check marks a group that is present in that bloc's shipped model, a dash marks a genuine absence. The last two columns record the last quarter shown and the source of the bloc's real GDP; an italic tag *k est.* counts the bloc's series that end before the panel quarter, whose remaining quarters are model estimates (amber in the application). Every bloc is built from its source registry with no fills.

Table 5: Country × variable-group coverage matrix for all 52 deployed blocs. Columns: **GDP** real activity; **Price** HICP/CPI; **Unemp** unemployment rate; **Pol** policy rate; **Long** 10-year/long government yield; **Cred** credit-to-GDP; **Lend** lending/short rate; **FX** nominal effective exchange rate; **Fisc** the four-variable fiscal block (`FREV/FPEXP/EFFI/SFA`). ✓ = carried; `--` = absent in that bloc. *Last obs* is the last quarter shown (the panel quarter); *Source* is the provider of the bloc's real GDP (*national*: the statistics office or central bank; *derived*: built from other series). An italic tag *k est.* counts the bloc's series that end before the panel quarter: their remaining quarters are model estimates, shown in amber. Every bloc is built from its source registry with no fills.

Bloc	GDP	Price	Unemp	Pol	Long	Cred	Lend	FX	Fisc	Last obs	Source
AT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
AU	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q2	national *5 est.*
BE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *6 est.*
BR	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	OECD *4 est.*
CA	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	national *5 est.*
CH	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q2	Eurostat *4 est.*
CL	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	OECD *6 est.*
CN	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q2	derived *8 est.*
CO	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	OECD *10 est.*
CR	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q2	OECD *8 est.*
CZ	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *9 est.*
DE	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
DK	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *9 est.*
EA	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *8 est.*
EE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
ES	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *6 est.*
FI	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *6 est.*
FR	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
GR	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
HK	✓	✓	–	–	✓	✓	✓	✓	–	2026Q2	national *5 est.*
HU	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *9 est.*
ID	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q2	OECD *9 est.*
IE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
IN	✓	✓	–	✓	✓	✓	✓	✓	–	2026Q2	OECD *5 est.*
IS	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q2	OECD *3 est.*
IT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*

Bloc	GDP	Price	Unemp	Pol	Long	Cred	Lend	FX	Fisc	Last obs	Source
JP	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	national *9 est.*
KR	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	national *6 est.*
LT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
LU	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *14 est.*
LV	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
MX	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	OECD *9 est.*
MY	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q2	national *4 est.*
NL	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *6 est.*
NO	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *8 est.*
NZ	✓	✓	✓	–	✓	✓	✓	✓	✓	2026Q2	OECD *11 est.*
PE	✓	✓	–	–	✓	–	✓	✓	–	2026Q2	national *4 est.*
PH	✓	✓	–	–	✓	–	✓	✓	–	2026Q2	national *4 est.*
PL	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *9 est.*
PT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *6 est.*
RO	✓	✓	–	–	✓	–	✓	✓	–	2026Q2	Eurostat *4 est.*
SA	✓	✓	✓	✓	✓	✓	–	✓	–	2026Q2	OECD *5 est.*
SE	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	Eurostat *8 est.*
SG	✓	✓	–	–	✓	✓	–	✓	–	2026Q2	national *4 est.*
SI	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
SK	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q2	Eurostat *5 est.*
TH	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	national *9 est.*
TR	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	OECD *5 est.*
TW	✓	✓	✓	–	✓	✓	✓	✓	–	2026Q2	national *3 est.*
UK	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	national *4 est.*
US	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	FRED *3 est.*
ZA	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q2	OECD *9 est.*

B.1 The deployed roster at a glance

Generated from the shipped `spec.json`, `fiscal_meta.json`, `external_meta.json`, the trade-weights file, and the coverage manifest (`coverage_manifest.json`); the table therefore reflects the deployed data exactly. n counts the variables of the estimated VAR (domestic, global, fiscal, and star columns included). *Coupled* marks membership of the cross-country fixed point (Section 5); the remaining blocs are standalone models. *Fiscal* and *External* mark the accounting blocks of Section 6. *Last obs.* is the bloc's last historical quarter.

Table 6: Deployed economies: dimension, coupling, accounting blocks, last observation.

Code	Economy	n	Coupled	Fiscal	External	Last obs.
US	United States	40	✓	✓	✓	2026Q2
EA	Euro area (aggregate)	26	✓	✓	✓	2026Q2
JP	Japan	27	✓	✓	—	2026Q2
UK	United Kingdom	28	✓	✓	✓	2026Q2
CA	Canada	28	✓	✓	✓	2026Q2
CN	China	22	✓	✓	✓	2026Q2
AT	Austria	24	—	✓	✓	2026Q2
AU	Australia	26	✓	✓	✓	2026Q2
BE	Belgium	24	—	✓	✓	2026Q2
BR	Brazil	25	✓	✓	✓	2026Q2
CH	Switzerland	20	✓	—	✓	2026Q2
CL	Chile	26	✓	✓	✓	2026Q2
CO	Colombia	27	✓	✓	✓	2026Q2
CR	Costa Rica	21	✓	✓	—	2026Q2
CZ	Czechia	26	✓	✓	✓	2026Q2
DE	Germany	23	—	✓	✓	2026Q2
DK	Denmark	27	✓	✓	✓	2026Q2
EE	Estonia	24	—	✓	✓	2026Q2
ES	Spain	24	—	✓	✓	2026Q2
FI	Finland	24	—	✓	✓	2026Q2
FR	France	24	—	✓	✓	2026Q2
GR	Greece	24	—	✓	✓	2026Q2
HK	HK	17	✓	—	✓	2026Q2
HU	Hungary	26	✓	✓	✓	2026Q2
ID	Indonesia	25	✓	✓	✓	2026Q2
IE	Ireland	24	—	✓	✓	2026Q2
IN	India	20	✓	—	✓	2026Q2
IS	Iceland	21	✓	—	✓	2026Q2
IT	Italy	24	—	✓	✓	2026Q2
KR	Korea	27	✓	✓	—	2026Q2
LT	Lithuania	24	—	✓	✓	2026Q2
LU	Luxembourg	24	—	✓	✓	2026Q2

Code	Economy	n	Coupled	Fiscal	External	Last obs.
LV	Latvia	24	—	✓	✓	2026Q2
MX	Mexico	27	✓	✓	✓	2026Q2
MY	Malaysia	21	✓	—	✓	2026Q2
NL	Netherlands	24	—	✓	✓	2026Q2
NO	Norway	25	✓	✓	✓	2026Q2
NZ	New Zealand	25	✓	✓	✓	2026Q2
PE	PE	20	✓	—	✓	2026Q2
PH	PH	17	✓	—	✓	2026Q2
PL	Poland	26	✓	✓	✓	2026Q2
PT	Portugal	24	—	✓	✓	2026Q2
RO	RO	18	✓	—	✓	2026Q2
SA	Saudi Arabia	18	✓	—	✓	2026Q2
SE	Sweden	27	✓	✓	✓	2026Q2
SG	SG	15	✓	—	✓	2026Q2
SI	Slovenia	24	—	✓	✓	2026Q2
SK	Slovakia	24	—	✓	✓	2026Q2
TH	Thailand	27	✓	✓	✓	2026Q2
TR	Türkiye	25	✓	✓	—	2026Q2
TW	Taiwan Province of China	21	✓	—	—	2026Q2
ZA	South Africa	26	✓	✓	✓	2026Q2

B.2 Variables, transformations, and priors by economy

One block per economy, generated from the shipped `spec.json`. *Trans.*: `log` = the variable enters the VAR as $100\ln(\cdot)$ (growth displayed); `lin` = the variable enters in levels. *Prior*: RW = random-walk own-lag prior mean ($\delta_i = 1$), WN = stationary white-noise own-lag prior mean ($\delta_i = 0$; Section 3.2). *Fin.*: the `isFinancial` tag — financial variables stay *observed* through the COVID window (Section 3.4). *Block*: dom. = domestic, glob. = US-determined global common, fisc. = fiscal driver, row = trade-weighted star, ext. = area-wide instrument carried as weakly exogenous (euro members). The dates give the *shipped* history window the application loads (the recent span the charts display and the state initializes from); the estimation sample is longer and ends at the 2025Q4 estimation vintage (Section 11.3).

Table 7: United States (US): $n = 40$, shipped history from 1998Q3, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDPH	Real Gross Domestic Product	log	WN	—	dom.
CH	Real Personal Consumption Expenditures	log	RW	—	dom.
FRH	Real Private Residential Fixed Investment (nom/deflator)	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
FNH	Real Private Nonresidential Fixed Investment (nom/deflator)	log	RW	—	dom.
XH	Real Exports of Goods & Services	log	RW	—	dom.
MH	Real Imports of Goods & Services	log	RW	—	dom.
GH	Real Government Consumption Expenditures & Gross Investment	log	RW	—	dom.
JGDP	Gross Domestic Product: Chain Price Index	log	WN	—	dom.
JC	Personal Consumption Expenditures: Chain Price Index	log	RW	—	dom.
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW	—	dom.
UI	CPI-U: All Items	log	RW	—	dom.
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW	—	dom.
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW	—	dom.
LANAGRA	All Employees: Total Nonfarm	log	RW	—	dom.
LR	Civilian Unemployment Rate: 16 yr +	lin	RW	—	dom.
IP	Industrial Production Index	log	RW	—	dom.
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW	—	dom.
HST	Housing Starts	log	RW	—	dom.
YPDH	Real Disposable Personal Income	log	RW	—	dom.
CSENT	University of Michigan: Consumer Sentiment	lin	RW	—	dom.
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW	✓	dom.
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	dom.
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW	✓	dom.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW	✓	dom.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
FXTWM	US Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
GSNE	IMF global price index of non-fuel commodities (PNFUELINDEXM)	log	RW	✓	dom.
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	RW	✓	dom.
HP	House Price	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
LEV	Real nonfinancial business debt	log	RW	✓	dom.
FREV_GDP	Federal Receipts (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Federal Effective Interest Rate (% p.a.)	lin	RW	✓	fisc.
SFA	Stock-Flow Adjustment (% of GDP)	lin	WN	—	fisc.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
ROWDEM_US	US RoW Demand	lin	RW	✓	row
ROWINF_US	US RoW Inflation	lin	RW	✓	row
ROWFIN_US	US RoW Fin	lin	RW	✓	row

Table 8: Euro area (aggregate) (EA): $n = 26$, shipped history from 1999Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_EA	EA Real GDP	log	WN	—	dom.
CONS_EA	EA Consumption	log	RW	—	dom.
INV_EA	EA Investment	log	RW	—	dom.
GOV_EA	EA Gov Spending	log	RW	—	dom.
EXP_EA	EA Exports	log	RW	—	dom.
IMP_EA	EA Imports	log	RW	—	dom.
HICP_EA	EA HICP Index	log	WN	—	dom.
DEFL_EA	EA GDP implicit price deflator (2015=100, SCA)	log	WN	—	dom.
RATE_EA	EA 3M Interbank Rate	lin	RW	✓	dom.
YIELD10_EA	EA 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_EA	EA NEER (BIS broad nominal effective exchange rate, 2020=100) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_EA	EA ECB Deposit Facility Rate (policy)	lin	RW	✓	dom.
HPI_EA	EA Real House Price Index (Eurostat HPI 2015=100 / HICP 2025=100, x100)	log	RW	✓	dom.
EACORP	Corporate bond yield proxy: German non-financial-corporate issuers (Bundesbank, %)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	EA Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	EA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_EA	EA Credit to NFCs	log	RW	✓	dom.
ROWDEM_EA	EA RoW Demand	lin	RW	✓	row
ROWINF_EA	EA RoW Inflation	lin	RW	✓	row
ROWFIN_EA	EA RoW Fin	lin	RW	✓	row

Table 9: Japan (JP): $n = 27$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_JP	JP Real GDP	log	WN	—	dom.
CONS_JP	JP Consumption	log	RW	—	dom.
INV_JP	JP Investment	log	RW	—	dom.
GOV_JP	JP Gov Spending	log	RW	—	dom.
EXP_JP	JP Exports	log	RW	—	dom.
IMP_JP	JP Imports	log	RW	—	dom.
LR_JP	JP Unemployment Rate	lin	RW	—	dom.
CPI_JP	JP CPI Index	log	WN	—	dom.
DEFL_JP	JP GDP implicit price deflator (2020=100, SA)	log	WN	—	dom.
RATE_JP	JP 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_JP	JP uncollateralized overnight call rate (BoJ, monthly average)	lin	RW	✓	dom.
YIELD10_JP	JP 10Y JGB Yield	lin	RW	✓	dom.
HP_JP	JP Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	JP Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	JP Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_JP	JP Credit to NFCs	log	RW	✓	dom.
LENDR_JP	JP Avg Contract Interest Rate on New Loans (% p.a.)	lin	RW	✓	dom.
NEER_JP	JP NEER (index; rise = appreciation)	log	RW	✓	dom.
ROWDEM_JP	JP RoW Demand	lin	RW	✓	row
ROWINF_JP	JP RoW Inflation	lin	RW	✓	row
ROWFIN_JP	JP RoW Fin	lin	RW	✓	row

Table 10: United Kingdom (UK): $n = 28$, shipped history from 1997Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_UK	UK Real GDP	log	WN	—	dom.
CONS_UK	UK Consumption	log	RW	—	dom.
INV_UK	UK Investment	log	RW	—	dom.
GOV_UK	UK Gov Spending	log	RW	—	dom.
EXP_UK	UK Exports	log	RW	—	dom.
IMP_UK	UK Imports	log	RW	—	dom.
IP_UK	UK Ind Production Index	log	RW	—	dom.
LR_UK	UK Unemployment Rate	lin	RW	—	dom.
CPI_UK	UK CPI Index	log	WN	—	dom.
DEFL_UK	UK GDP implicit price deflator (market prices, SA)	log	WN	—	dom.
RATE_UK	UK SONIA (Sterling Overnight Index Average, BoE)	lin	RW	✓	dom.
POLRATE_UK	UK Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_UK	UK 10Y Gilt Yield	lin	RW	✓	dom.
HP_UK	UK Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_UK	UK NEER (BIS broad nominal effective exchange rate, 2020=100) (index; rise = appreciation)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	UK Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	UK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	UK Effective Rate (% p.a.)	lin	RW	✓	fisc.
SFA	UK Stock-Flow Adj (% of GDP)	lin	WN	—	fisc.
CREDIT_GDP_UK	UK Credit to NFCs	log	RW	✓	dom.
LENDR_UK	UK Effective Rate on New Loans to PNFCs (% p.a., BoE CFMBJ82)	lin	RW	✓	dom.
ROWDEM_UK	UK RoW Demand	lin	RW	✓	row
ROWINF_UK	UK RoW Inflation	lin	RW	✓	row
ROWFIN_UK	UK RoW Fin	lin	RW	✓	row

Table 11: Canada (CA): $n = 28$, shipped history from 1998Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CA	CA Real GDP	log	WN	—	dom.
CONS_CA	CA Consumption	log	RW	—	dom.
INV_CA	CA Investment	log	RW	—	dom.
GOV_CA	CA Gov Spending	log	RW	—	dom.
EXP_CA	CA Exports	log	RW	—	dom.
IMP_CA	CA Imports	log	RW	—	dom.
IP_CA	CA Industrial Production Index (OECD, industry B-E, SA, 2015=100)	log	RW	—	dom.
LR_CA	CA Unemployment Rate	lin	RW	—	dom.
CPI_CA	CA CPI Index	log	WN	—	dom.
DEFL_CA	CA GDP implicit price deflator (2017=100)	log	WN	—	dom.
RATE_CA	CA 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CA	CA Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CA	CA 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CA	CA Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_CA	CA NEER (BIS broad nominal effective exchange rate, 2020=100) (index; rise = appreciation)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	CA Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment (% of GDP)	lin	WN	—	fisc.
CREDIT_GDP_CA	CA Credit to NFCs	log	RW	✓	dom.
LENDR_CA	CA Bank Lending Rate (effective business interest rate, % p.a.)	lin	RW	✓	dom.
ROWDEM_CA	CA RoW Demand	lin	RW	✓	row
ROWINF_CA	CA RoW Inflation	lin	RW	✓	row
ROWFIN_CA	CA RoW Fin	lin	RW	✓	row

Table 12: China (CN): $n = 22$, shipped history from 1999Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CN	CN Real GDP (quarterly, chain-linked SA volume index, base 2011Q1=100)	log	WN	—	dom.
CPI_CN	CN CPI	log	WN	—	dom.
RATE_CN	CN 3M Interbank Rate	lin	RW	✓	dom.
EXP_CN	CN Exports (USD)	log	RW	—	dom.
IMP_CN	CN Imports (USD)	log	RW	—	dom.
REER_CN	CN Real Effective FX (index; rise = appreciation)	log	RW	✓	dom.
EQ_CN	Shanghai Share Prices	log	RW	✓	dom.
POLRATE_CN	CN Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CN	CN Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	CN Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CN Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_CN	CN Credit to NFCs	log	RW	✓	dom.
YIELD10_CN	CN 10Y government bond yield (%)	lin	RW	✓	dom.
ROWDEM_CN	CN RoW Demand	lin	RW	✓	row
ROWINF_CN	CN RoW Inflation	lin	RW	✓	row
ROWFIN_CN	CN RoW Fin	lin	RW	✓	row

Table 13: Austria (AT): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AT	AT GDP	log	WN	—	dom.
CONS_AT	AT CONS	log	RW	—	dom.
INV_AT	AT INV	log	RW	—	dom.
GOV_AT	AT GOV	log	RW	—	dom.
EXP_AT	AT EXP	log	RW	—	dom.
IMP_AT	AT IMP	log	RW	—	dom.
HICP_AT	AT HICP Index	log	WN	—	dom.
DEFL_AT	AT GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_AT	AT Unemployment Rate	lin	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
HPI_AT	AT Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_AT	AT NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_AT	AT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_AT	AT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_AT	AT 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_AT	AT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_AT	AT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	AT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	AT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_AT	AT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_AT	AT NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 14: Australia (AU): $n = 26$, shipped history from 1999Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_AU	AU Real GDP	log	WN	—	dom.
CONS_AU	AU Real Household Consumption	log	RW	—	dom.
GOV_AU	AU Real Govt Consumption	log	RW	—	dom.
INV_AU	AU Real Investment (GFCF)	log	RW	—	dom.
EXP_AU	AU Real Exports	log	RW	—	dom.
IMP_AU	AU Real Imports	log	RW	—	dom.
CPI_AU	AU CPI All-items Index	log	WN	—	dom.
DEFL_AU	AU GDP implicit price deflator (SA)	log	WN	—	dom.
LR_AU	AU Unemployment Rate	lin	RW	—	dom.
NEER_AU	AU NEER Index (index; rise = appreciation)	log	RW	✓	dom.
RATE_AU	AU Cash Rate (Interbank Overnight)	lin	RW	✓	dom.
YIELD10_AU	AU 10Y Govt Bond Yield	lin	RW	✓	dom.
HP_AU	AU Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	AU Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	AU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	AU Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	AU Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_AU	AU Credit to NFCs	log	RW	✓	dom.
LENDR_AU	AU small-business variable term lending INDICATOR rate (RBA F5, % p.a.)	lin	RW	✓	dom.
ROWDEM_AU	AU RoW Demand	lin	RW	✓	row
ROWINF_AU	AU RoW Inflation	lin	RW	✓	row
ROWFIN_AU	AU RoW Fin	lin	RW	✓	row

Table 15: Belgium (BE): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_BE	BE GDP	log	WN	—	dom.
CONS_BE	BE CONS	log	RW	—	dom.
INV_BE	BE INV	log	RW	—	dom.
GOV_BE	BE GOV	log	RW	—	dom.
EXP_BE	BE EXP	log	RW	—	dom.
IMP_BE	BE IMP	log	RW	—	dom.
HICP_BE	BE HICP Index	log	WN	—	dom.
DEFL_BE	BE GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_BE	BE Unemployment Rate	lin	RW	—	dom.
HPI_BE	BE Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_BE	BE NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_BE	BE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_BE	BE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_BE	BE 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
ROWEA_GDP_BE	BE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_BE	BE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	BE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	BE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_BE	BE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_BE	BE NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 16: Brazil (BR): $n = 25$, shipped history from 2012Q2, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_BR	BR Real GDP	log	WN	—	dom.
CONS_BR	BR Real Household Consumption	log	RW	—	dom.
INV_BR	BR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_BR	BR Real Government Consumption	log	RW	—	dom.
EXP_BR	BR Real Exports	log	RW	—	dom.
IMP_BR	BR Real Imports	log	RW	—	dom.
CPI_BR	BR CPI Index	log	WN	—	dom.
DEFL_BR	BR GDP Deflator (implicit, OECD reference year=100)	log	WN	—	dom.
LR_BR	BR Unemployment Rate	lin	RW	—	dom.
NEER_BR	BR Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_BR	BR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_BR	BR Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	BR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	BR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	BR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	BR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_BR	BR Credit to NFCs	log	RW	✓	dom.
LENDR_BR	BR Corporate Bank Lending Rate (BCB SGS 20718, % p.a.)	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
ROWDEM_BR	BR RoW Demand	lin	RW	✓	row
ROWINF_BR	BR RoW Inflation	lin	RW	✓	row
ROWFIN_BR	BR RoW Fin	lin	RW	✓	row

Table 17: Switzerland (CH): $n = 20$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CH	CH GDP	log	WN	—	dom.
CONS_CH	CH CONS	log	RW	—	dom.
INV_CH	CH INV	log	RW	—	dom.
GOV_CH	CH GOV	log	RW	—	dom.
EXP_CH	CH EXP	log	RW	—	dom.
IMP_CH	CH IMP	log	RW	—	dom.
HICP_CH	CH HICP Index	log	WN	—	dom.
LR_CH	CH Unemployment Rate	lin	RW	—	dom.
NEER_CH	CH NEER (BIS broad nominal effective exchange rate, 2020=100) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_CH	CH Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CH	CH Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
CREDIT_GDP_CH	CH Credit to NFCs	log	RW	✓	dom.
LENDR_CH	CH Corporate Lending Rate (SNB new investment loans, fixed rate, % p.a.)	lin	RW	✓	dom.
ROWDEM_CH	CH RoW Demand	lin	RW	✓	row
ROWINF_CH	CH RoW Inflation	lin	RW	✓	row
ROWFIN_CH	CH RoW Fin	lin	RW	✓	row

Table 18: Chile (CL): $n = 26$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CL	CL Real GDP	log	WN	—	dom.
CONS_CL	CL Real Household Consumption	log	RW	—	dom.
INV_CL	CL Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_CL	CL Real Government Consumption	log	RW	—	dom.
EXP_CL	CL Real Exports	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
IMP_CL	CL Real Imports	log	RW	—	dom.
CPI_CL	CL CPI Index	log	WN	—	dom.
DEFL_CL	CL GDP Deflator (implicit, OECD reference year=100)	log	WN	—	dom.
LR_CL	CL Unemployment Rate	lin	RW	—	dom.
RATE_CL	CL 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CL	CL Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CL	CL 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CL	CL Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_CL	CL NEER (index; rise = appreciation)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	CL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CL	CL Credit to NFCs	log	RW	✓	dom.
ROWDEM_CL	CL RoW Demand	lin	RW	✓	row
ROWINF_CL	CL RoW Inflation	lin	RW	✓	row
ROWFIN_CL	CL RoW Fin	lin	RW	✓	row

Table 19: Colombia (CO): $n = 27$, shipped history from 2005Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CO	CO Real GDP	log	WN	—	dom.
CONS_CO	CO Real Household Consumption	log	RW	—	dom.
INV_CO	CO Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_CO	CO Real Government Consumption	log	RW	—	dom.
EXP_CO	CO Real Exports	log	RW	—	dom.
IMP_CO	CO Real Imports	log	RW	—	dom.
CPI_CO	CO CPI Index	log	WN	—	dom.
DEFL_CO	CO GDP Deflator (implicit, OECD reference year=100)	log	WN	—	dom.
LR_CO	CO Unemployment Rate	lin	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
RATE_CO	CO 3M Interbank Rate	lin	RW	✓	dom.
POLRATE_CO	CO Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_CO	CO 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_CO	CO Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_CO	CO NEER (BIS broad, nominal, monthly avg) (index; rise = appreciation)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	CO Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CO Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CO Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CO Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CO	CO Credit to NFCs (BIS, all sectors)	log	RW	✓	dom.
LENDR_CO	CO Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_CO	CO RoW Demand	lin	RW	✓	row
ROWINF_CO	CO RoW Inflation	lin	RW	✓	row
ROWFIN_CO	CO RoW Fin	lin	RW	✓	row

Table 20: Costa Rica (CR): $n = 21$, shipped history from 2010Q3, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CR	CR Real GDP	log	WN	—	dom.
EXP_CR	CR Real Exports	log	RW	—	dom.
IMP_CR	CR Real Imports	log	RW	—	dom.
CPI_CR	CR CPI Index	log	WN	—	dom.
DEFL_CR	CR GDP Deflator (implicit, OECD reference year=100)	log	WN	—	dom.
LR_CR	CR Unemployment Rate	lin	RW	—	dom.
RATE_CR	CR Monetary Policy Rate	lin	RW	✓	dom.
NEER_CR	CR NEER (index; rise = appreciation)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	CR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	CR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CR	CR Credit to NFCs	log	RW	✓	dom.
LENDR_CR	CR Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_CR	CR RoW Demand	lin	RW	✓	row
ROWINF_CR	CR RoW Inflation	lin	RW	✓	row
ROWFIN_CR	CR RoW Fin	lin	RW	✓	row

Table 21: Czechia (CZ): $n = 26$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_CZ	CZ GDP	log	WN	—	dom.
CONS_CZ	CZ CONS	log	RW	—	dom.
INV_CZ	CZ INV	log	RW	—	dom.
GOV_CZ	CZ GOV	log	RW	—	dom.
EXP_CZ	CZ EXP	log	RW	—	dom.
IMP_CZ	CZ IMP	log	RW	—	dom.
HICP_CZ	CZ HICP Index	log	WN	—	dom.
DEFL_CZ	CZ GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_CZ	CZ Unemployment Rate	lin	RW	—	dom.
YIELD10_CZ	CZ 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_CZ	CZ NEER (BIS broad, nominal, monthly avg) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_CZ	CZ Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_CZ	CZ Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	CZ Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	CZ Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	CZ Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
SFA	CZ Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_CZ	CZ Credit to NFCs	log	RW	✓	dom.
LENDR_CZ	CZ NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_CZ	CZ RoW Demand	lin	RW	✓	row
ROWINF_CZ	CZ RoW Inflation	lin	RW	✓	row
ROWFIN_CZ	CZ RoW Fin	lin	RW	✓	row

Table 22: Germany (DE): $n = 23$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_DE	DE GDP	log	WN	—	dom.
CONS_DE	DE CONS	log	RW	—	dom.
INV_DE	DE INV	log	RW	—	dom.
GOV_DE	DE GOV	log	RW	—	dom.
EXP_DE	DE EXP	log	RW	—	dom.
IMP_DE	DE IMP	log	RW	—	dom.
HICP_DE	DE HICP Index	log	WN	—	dom.
DEFL_DE	DE GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_DE	DE Unemployment Rate	lin	RW	—	dom.
HPI_DE	DE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_DE	DE NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund)	lin	RW	✓	dom.
POLRATE_DE	DE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_DE	DE 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
ROWEA_GDP_DE	DE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_DE	DE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	DE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	DE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_DE	DE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_DE	DE NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 23: Denmark (DK): $n = 27$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_DK	DK GDP	log	WN	—	dom.
CONS_DK	DK CONS	log	RW	—	dom.
INV_DK	DK INV	log	RW	—	dom.
GOV_DK	DK GOV	log	RW	—	dom.
EXP_DK	DK EXP	log	RW	—	dom.
IMP_DK	DK IMP	log	RW	—	dom.
HICP_DK	DK HICP Index	log	WN	—	dom.
DEFL_DK	DK GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_DK	DK Unemployment Rate	lin	RW	—	dom.
RATE_DK	DK 3M Rate	lin	RW	✓	dom.
YIELD10_DK	DK 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_DK	DK NEER (BIS broad, nominal, monthly avg) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_DK	DK Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_DK	DK Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	DK Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	DK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	DK Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	DK Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_DK	DK Credit to NFCs	log	RW	✓	dom.
LENDR_DK	DK Corporate Lending Rate (ECB MIR loans to NFCs, new business, % p.a.)	lin	RW	✓	dom.
ROWDEM_DK	DK RoW Demand	lin	RW	✓	row
ROWINF_DK	DK RoW Inflation	lin	RW	✓	row
ROWFIN_DK	DK RoW Fin	lin	RW	✓	row

Table 24: Estonia (EE): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_EE	EE GDP	log	WN	—	dom.
CONS_EE	EE CONS	log	RW	—	dom.
INV_EE	EE INV	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
GOV_EE	EE GOV	log	RW	—	dom.
EXP_EE	EE EXP	log	RW	—	dom.
IMP_EE	EE IMP	log	RW	—	dom.
HICP_EE	EE HICP Index	log	WN	—	dom.
DEFL_EE	EE GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_EE	EE Unemployment Rate	lin	RW	—	dom.
HPI_EE	EE Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_EE	EE NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_EE	EE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_EE	EE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_EE	EE 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_EE	EE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_EE	EE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	EE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	EE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_EE	EE Credit to NFCs (% of GDP; loans + debt securities)	lin	RW	✓	dom.
LENDR_EE	EE ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 25: Spain (ES): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ES	ES GDP	log	WN	—	dom.
CONS_ES	ES CONS	log	RW	—	dom.
INV_ES	ES INV	log	RW	—	dom.
GOV_ES	ES GOV	log	RW	—	dom.
EXP_ES	ES EXP	log	RW	—	dom.
IMP_ES	ES IMP	log	RW	—	dom.
HICP_ES	ES HICP Index	log	WN	—	dom.
DEFL_ES	ES GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_ES	ES Unemployment Rate	lin	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
HPI_ES	ES Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_ES	ES NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_ES	ES 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_ES	ES ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_ES	ES 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_ES	ES Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_ES	ES Rest-of-EA HICP	log	RW	—	row
FREV_GDP	ES Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ES Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_ES	ES Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_ES	ES NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 26: Finland (FI): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_FI	FI GDP	log	WN	—	dom.
CONS_FI	FI CONS	log	RW	—	dom.
INV_FI	FI INV	log	RW	—	dom.
GOV_FI	FI GOV	log	RW	—	dom.
EXP_FI	FI EXP	log	RW	—	dom.
IMP_FI	FI IMP	log	RW	—	dom.
HICP_FI	FI HICP Index	log	WN	—	dom.
DEFL_FI	FI GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_FI	FI Unemployment Rate	lin	RW	—	dom.
HPI_FI	FI Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_FI	FI NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_FI	FI 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_FI	FI ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_FI	FI 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_FI	FI Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_FI	FI Rest-of-EA HICP	log	RW	—	row
FREV_GDP	FI Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	FI Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_FI	FI Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_FI	FI ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 27: France (FR): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_FR	FR GDP	log	WN	—	dom.
CONS_FR	FR CONS	log	RW	—	dom.
INV_FR	FR INV	log	RW	—	dom.
GOV_FR	FR GOV	log	RW	—	dom.
EXP_FR	FR EXP	log	RW	—	dom.
IMP_FR	FR IMP	log	RW	—	dom.
HICP_FR	FR HICP Index	log	WN	—	dom.
DEFL_FR	FR GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_FR	FR Unemployment Rate	lin	RW	—	dom.
HPI_FR	FR Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_FR	FR NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_FR	FR 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_FR	FR ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_FR	FR 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_FR	FR Rest-of-EA Real GDP	log	RW	—	row

ID	Description	Trans.	Prior	Fin.	Block
ROWEA_HICP_FR	FR Rest-of-EA HICP	log	RW	—	row
FREV_GDP	FR Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	FR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_FR	FR Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_FR	FR NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 28: Greece (GR): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_GR	GR GDP	log	WN	—	dom.
CONS_GR	GR CONS	log	RW	—	dom.
INV_GR	GR INV	log	RW	—	dom.
GOV_GR	GR GOV	log	RW	—	dom.
EXP_GR	GR EXP	log	RW	—	dom.
IMP_GR	GR IMP	log	RW	—	dom.
HICP_GR	GR HICP Index	log	WN	—	dom.
DEFL_GR	GR GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_GR	GR Unemployment Rate	lin	RW	—	dom.
HPI_GR	GR Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_GR	GR NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_GR	GR 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_GR	GR ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_GR	GR 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_GR	GR Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_GR	GR Rest-of-EA HICP	log	RW	—	row
FREV_GDP	GR Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	GR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_GR	GR Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_GR	GR NFC Bank Lending Rate (ECB MIR new-business floating, %)	lin	RW	✓	dom.

Table 29: HK (HK): $n = 17$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_HK	HK GDP	log	WN	—	dom.
EXP_HK	HK EXP	log	RW	—	dom.
IMP_HK	HK IMP	log	RW	—	dom.
CPI_HK	HK CPI	log	WN	—	dom.
RATE_HK	HK RATE	lin	RW	✓	dom.
YIELD10_HK	HK YIELD10	lin	RW	✓	dom.
NEER_HK	HK NEER (index; rise = appreciation)	log	RW	✓	dom.
HP_HK	HK HP	log	RW	✓	dom.
EQ_HK	HK EQ	log	RW	✓	dom.
CREDIT_GDP_HK	HK Credit to Non-Financial Corporations (BIS, level, HKD bn)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
ROWDEM_HK	HK RoW Demand	lin	RW	✓	row
ROWINF_HK	HK RoW Inflation	lin	RW	✓	row
ROWFIN_HK	HK RoW Fin	lin	RW	✓	row

Table 30: Hungary (HU): $n = 26$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_HU	HU GDP	log	WN	—	dom.
CONS_HU	HU CONS	log	RW	—	dom.
INV_HU	HU INV	log	RW	—	dom.
GOV_HU	HU GOV	log	RW	—	dom.
EXP_HU	HU EXP	log	RW	—	dom.
IMP_HU	HU IMP	log	RW	—	dom.
HICP_HU	HU HICP Index	log	WN	—	dom.
DEFL_HU	HU GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_HU	HU Unemployment Rate	lin	RW	—	dom.
YIELD10_HU	HU 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_HU	HU NEER (BIS broad, nominal, monthly avg) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_HU	HU Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_HU	HU Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	HU Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	HU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	HU Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	HU Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_HU	HU Credit to NFCs	log	RW	✓	dom.
LENDR_HU	HU NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_HU	HU RoW Demand	lin	RW	✓	row
ROWINF_HU	HU RoW Inflation	lin	RW	✓	row
ROWFIN_HU	HU RoW Fin	lin	RW	✓	row

Table 31: Indonesia (ID): $n = 25$, shipped history from 2005Q3, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ID	ID Real GDP (SA, constant prices)	log	WN	—	dom.
CONS_ID	ID Real Private Consumption (SA)	log	RW	—	dom.
INV_ID	ID Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_ID	ID Real Government Consumption (SA)	log	RW	—	dom.
EXP_ID	ID Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_ID	ID Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_ID	ID CPI (all items, index)	log	WN	—	dom.
DEFL_ID	ID GDP Deflator (OECD QNA)	log	WN	—	dom.
RATE_ID	ID 3-month interbank rate (JIBOR, %)	lin	RW	✓	dom.
NEER_ID	ID Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_ID	ID Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_ID	ID Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	ID Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ID Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	ID Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
SFA	ID Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_ID	ID Credit to NFCs	log	RW	✓	dom.
LENDR_ID	ID IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_ID	ID RoW Demand	lin	RW	✓	row
ROWINF_ID	ID RoW Inflation	lin	RW	✓	row
ROWFIN_ID	ID RoW Fin	lin	RW	✓	row

Table 32: Ireland (IE): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IE	IE GDP	log	WN	—	dom.
CONS_IE	IE CONS	log	RW	—	dom.
INV_IE	IE INV	log	RW	—	dom.
GOV_IE	IE GOV	log	RW	—	dom.
EXP_IE	IE EXP	log	RW	—	dom.
IMP_IE	IE IMP	log	RW	—	dom.
HICP_IE	IE HICP Index	log	WN	—	dom.
DEFL_IE	IE GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_IE	IE Unemployment Rate	lin	RW	—	dom.
HPI_IE	IE Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_IE	IE NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_IE	IE 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_IE	IE ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_IE	IE 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_IE	IE Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_IE	IE Rest-of-EA HICP	log	RW	—	row
FREV_GDP	IE Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	IE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_IE	IE Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_IE	IE NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 33: India (IN): $n = 20$, shipped history from 2004Q2, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IN	IN Real GDP (chain volume, SA)	log	WN	—	dom.
CONS_IN	IN Real Household Consumption (chain vol, SA)	log	RW	—	dom.
INV_IN	IN Real Gross Fixed Capital Formation (chain vol, SA)	log	RW	—	dom.
GOV_IN	IN Real Government Consumption (chain vol, SA)	log	RW	—	dom.
EXP_IN	IN Real Exports of Goods & Services (chain vol, SA)	log	RW	—	dom.
IMP_IN	IN Real Imports of Goods & Services (chain vol, SA)	log	RW	—	dom.
CPI_IN	IN CPI (all items, index)	log	WN	—	dom.
RATE3M_IN	IN short-term market rate (call money to 2011Q3, OECD 3-month interbank from 2011Q4)	lin	RW	✓	dom.
YIELD10_IN	IN Long-term 10Y Government Bond Yield	lin	RW	✓	dom.
NEER_IN	IN Nominal Effective Exchange Rate (BIS broad, index) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_IN	IN Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_IN	IN Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
CREDIT_GDP_IN	IN Credit to NFCs	log	RW	✓	dom.
ROWDEM_IN	IN RoW Demand	lin	RW	✓	row
ROWINF_IN	IN RoW Inflation	lin	RW	✓	row
ROWFIN_IN	IN RoW Fin	lin	RW	✓	row

Table 34: Iceland (IS): $n = 21$, shipped history from 2000Q2, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IS	IS Real GDP	log	WN	—	dom.
CONS_IS	IS Real Private Consumption	log	RW	—	dom.
INV_IS	IS Real Investment	log	RW	—	dom.
GOV_IS	IS Real Govt Consumption	log	RW	—	dom.
EXP_IS	IS Real Exports	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
IMP_IS	IS Real Imports	log	RW	—	dom.
CPI_IS	IS CPI Index	log	WN	—	dom.
LR_IS	IS Unemployment Rate	lin	RW	—	dom.
RATE_IS	IS 3-Month Interbank Rate	lin	RW	✓	dom.
NEER_IS	IS Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_IS	IS Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_IS	IS Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
CREDIT_GDP_IS	IS Credit to NFCs	log	RW	✓	dom.
LENDR_IS	IS Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_IS	IS RoW Demand	lin	RW	✓	row
ROWINF_IS	IS RoW Inflation	lin	RW	✓	row
ROWFIN_IS	IS RoW Fin	lin	RW	✓	row

Table 35: Italy (IT): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_IT	IT GDP	log	WN	—	dom.
CONS_IT	IT CONS	log	RW	—	dom.
INV_IT	IT INV	log	RW	—	dom.
GOV_IT	IT GOV	log	RW	—	dom.
EXP_IT	IT EXP	log	RW	—	dom.
IMP_IT	IT IMP	log	RW	—	dom.
HICP_IT	IT HICP Index	log	WN	—	dom.
DEFL_IT	IT GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_IT	IT Unemployment Rate	lin	RW	—	dom.
HPI_IT	IT Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_IT	IT NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_IT	IT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_IT	IT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
RATE_IT	IT 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_IT	IT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_IT	IT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	IT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	IT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_IT	IT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_IT	IT NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 36: Korea (KR): $n = 27$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_KR	KR Real GDP	log	WN	—	dom.
CONS_KR	KR Real Household Consumption	log	RW	—	dom.
INV_KR	KR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_KR	KR Real Government Consumption	log	RW	—	dom.
EXP_KR	KR Real Exports	log	RW	—	dom.
IMP_KR	KR Real Imports	log	RW	—	dom.
CPI_KR	KR CPI (all items, index)	log	WN	—	dom.
DEFL_KR	KR GDP implicit price deflator (SA)	log	WN	—	dom.
LR_KR	KR Unemployment Rate	lin	RW	—	dom.
RATE_KR	KR money market rate (overnight call rate, IMF MFS)	lin	RW	✓	dom.
POLRATE_KR	KR Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_KR	KR government bond yield (IMF MFS)	lin	RW	✓	dom.
HP_KR	KR Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	KR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	KR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	KR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.

ID	Description	Trans.	Prior	Fin.	Block
SFA	KR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_KR	KR Credit to NFCs	log	RW	✓	dom.
CORP_KR	KR Corporate Bond Yield (3Y, AA-, BoK ECOS)	lin	RW	✓	dom.
NEER_KR	KR NEER (index; rise = appreciation)	log	RW	✓	dom.
ROWDEM_KR	KR RoW Demand	lin	RW	✓	row
ROWINF_KR	KR RoW Inflation	lin	RW	✓	row
ROWFIN_KR	KR RoW Fin	lin	RW	✓	row

Table 37: Lithuania (LT): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LT	LT GDP	log	WN	—	dom.
CONS_LT	LT CONS	log	RW	—	dom.
INV_LT	LT INV	log	RW	—	dom.
GOV_LT	LT GOV	log	RW	—	dom.
EXP_LT	LT EXP	log	RW	—	dom.
IMP_LT	LT IMP	log	RW	—	dom.
HICP_LT	LT HICP Index	log	WN	—	dom.
DEFL_LT	LT GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_LT	LT Unemployment Rate	lin	RW	—	dom.
HPI_LT	LT Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_LT	LT NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_LT	LT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LT	LT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_LT	LT 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LT	LT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LT	LT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LT	LT Credit to NFCs (% of GDP; loans + debt securities)	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
LENDR_LT	LT ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 38: Luxembourg (LU): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LU	LU GDP	log	WN	—	dom.
CONS_LU	LU CONS	log	RW	—	dom.
INV_LU	LU INV	log	RW	—	dom.
GOV_LU	LU GOV	log	RW	—	dom.
EXP_LU	LU EXP	log	RW	—	dom.
IMP_LU	LU IMP	log	RW	—	dom.
HICP_LU	LU HICP Index	log	WN	—	dom.
DEFL_LU	LU GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_LU	LU Unemployment Rate	lin	RW	—	dom.
HPI_LU	LU Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LU	LU NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_LU	LU 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LU	LU ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_LU	LU 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LU	LU Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LU	LU Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LU Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LU Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LU	LU Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_LU	LU ECB MIR Bank Lending Rate to NFCs (outstanding amounts, % p.a.)	lin	RW	✓	dom.

Table 39: Latvia (LV): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_LV	LV GDP	log	WN	—	dom.
CONS_LV	LV CONS	log	RW	—	dom.
INV_LV	LV INV	log	RW	—	dom.
GOV_LV	LV GOV	log	RW	—	dom.
EXP_LV	LV EXP	log	RW	—	dom.
IMP_LV	LV IMP	log	RW	—	dom.
HICP_LV	LV HICP Index	log	WN	—	dom.
DEFL_LV	LV GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_LV	LV Unemployment Rate	lin	RW	—	dom.
HPI_LV	LV Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_LV	LV NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_LV	LV 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_LV	LV ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_LV	LV 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_LV	LV Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_LV	LV Rest-of-EA HICP	log	RW	—	row
FREV_GDP	LV Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	LV Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_LV	LV Credit to NFCs (% of GDP; loans + debt securities)	lin	RW	✓	dom.
LENDR_LV	LV NFC Bank Lending Rate (ECB MIR new-business floating, %)	lin	RW	✓	dom.

Table 40: Mexico (MX): $n = 27$, shipped history from 2001Q4, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_MX	MX Real GDP (chain volume, SA)	log	WN	—	dom.
CONS_MX	MX Real Household Consumption (chain vol, SA)	log	RW	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
INV_MX	MX Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_MX	MX Real Government Consumption (chain vol, SA)	log	RW	—	dom.
EXP_MX	MX Real Exports (chain volume, SA)	log	RW	—	dom.
IMP_MX	MX Real Imports (chain volume, SA)	log	RW	—	dom.
CPI_MX	MX CPI (all items, index)	log	WN	—	dom.
DEFL_MX	MX GDP Deflator (implicit, OECD reference year=100)	log	WN	—	dom.
LR_MX	MX Unemployment Rate	lin	RW	—	dom.
RATE_MX	MX Money Market Rate (short-term)	lin	RW	✓	dom.
POLRATE_MX	MX Central Bank Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
YIELD10_MX	MX Government Bond Yield (long-term, S13)	lin	RW	✓	dom.
HP_MX	MX Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_MX	MX Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	MX Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	MX Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	MX Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	MX Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_MX	MX Credit to NFCs	log	RW	✓	dom.
LENDR_MX	MX Bank Lending Rate (IMF IFS FILR, % p.a.)	lin	RW	✓	dom.
ROWDEM_MX	MX RoW Demand	lin	RW	✓	row
ROWINF_MX	MX RoW Inflation	lin	RW	✓	row
ROWFIN_MX	MX RoW Fin	lin	RW	✓	row

Table 41: Malaysia (MY): $n = 21$, shipped history from 2015Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_MY	MY Real GDP (chain volume, SA)	log	WN	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
CONS_MY	MY Real Private Consumption (SA)	log	RW	—	dom.
INV_MY	MY Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_MY	MY Real Government Consumption (SA)	log	RW	—	dom.
EXP_MY	MY Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_MY	MY Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_MY	MY CPI Index (all items)	log	WN	—	dom.
LR_MY	MY Unemployment Rate	lin	RW	—	dom.
RATE_MY	MY Overnight Interbank Rate (IMF MFS_IR money-market rate)	lin	RW	✓	dom.
NEER_MY	MY Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_MY	MY Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_MY	MY Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
CREDIT_GDP_MY	MY Credit to NFCs	log	RW	✓	dom.
LENDR_MY	MY IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_MY	MY RoW Demand	lin	RW	✓	row
ROWINF_MY	MY RoW Inflation	lin	RW	✓	row
ROWFIN_MY	MY RoW Fin	lin	RW	✓	row

Table 42: Netherlands (NL): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NL	NL GDP	log	WN	—	dom.
CONS_NL	NL CONS	log	RW	—	dom.
INV_NL	NL INV	log	RW	—	dom.
GOV_NL	NL GOV	log	RW	—	dom.
EXP_NL	NL EXP	log	RW	—	dom.
IMP_NL	NL IMP	log	RW	—	dom.
HICP_NL	NL HICP Index	log	WN	—	dom.
DEFL_NL	NL GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_NL	NL Unemployment Rate	lin	RW	—	dom.
HPI_NL	NL Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
NEER_NL	NL NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_NL	NL 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_NL	NL ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_NL	NL 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_NL	NL Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_NL	NL Rest-of-EA HICP	log	RW	—	row
FREV_GDP	NL Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	NL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_NL	NL Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_NL	NL NFC Bank Lending Rate (ECB MIR new business, % p.a.)	lin	RW	✓	dom.

Table 43: Norway (NO): $n = 25$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NO	NO GDP	log	WN	—	dom.
CONS_NO	NO CONS	log	RW	—	dom.
INV_NO	NO INV	log	RW	—	dom.
GOV_NO	NO GOV	log	RW	—	dom.
EXP_NO	NO EXP	log	RW	—	dom.
IMP_NO	NO IMP	log	RW	—	dom.
HICP_NO	NO HICP Index	log	WN	—	dom.
DEFL_NO	NO GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_NO	NO Unemployment Rate	lin	RW	—	dom.
NEER_NO	NO NEER (BIS broad, nominal, monthly avg) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_NO	NO Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_NO	NO Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
FREV_GDP	NO Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	NO Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	NO Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	NO Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_NO	NO Credit to NFCs	log	RW	✓	dom.
LENDR_NO	NO NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_NO	NO RoW Demand	lin	RW	✓	row
ROWINF_NO	NO RoW Inflation	lin	RW	✓	row
ROWFIN_NO	NO RoW Fin	lin	RW	✓	row

Table 44: New Zealand (NZ): $n = 25$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_NZ	NZ Real GDP	log	WN	—	dom.
CONS_NZ	NZ Real Consumption	log	RW	—	dom.
INV_NZ	NZ Real Investment	log	RW	—	dom.
GOV_NZ	NZ Real Govt Consumption	log	RW	—	dom.
EXP_NZ	NZ Real Exports	log	RW	—	dom.
IMP_NZ	NZ Real Imports	log	RW	—	dom.
CPI_NZ	NZ CPI Index	log	WN	—	dom.
LR_NZ	NZ Unemployment Rate	lin	RW	—	dom.
RATE_NZ	NZ Policy Rate (OCR)	lin	RW	✓	dom.
YIELD10_NZ	NZ 10Y Gov Bond Yield	lin	RW	✓	dom.
HP_NZ	NZ Real Residential House-Price Index (BIS RPP, 2010=100)	log	RW	✓	dom.
NEER_NZ	NZ NEER (index; rise = appreciation)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	NZ Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	NZ Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	NZ Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	NZ Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_NZ	NZ Credit to NFCs	log	RW	✓	dom.
LENDR_NZ	NZ business lending rate (% p.a.)	lin	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
ROWDEM_NZ	NZ RoW Demand	lin	RW	✓	row
ROWINF_NZ	NZ RoW Inflation	lin	RW	✓	row
ROWFIN_NZ	NZ RoW Fin	lin	RW	✓	row

Table 45: PE (PE): $n = 20$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PE	PE GDP	log	WN	—	dom.
CONS_PE	PE CONS	log	RW	—	dom.
INV_PE	PE INV	log	RW	—	dom.
GOV_PE	PE GOV	log	RW	—	dom.
EXP_PE	PE EXP	log	RW	—	dom.
IMP_PE	PE IMP	log	RW	—	dom.
CPI_PE	PE CPI	log	WN	—	dom.
RATE_PE	PE Policy Rate (BCRP reference rate, BIS CBPOL)	lin	RW	✓	dom.
YIELD10_PE	PE YIELD10	lin	RW	✓	dom.
NEER_PE	PE NEER (index; rise = appreciation)	log	RW	✓	dom.
HP_PE	PE HP	log	RW	✓	dom.
EQ_PE	PE EQ	log	RW	✓	dom.
CREDIT_PE	PE CREDIT	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
ROWDEM_PE	PE RoW Demand	lin	RW	✓	row
ROWINF_PE	PE RoW Inflation	lin	RW	✓	row
ROWFIN_PE	PE RoW Fin	lin	RW	✓	row

Table 46: PH (PH): $n = 17$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PH	PH GDP	log	WN	—	dom.
CONS_PH	PH CONS	log	RW	—	dom.
INV_PH	PH INV	log	RW	—	dom.
GOV_PH	PH GOV	log	RW	—	dom.
EXP_PH	PH EXP	log	RW	—	dom.
IMP_PH	PH IMP	log	RW	—	dom.
CPI_PH	PH CPI	log	WN	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
RATE_PH	PH RATE	lin	RW	✓	dom.
NEER_PH	PH NEER (index; rise = appreciation)	log	RW	✓	dom.
HP_PH	PH HP	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
ROWDEM_PH	PH RoW Demand	lin	RW	✓	row
ROWINF_PH	PH RoW Inflation	lin	RW	✓	row
ROWFIN_PH	PH RoW Fin	lin	RW	✓	row

Table 47: Poland (PL): $n = 26$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PL	PL GDP	log	WN	—	dom.
CONS_PL	PL CONS	log	RW	—	dom.
INV_PL	PL INV	log	RW	—	dom.
GOV_PL	PL GOV	log	RW	—	dom.
EXP_PL	PL EXP	log	RW	—	dom.
IMP_PL	PL IMP	log	RW	—	dom.
HICP_PL	PL HICP Index	log	WN	—	dom.
DEFL_PL	PL GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_PL	PL Unemployment Rate	lin	RW	—	dom.
YIELD10_PL	PL 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_PL	PL NEER (BIS broad, nominal, monthly avg) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_PL	PL Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_PL	PL Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	PL Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	PL Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	PL Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	PL Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_PL	PL Credit to NFCs	log	RW	✓	dom.

ID	Description	Trans.	Prior	Fin.	Block
LENDR_PL	PL NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_PL	PL RoW Demand	lin	RW	✓	row
ROWINF_PL	PL RoW Inflation	lin	RW	✓	row
ROWFIN_PL	PL RoW Fin	lin	RW	✓	row

Table 48: Portugal (PT): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_PT	PT GDP	log	WN	—	dom.
CONS_PT	PT CONS	log	RW	—	dom.
INV_PT	PT INV	log	RW	—	dom.
GOV_PT	PT GOV	log	RW	—	dom.
EXP_PT	PT EXP	log	RW	—	dom.
IMP_PT	PT IMP	log	RW	—	dom.
HICP_PT	PT HICP Index	log	WN	—	dom.
DEFL_PT	PT GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_PT	PT Unemployment Rate	lin	RW	—	dom.
HPI_PT	PT Real House Price Index (BIS nominal RPP rebased 2015=100 / HICP)	log	RW	✓	dom.
NEER_PT	PT NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_PT	PT 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_PT	PT ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_PT	PT 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_PT	PT Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_PT	PT Rest-of-EA HICP	log	RW	—	row
FREV_GDP	PT Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	PT Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_PT	PT Credit to NFCs (% of GDP)	lin	RW	✓	dom.
LENDR_PT	PT NFC Bank Lending Rate (ECB MIR composite, %)	lin	RW	✓	dom.

Table 49: RO (RO): $n = 18$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_RO	RO GDP	log	WN	—	dom.
CONS_RO	RO CONS	log	RW	—	dom.
INV_RO	RO INV	log	RW	—	dom.
GOV_RO	RO GOV	log	RW	—	dom.
EXP_RO	RO EXP	log	RW	—	dom.
IMP_RO	RO IMP	log	RW	—	dom.
CPI_RO	RO CPI	log	WN	—	dom.
RATE_RO	RO Policy Rate (NBR, BIS CBPOL)	lin	RW	✓	dom.
YIELD10_RO	RO YIELD10	lin	RW	✓	dom.
NEER_RO	RO NEER (index; rise = appreciation)	log	RW	✓	dom.
HP_RO	RO HP	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
ROWDEM_RO	RO RoW Demand	lin	RW	✓	row
ROWINF_RO	RO RoW Inflation	lin	RW	✓	row
ROWFIN_RO	RO RoW Fin	lin	RW	✓	row

Table 50: Saudi Arabia (SA): $n = 18$, shipped history from 2010Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SA	SA Real GDP	log	WN	—	dom.
CONS_SA	SA Real Household Consumption	log	RW	—	dom.
INV_SA	SA Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_SA	SA Real Government Consumption	log	RW	—	dom.
EXP_SA	SA Real Exports	log	RW	—	dom.
IMP_SA	SA Real Imports	log	RW	—	dom.
CPI_SA	SA CPI Index	log	WN	—	dom.
LR_SA	SA Unemployment Rate	lin	RW	—	dom.
NEER_SA	SA Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_SA	SA Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.

ID	Description	Trans.	Prior	Fin.	Block
CREDIT_GDP_SA	SA Credit to NFCs	log	RW	✓	dom.
ROWDEM_SA	SA RoW Demand	lin	RW	✓	row
ROWINF_SA	SA RoW Inflation	lin	RW	✓	row
ROWFIN_SA	SA RoW Fin	lin	RW	✓	row

Table 51: Sweden (SE): $n = 27$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SE	SE GDP	log	WN	—	dom.
CONS_SE	SE CONS	log	RW	—	dom.
INV_SE	SE INV	log	RW	—	dom.
GOV_SE	SE GOV	log	RW	—	dom.
EXP_SE	SE EXP	log	RW	—	dom.
IMP_SE	SE IMP	log	RW	—	dom.
HICP_SE	SE HICP Index	log	WN	—	dom.
DEFL_SE	SE GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_SE	SE Unemployment Rate	lin	RW	—	dom.
RATE_SE	SE 3-month T-bill rate (OECD MEI via FRED, monthly avg)	lin	RW	✓	dom.
YIELD10_SE	SE 10Y Gov Bond Yield	lin	RW	✓	dom.
NEER_SE	SE NEER (BIS broad, nominal, monthly avg) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_SE	SE Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_SE	SE Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	SE Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	SE Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	SE Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	SE Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_SE	SE Credit to NFCs	log	RW	✓	dom.
LENDR_SE	SE NFC bank lending rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_SE	SE RoW Demand	lin	RW	✓	row
ROWINF_SE	SE RoW Inflation	lin	RW	✓	row
ROWFIN_SE	SE RoW Fin	lin	RW	✓	row

Table 52: SG (SG): $n = 15$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SG	SG GDP	log	WN	—	dom.
EXP_SG	SG EXP	log	RW	—	dom.
IMP_SG	SG IMP	log	RW	—	dom.
CPI_SG	SG CPI	log	WN	—	dom.
NEER_SG	SG NEER (index; rise = appreciation)	log	RW	✓	dom.
HP_SG	SG HP	log	RW	✓	dom.
EQ_SG	SG EQ	log	RW	✓	dom.
CREDIT_GDP_SG	SG Credit to Non-Financial Corporations (BIS, level, SGD bn)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
ROWDEM_SG	SG RoW Demand	lin	RW	✓	row
ROWINF_SG	SG RoW Inflation	lin	RW	✓	row
ROWFIN_SG	SG RoW Fin	lin	RW	✓	row

Table 53: Slovenia (SI): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SI	SI GDP	log	WN	—	dom.
CONS_SI	SI CONS	log	RW	—	dom.
INV_SI	SI INV	log	RW	—	dom.
GOV_SI	SI GOV	log	RW	—	dom.
EXP_SI	SI EXP	log	RW	—	dom.
IMP_SI	SI IMP	log	RW	—	dom.
HICP_SI	SI HICP Index	log	WN	—	dom.
DEFL_SI	SI GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_SI	SI Unemployment Rate	lin	RW	—	dom.
HPI_SI	SI Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_SI	SI NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_SI	SI 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_SI	SI ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.

ID	Description	Trans.	Prior	Fin.	Block
RATE_SI	SI 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_SI	SI Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_SI	SI Rest-of-EA HICP	log	RW	—	row
FREV_GDP	SI Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	SI Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_SI	SI Credit to NFCs (% of GDP; loans + debt securities)	lin	RW	✓	dom.
LENDR_SI	SI ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 54: Slovakia (SK): $n = 24$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_SK	SK GDP	log	WN	—	dom.
CONS_SK	SK CONS	log	RW	—	dom.
INV_SK	SK INV	log	RW	—	dom.
GOV_SK	SK GOV	log	RW	—	dom.
EXP_SK	SK EXP	log	RW	—	dom.
IMP_SK	SK IMP	log	RW	—	dom.
HICP_SK	SK HICP Index	log	WN	—	dom.
DEFL_SK	SK GDP Deflator (implicit, 2015=100)	log	WN	—	dom.
LR_SK	SK Unemployment Rate	lin	RW	—	dom.
HPI_SK	SK Real House Price Index (HPI/HICP, 2015=100)	log	RW	✓	dom.
NEER_SK	SK NEER (trade-weighted, IC42 partners) (index; rise = appreciation)	log	RW	✓	dom.
SPREAD_SK	SK 10Y Sovereign Spread vs Bund	lin	RW	✓	dom.
POLRATE_SK	SK ECB Deposit Facility Rate (policy)	lin	RW	✓	ext.
EACORP	EA corporate bond yield proxy: German non-financial corporate issuers (Bundesbank BBSIS, % p.a.)	lin	RW	✓	ext.
RATE_SK	SK 3M Rate (EA 3-month EURIBOR)	lin	RW	✓	ext.
YIELD10_DE	DE 10Y Gov Bond Yield (Bund, spread base)	lin	RW	✓	ext.
ROWEA_GDP_SK	SK Rest-of-EA Real GDP	log	RW	—	row
ROWEA_HICP_SK	SK Rest-of-EA HICP	log	RW	—	row
FREV_GDP	SK Revenue (% of GDP)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
FPEXP_GDP	SK Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	Effective interest rate	lin	RW	✓	fisc.
SFA	Stock-flow adjustment	lin	WN	—	fisc.
CREDIT_GDP_SK	SK Credit to NFCs (% of GDP; loans + debt securities)	lin	RW	✓	dom.
LENDR_SK	SK ECB MIR Composite Cost of Borrowing, NFCs (new business, % p.a.)	lin	RW	✓	dom.

Table 55: Thailand (TH): $n = 27$, shipped history from 2003Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TH	TH Real GDP (SA, constant prices)	log	WN	—	dom.
CONS_TH	TH Real Private Consumption (SA)	log	RW	—	dom.
INV_TH	TH Real Gross Fixed Capital Formation (SA)	log	RW	—	dom.
GOV_TH	TH Real Government Consumption (SA)	log	RW	—	dom.
EXP_TH	TH Real Exports of Goods & Services (SA)	log	RW	—	dom.
IMP_TH	TH Real Imports of Goods & Services (SA)	log	RW	—	dom.
CPI_TH	TH CPI (all items, index)	log	WN	—	dom.
DEFL_TH	TH GDP deflator (IMF QNEA, SA, 2010=100)	log	WN	—	dom.
LR_TH	TH Unemployment Rate	lin	RW	—	dom.
RATE_TH	TH Treasury-bill yield (IMF MFS_IR GSTBILY, % p.a.)	lin	RW	✓	dom.
YIELD10_TH	TH 10-Year Govt Bond Yield	lin	RW	✓	dom.
NEER_TH	TH Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_TH	TH Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_TH	TH Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	TH Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	TH Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	TH Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	TH Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.

ID	Description	Trans.	Prior	Fin.	Block
CREDIT_GDP_TH	TH Credit to NFCs	log	RW	✓	dom.
LENDR_TH	TH IFS Lending Rate (% p.a.)	lin	RW	✓	dom.
ROWDEM_TH	TH RoW Demand	lin	RW	✓	row
ROWINF_TH	TH RoW Inflation	lin	RW	✓	row
ROWFIN_TH	TH RoW Fin	lin	RW	✓	row

Table 56: Türkiye (TR): $n = 25$, shipped history from 2000Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TR	TR Real GDP	log	WN	—	dom.
CONS_TR	TR Real Household Consumption	log	RW	—	dom.
INV_TR	TR Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_TR	TR Real Government Consumption	log	RW	—	dom.
EXP_TR	TR Real Exports	log	RW	—	dom.
IMP_TR	TR Real Imports	log	RW	—	dom.
CPI_TR	TR CPI (all items, index)	log	WN	—	dom.
DEFL_TR	TR GDP deflator (OECD QNA, index)	log	WN	—	dom.
LR_TR	TR Unemployment Rate (OECD harmonised, NSA)	lin	RW	—	dom.
RATE_TR	TR Overnight Interbank Repo Rate (Eurostat day-to-day money rate, source CBRT)	lin	RW	✓	dom.
POLRATE_TR	TR Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_TR	TR Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	TR Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	TR Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	TR Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	TR Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_TR	TR Credit to NFCs (BIS total credit, TRY bn, level)	log	RW	✓	dom.
NEER_TR	TR NEER (index; rise = appreciation)	log	RW	✓	dom.
ROWDEM_TR	TR RoW Demand	lin	RW	✓	row
ROWINF_TR	TR RoW Inflation	lin	RW	✓	row
ROWFIN_TR	TR RoW Fin	lin	RW	✓	row

Table 57: Taiwan Province of China (TW): $n = 21$, shipped history from 1997Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_TW	TW Real GDP	log	WN	—	dom.
CONS_TW	TW Real Household Consumption	log	RW	—	dom.
GOV_TW	TW Real Government Consumption	log	RW	—	dom.
INV_TW	TW Real Gross Capital Formation incl. inventories (DGBAS SA; proxy for GFCF)	log	RW	—	dom.
EXP_TW	TW Real Exports of Goods and Services	log	RW	—	dom.
IMP_TW	TW Real Imports of Goods and Services	log	RW	—	dom.
CPI_TW	TW CPI (all items, index, 2021=100)	log	WN	—	dom.
LR_TW	TW Unemployment Rate	lin	RW	—	dom.
RATE_TW	TW CBC Discount Rate (policy)	lin	RW	✓	dom.
MMRATE_TW	TW Interbank Call-Loan Rate (money market)	lin	RW	✓	dom.
YIELD10_TW	TW 10-Year Government Bond Yield	lin	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
CREDIT_GDP_TW	TW bank loans to private enterprises (CBC, q-avg of month-end stocks; proxy for credit to NFCs)	log	RW	✓	dom.
LENDR_TW	TW 5-Major-Banks New Loan Rate (CBC EH45 total, % p.a.)	lin	RW	✓	dom.
NEER_TW	TW NEER (index; rise = appreciation)	log	RW	✓	dom.
ROWDEM_TW	TW RoW Demand	lin	RW	✓	row
ROWINF_TW	TW RoW Inflation	lin	RW	✓	row
ROWFIN_TW	TW RoW Fin	lin	RW	✓	row

Table 58: South Africa (ZA): $n = 26$, shipped history from 2002Q1, last observation 2026Q2.

ID	Description	Trans.	Prior	Fin.	Block
GDP_ZA	ZA Real GDP	log	WN	—	dom.
CONS_ZA	ZA Real Household Consumption	log	RW	—	dom.
INV_ZA	ZA Real Gross Fixed Capital Formation	log	RW	—	dom.
GOV_ZA	ZA Real Government Consumption	log	RW	—	dom.
EXP_ZA	ZA Real Exports	log	RW	—	dom.
IMP_ZA	ZA Real Imports	log	RW	—	dom.
CPI_ZA	ZA CPI Index	log	WN	—	dom.

ID	Description	Trans.	Prior	Fin.	Block
DEFL_ZA	ZA GDP Deflator (OECD QNA, 2020=100)	log	WN	—	dom.
LR_ZA	ZA Unemployment Rate	lin	RW	—	dom.
RATE_ZA	ZA 3-month interbank rate (JIBAR, %)	lin	RW	✓	dom.
YIELD10_ZA	ZA 10-Year Govt Bond Yield	lin	RW	✓	dom.
NEER_ZA	ZA Nominal Effective Exchange Rate (BIS broad) (index; rise = appreciation)	log	RW	✓	dom.
POLRATE_ZA	ZA Policy Rate (BIS CBPOL)	lin	RW	✓	dom.
HP_ZA	ZA Real Residential Property Price (BIS)	log	RW	✓	dom.
GLB_OIL	WTI crude oil, WTISPLC q-avg ($/bbl)	log	RW	✓	glob.
GLB_USLR	US 10Y Treasury yield (%)	lin	RW	✓	glob.
GLB_USBAA	US BAA corporate yield (%)	lin	RW	✓	glob.
GLB_USEQ	US share prices (OECD total share-price index, SPASTT01USM661N)	log	RW	✓	glob.
FREV_GDP	ZA Government Revenue (% of GDP)	lin	WN	—	fisc.
FPEXP_GDP	ZA Primary Expenditure (% of GDP)	lin	WN	—	fisc.
EFFI	ZA Effective Interest Rate on Gov Debt (% p.a.)	lin	RW	✓	fisc.
SFA	ZA Stock-Flow Adjustment (% of GDP, ann.)	lin	WN	—	fisc.
CREDIT_GDP_ZA	ZA Credit to NFCs	log	RW	✓	dom.
ROWDEM_ZA	ZA RoW Demand	lin	RW	✓	row
ROWINF_ZA	ZA RoW Inflation	lin	RW	✓	row
ROWFIN_ZA	ZA RoW Fin	lin	RW	✓	row

C Glossary

BVAR

Bayesian vector autoregression; here estimated under the hierarchical Minnesota prior of Giannone et al. (2015) (Section 3).

GVAR, bloc, star

The coupled cross-country system; one economy's model; a bloc's trade-weighted foreign aggregate (Section 5).

Endogenous dominant unit / frozen anchor

The US as a full member of the fixed point, with its own stars and weights row, versus the legacy one-shot exogenous US.

Global commons

The four US-determined series (oil, US 10y, US Baa, US equity) carried by every coupled bloc; the euro members do not carry them (Section 5.4).

COVID-as-missing
The 2020Q2–Q4 macro cells (2020Q1–Q4 in the euro members) treated as missing and drawn inside estimation; financial columns stay observed (Section 3.4).

Nowcast, amber
The model's estimate of a history cell that is not yet published (or that the publisher skipped), made by one coupled solve given everything observed; always drawn amber (Section 11.3).

NC
The label of the first forecast quarter in the conditioning list, the quarter after the panel quarter; a forecast, not an amber cell.

Hard / soft pin
An exact conditioning value versus an outside forecast held with trust τ, i.e. measurement variance (15) (Sections 4.2, 4.5).

Drift setter (δc)
The long-run anchor mechanism: an exact constant shift of a variable's companion drift (Section 8).

SFA, EFFI
The stock-flow adjustment (deficit–debt discrepancy) and the effective interest rate on the debt stock (Section 6).

Snowball
The automatic interest–growth term of the debt identity (25).

Coupling artifact, model signature
The precomputed $(I - M)^{-1}$ and the fingerprint that refuses it when stale (Section 5.3).

Data pack, manifest
The folder `app/public/data/` that holds everything the tool shows, and its `manifest.json`: the pack id, the panel quarter, each economy's dating, and every file's size and SHA-256 fingerprint (Section 11.2).

Update pack
A pack that carries only the files changed since a built-in pack and reuses the rest; desktop only (Section 11.2).

Panel quarter
The common last historical quarter of the coupled blocs (2026Q2 at this writing).

Data shift, model update
The quarterly advance of the panel by one quarter with the model unchanged, and the yearly re-estimation (Section 11.3).

Estimation mask
`estimation_mask.json`: per bloc, the cells that were missing in estimation and drawn by the

sampler, including the withheld COVID quarters.

CRPS, PIT, MCS
The continuous ranked probability score (41); the probability integral transform; the model confidence set (Section 12).

Drivers
The experimental, model-conditional shock decomposition of a scenario path: the mix of a chosen DSGE model's own structural shocks that reproduces the path over twelve quarters (Section 10).

Naming budget
The rule that at most q shocks may be named when q observables are watched; observables that are invertible transforms of one already watched do not raise it (Section 10.5).

Collinearity flag
The build-time warning on a shock pair whose stacked responses satisfy $|\cos| > 0.95$; the sum over such a pair is identified, the split within it is not (Section 10.5).

Class view and its licence
Aggregation of contributions into Monetary, Demand and Supply, permitted only at the one (model, observables, named shocks) configuration the recovery bench validated (Section 10.6).

Collinearity cluster
A connected component of the $|\cos| > 0.95$ graph, used in place of individual shocks on the FRB/US lenses; members are chained, not mutually parallel (Section 10.7).

Exact-fit / residual-able
The two decomposition regimes: named shocks spanning the watched observable paths, so the residual is zero by construction, versus a genuine unexplained dimension (Section 10.2).

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